

Verification of a Six Degree of Freedom Simulation Model
for the REMUS Autonomous Underwater Vehicle

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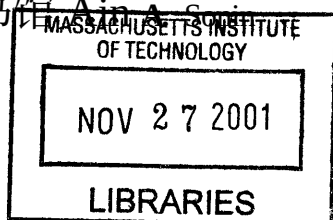
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Verification of a Six-DOF REMUS Autonomous Underwater Vehicle Simulation Model REMUS Autonomous Underwater Vehicle

by
Timothy Prestero

Submitted in partial fulfillment of the requirements for the degrees of
Master of Science in Ocean Engineering
and
Master of Science in Mechanical Engineering

Abstract

Improving the performance of a modular Autonomous Underwater Vehicle (AUV) for long range oceanographic survey, autonomous docking, and shallow water mine countermeasures requires improving the vehicles' maneuvering precision and battery life. These goals can be achieved through the improvement of the vehicle control system. A vehicle dynamics model based on a combination of theory and empirical data would provide an efficient platform for vehicle control system development, and an alternative to the typical trial-and-error method of vehicle control system field tuning. As there exists no standard procedure for vehicle modeling in industry, the simulation of each vehicle system represents a new challenge.

Developed by von Ahn Associates and the Woods Hole Oceanographic Institute, the REMUS AUV is a small, low-cost platform serving in a range of oceanographic applications. This thesis describes the development and verification of a six degree of freedom, non-linear simulation model for the REMUS vehicle, the first such model for this platform. In this model, the external forces and moments resulting from hydrostatics, hydrodynamic lift and drag, added mass, and the control inputs of the vehicle propeller and fins are all defined in terms of vehicle coefficients. This thesis describes the vehicle's inherent non-linear behavior, and how the equations determining the coefficients, as well as those describing the vehicle rigid-body dynamics, are left in non-linear form to better simulate the inherently non-linear behavior of the vehicle. Simulation of the vehicle motion is achieved through numeric integration of the equations of motion. The simulator output is then checked against vehicle dynamics data collected in experiments performed at sea. The simulator is shown to accurately model the motion of the vehicle.

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Candide had been wounded by some splinters of stone; he was stretched out in the street and covered with debris. He said to Pangloss: "Alas, get me a little wine and oil, I am dying."

"This earthquake is not a new thing," replied Pangloss. "The town of Lima suffered the same shocks in America last year; same causes, same effects; there is certainly a vein of sulfur underground from Lima to Lisbon."

"Nothing is more probable," said Candide, "but for the love of God, a little oil and wine."

"What do you mean, probable?" replied the philosopher. "I maintain that the matter is proved." Candide lost consciousness.

—《Candide》, Voltaire

Did I possess all the knowledge in the world, but had no love, how would this help me before God, who will judge me by my deeds?

—《效法基督》, Thomas à Kempis

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Contents

1 Introduction	12
1.1 Motivation	12
1.2 Vehicle Model Development	12
1.3 Research Platform	12
1.4 Model Code	13
1.5 Modeling Assumptions	13
1.5.1 Environmental Assumptions	13
1.5.2 Vehicle/Dynamics Assumptions	13
2 REMUS Autonomous Underwater Vehicle	14
2.1 Vehicle Profile	14
2.2 Sonar Transducer	15
2.3 Control Fins	15
2.4 Vehicle Weight and Buoyancy	15
2.5 Centers of Buoyancy and Gravity	17
2.6 Inertia Tensor	17
2.7 Final Vehicle Profile	17
3 Governing Equations	20
3.1 Body-Fixed Coordinate System Origin	20
3.2 Vehicle Kinematics	20
3.3 Vehicle Rigid-Body Dynamics	22
3.4 Vehicle Mechanics	23
4 Coefficient Derivation	24
4.1 Statics	24
4.2 Hydrodynamic Damping	25
4.2.1 Axial Drag	25
4.2.2 Crossflow Drag	26
4.2.3 Rolling Drag	26
4.3 Added Mass	27
4.3.1 Axial Added Mass	27
4.3.2 Crossflow Added Mass	28
4.3.3 Rolling Added Mass	29
4.3.4 Added Mass Cross-terms	29
4.4 Body Lift	29
4.4.1 Body Lift Force	31
4.4.2 Body Lift Moment	31
4.5 Fin Lift	31
4.6 Propulsion Model	33
4.6.1 Propeller Thrust	33
4.6.2 Propeller Torque	33
4.7 Combined Terms	33

4.8 总力和力矩 Forces and Moments	33	33
5 车辆拖曳 Tank Experiments	36	十六
5.1 动机 Motivation	36	36
5.2 实验设施与设备 Facilities and Equipment	36	36
5.2.1 拖曳水槽拖曳装置 Tow Tank Carriage	37	37
5.2.2 拖曳水槽拖曳装置 Tow Tank Carriage	37	37
5.3 阻力测试程序 Experimental Procedure	37	37
5.3.1 阻力运行 Instructions for Running	40	41
5.3.2 阻力结果 Drag Results	41	41
5.3.3 信号处理 Signal Processing	42	41
5.4 实验结果 Experimental Results	42	42
5.5 基于组件的阻力模型 Component-Based Drag Model	42	42
6 车辆模拟 Simulation	44	十四
6.1 耦合运动方程 Coupled Equations of Motion	44	44
6.2 数值积分 Numerical Integration of the Equations of Motion	46	47
6.2.1 欧拉方法 Euler's Method	46	46
6.2.2 改进欧拉方法 Improved Euler's Method	46	46
6.2.3 龙格-库塔方法 Runge-Kutta Method	46	46
6.3 计算机仿真 Computer Simulation	47	47
7 田野实验 Field Experiments	48	
7.1 动机 Motivation	48	48
7.2 测量状态 Measured States	48	48
7.3 传感器 Sensors	49	49
7.3.1 航向: 磁罗盘 Heading: Magnetic Compass	49	49
7.3.2 偏航: 音叉陀螺仪 Yaw Rate: Tuning Fork Gyro	49	49
7.3.3 姿态: 倾斜传感器 Attitude: Tilt Sensor	49	49
7.3.4 深度: 压力传感器 Depth: Pressure Sensor	49	49
7.4 实验程序 Experimental Procedure	49	49
7.4.1 发射前检查清单 Check List	50	50
7.4.2 修整和压载 Trim and Ballast	50	50
7.4.3 车辆任务编程 Vehicle Mission Programming	50	50
7.4.4 指南针校准 Compass Calibration	52	52
7.4.5 车辆跟踪 Vehicle Tracking	52	52
7.5 实验结果 Experimental Results	52	52
7.5.1 水平面动力学 Horizontal-Plane Dynamics	52	52
7.5.2 垂直面动力学 Vertical-Plane Dynamics	52	52
8 模型输出比较 Comparison of Simulator Output and Experimental Data	59	
8.1 模型准备 Model Preparation	59	59
8.1.1 初始条件 Initial Conditions	59	59
8.1.2 系数调整 Coefficient Adjustments	59	59
8.2 模型比较中的不确定性 Uncertainties in Model Comparison	60	60
8.3 水平面动力学 Horizontal Plane Dynamics	60	60
8.4 垂直面动力学 Vertical Plane Dynamics	60	60
8.4.1 车辆抬头 Vehicle Pitching Up	60	60
8.4.2 车辆下俯 Vehicle Pitching Down	71	71

9 线性化深度平面模型与控制器	77
9.1 车辆运动方程的线性化 Equations of Motion	77
9.1.1 车辆运动学 Kinematics	77
9.1.2 Vehicle Rigid-Body Dynamics 车辆刚体动力学	77
9.1.3 Vehicle Mechanics 车辆力学	78
9.2 线性化系数推导 Coefficient Derivation	79
9.2.1 静水力学 Statics	79
9.2.2 轴向阻力 Axial Drag	79
9.2.3 横流阻力 Crossflow Drag	79
9.2.4 附加质量 Added Mass	80
9.2.5 机体升力和力矩 Body Lift Force and Moment	81
9.2.6 鳍升力 Fin Lift	81
9.2.7 组合项 Combined Terms	82
9.2.8 线性化系数 Linearized Coefficients	82
9.3 线性化运动方程 Linearized Equations of Motion	82
9.3.1 运动方程 Equations of Motion	83
9.3.2 四项状态向量 Four-term State Vector	83
9.3.3 三项状态向量 Three-term State Vector	85
9.4 控制系统设计 Control System Design	85
9.4.1 车辆传递函数 Transfer Functions	85
9.4.2 控制律 Control Law	86
9.4.3 控制器设计程序 Controller Design Procedure	87
9.4.4 俯仰回路控制器增益 Pitch Loop Controller Gains	87
9.4.5 深度回路控制器增益 Depth Loop Controller Gains	88
9.5 现实世界现象 Real World Phenomena	88
9.6 控制器实现 Controller Implementation	88
10 结论 Conclusion	98
10.1 扩展拖车测量 Expanded Tank Measurements	98
10.2 海上未来实验 Future Experiments at Sea	98
10.2.1 改进的车辆仪器 Improved Vehicle Instrumentation	99
10.2.2 车辆参数测量 Measurement of Vehicle Parameters	99
10.2.3 车辆运动隔离 Isolation of Vehicle Motion	99
10.3 基于控制器的模型比较 Comparison of Controller-Based Models	99
10.4 车辆传感器模型 Vehicle Sensor Model	99
10.5 改进的基于系数的模型 Improved Coefficient-Based Model	100
A 参数表 Tables of Parameters	101
B Tables of Combined Non-Linear Coefficients	103
C Tables of Non-Linear Coefficients by Type	105
D 线性化模型参数表 Linearized Model Parameters	111
E MATLAB 代码 MATLAB Code	113
E.1 车辆仿真 Vehicle Simulation	113
E.1.1 REMUSSIM.m	113
E.1.2 REMUS.m	116
F 示例 REMUS Mission File	119
F.1 REMUS 任务代码 REMUS Mission Code	119

图表列表 Figures

2-1 我的轮廓 Profile	15
2-2 REMUS Low-Frequency Sonar Transducer (XZ-plane) 和 XZ 平面	16
2-3 REMUS Tail Fins (XY- and XZ-plane)	16
2-4 REMUS 车体固定坐标系和惯性坐标系	19
2-5 STD REMUS Profile (XZ-plane)	19
2-5 The REMUS Autonomous Underwater Vehicle	19
3-1 REMUS Body-Fixed and Inertial Coordinate Systems	21
4-1 有效舵角攻角 Rudder Angle of Attack	32
4-2 有效尾舵面角攻角 Effective Stern Plane Angle of Attack	32
5-1 URI 拖曳水箱布局	37
5-2 设置和车辆安装	38
5-3 Carriage Setup and Vehicle Mounting	39
5-4 URI Tow Tank Carriage	39
5-5 URI Tow Tank Carriage	40
5-6 碰撞图	41
5-7 REMUS 任务数据: 闭环控制	42
7-1 Vehicle Experiments at the Rutgers Marine Field Station	48
7-2 REMUS Pre-Launch Checklist (Page One)	51
7-3 REMUS Mission Data: Crash Plot	53
7-4 Vehicle Experiments at WHOI	54
7-5 The REMUS Ranger	54
7-6 REMUS Mission Data: Closed-Loop Control	55
7-7 REMUS Mission Data: Rudder	56
7-8 REMUS Mission Data: Pitching Up	57
7-9 REMUS Mission Data: Pitching Down	58
8-1 Horizontal Plane Simulation: Linear	61
8-2 Horizontal Plane Simulation: Angular	62
8-3 Horizontal Plane Simulation: Forces and Moments	63
8-4 Horizontal Plane Simulation: Vehicle Trajectory	64
8-5 Horizontal Plane Simulation: Model Comparison	65
8-6 Vertical Plane Simulation: Linear	66
8-7 Vertical Plane Simulation: Angular	67
8-8 Vertical Plane Simulation: Forces and Moments	68
8-9 Vertical Plane Simulation: Vehicle Trajectory	69
8-10 Vertical Plane Simulation: Model Comparison	70
8-11 Vertical Plane Simulation: Linear	72
8-12 Vertical Plane Simulation: Angular	73
8-13 Vertical Plane Simulation: Forces and Moments	74
8-14 Vertical Plane Simulation: Vehicle Trajectory	75
8-15 Vertical Plane Simulation: Model Comparison	76

9-1 干扰速度线性化 Velocity Linearization	80	9-2 深度平面控制系统框图	80
9-2 Depth-Plane Control System Block Diagram	80	9-3 G_θ 极点零点图	89
9-3 G_θ Pole-Zero Plot	89	9-4 G_θ 开环阶跃响应	89
9-4 G_θ Open-Loop Step Response	89	9-5 G_θ 根轨迹图	90
9-5 G_θ Root-Locus Plot	90	9-6 G_θ 闭环阶跃响应	90
9-6 G_θ Closed-Loop Step Response	90	9-7 $G_z * H_\theta$ 极点零点图	90
9-7 $G_z * H_\theta$ Pole-Zero Plot	90	9-8 $G_z * H_\theta$ 根轨迹图	91
9-8 $G_z * H_\theta$ Root-Locus Plot	91	9-9 $G_z * H_\theta$ 闭环阶跃响应	91
9-9 $G_z * H_\theta$ Closed-Loop Step Response	91	9-10 改进的深度平面控制系统框图	93
9-10 Modified Depth Plane Control System Block Diagram	93	9-11 Vehicle Simulation: Case One	94
9-11 Vehicle Simulation: Case One	94	9-12 Vehicle Simulation: Case Two	95
9-12 Vehicle Simulation: Case Two	95	9-13 Vehicle Simulation: Case Three	96
9-13 Vehicle Simulation: Case Three	96	9-14 Vehicle Simulation: Case Four	97
9-14 Vehicle Simulation: Case Four	97	10-1 在迎角下作用于车辆的力量	98
10-1 Force on vehicle at an angle of attack	98	10-2 随深度和海况变化的车辆性能极限	100
10-2 Vehicle performance limits as a function of depth and sea state	100		

表格列表Tables

2.1	Myring 参数用于STD REMUS REMUS	15
2.2	BEM 鳍片参数Parameters STD REMUS 重量和	17
2.3	STD REMUS Weight and Buoyancy	17
2.4	STD REMUS Center of Buoyancy	17
2.5	STD REMUS Center of Gravity	18
2.6	STD REMUS Moments of Inertia	18
2.7	STD REMUS Hull Parameters	18
4.1	轴向附加质量参数Parameters α and β STD REMUS 非线性操	28
4.2	非线性系数Non-Linear Maneuvering Coefficients: Forces	35
4.3	STD REMUS Non-Linear Maneuvering Coefficients: Moments	35
5.1	REMUS阻力分析	41
5.2	REMUS 基于组件的	41
5.3	REMUS 基于组件的阻力	43
5.4	REMUS 基于组件的阻力	43
7.1	车辆现场实验Experiments	50
8.1	REMUS 模拟器初始条件Initial Conditions	59
8.2	Vehicle Coefficient Adjustment Factors	60
9.1	线性化速度参数city.Parameters	79
9.2	线性化组合系数	79
9.3	线性化组合系数	83
9.4	Percent Overshoot and Damping Ratio	84
A.1	STD REMUS 船体参数Parameters	101
A.2	积分范围的船体坐标Limits of Integration	101
A.3	STD REMUS Center of Buoyancy	101
A.4	STD REMUS Center of Gravity	102
A.5	REMUS Fin Parameters	102
B.1	非线性力系数Force Coefficients	103
B.2	Non-Linear Moment Coefficients	104
C.1	轴向阻力系数Coefficient	105
C.2	Crossflow Drag Coefficients	105
C.3	滚动阻力系数Rolling Resistance Coefficient	105
C.4	船体升力及力矩系数Body Lift and Moment Coefficients	106
C.5	附加质量系数Added Mass Coefficients	106
C.6	附加质量力交叉项系数Added Mass Force Cross-term Coefficients	107
C.7	附加质量力交叉项系数Added Mass Force Cross-term Coefficients	107
C.8	附加质量M-, N-Moment Cross-term Coefficients	107
C.9	螺旋桨项Propeller Terms	109

C.10 控制系数	110
D.1 线性化组合系数	111
D.2 线性化机动系数	112

Chapter 1

Introduction

1.1 Motivation

在远程海洋调查无人水下载具和浅水水雷反制等应用中, 提高模块化、低成本自主水下航行器(UAV)的性能需要提升车辆的操纵精度和电池寿命。这些目标可以通过改进车辆控制系统来实现。基于理论和经验数据相结合的车辆动力学模型将帮助车辆控制系统开发提供高效的平台, 并且对替代典型的车辆控制系统现场调试的试错方法。由于工业中不存在车辆建模的标准程序, 每个车辆系统的仿真都代表着新的挑战。

Based on a combination of theory and empirical data would provide an efficient platform for vehicle control system development, and an alternative to the typical trial-and-error method of vehicle control system field tuning. As there exists no standard procedure for vehicle modeling in industry, the simulation of each vehicle system represents a new challenge.

1.2 车辆模型开发 Model Development

本文描述了描述用于 REMUS 车辆自由运动运动的仿真模型的开发与验证。在该模型中, 静水力学、流体升力和阻力、附加质量、流体螺旋桨和舵的控制输入、外部扰动等, 以及车辆系数的形成。

hydrostatics, hydrodynamic lift and drag, added mass, and the control inputs of the vehicle propeller and fins are all defined in terms of vehicle coefficients.

本文还描述了这些系数的推导, 并描述了车辆轴向阻力的实验测量, 并描述了实验测量和确定系数的方程。

This article describes the derivation of these coefficients and describes the experimental measurement of the vehicle axial drag.

确定系数的方程以及描述车辆运动方程的方程都保持非线性形式, 以更好地模拟车辆具有非线性动力学。车辆运动仿真通过求解运动方程进行数值积分来实现。然后, 将模拟器的输出与在现场收集的开环数据进行对比检查。这些现场数据侧重于车辆对控制鳍角阶跃变化的响应。结果表明, 模拟器能够准确地模拟车辆在六自由度下的运动。

The simulator output is then checked against open-loop data collected in the field. This field data measured the vehicle response to step changes in control fin angle. The simulator is shown to accurately model the vehicle motion in six degrees of freedom.

为了展示这项工作的前期应用, 本文展示了使用车辆模型的线性化版本来开发车辆深度平面控制系统。

To demonstrate the early application of this work, a linearized version of the vehicle model to develop a vehicle depth-plane control system.

总之, 本文讨论了进一步实验验证车辆系数的计划, 包括牵引水池升力和阻力测量, 以及在车辆开环运动传感器动态的精密惯性测量。

In closing, this article discusses the plan for further experimental validation of the vehicle coefficients, including towing tank lift and drag measurements, and precision inertial measurements of the vehicle open-water motion and sensor dynamics.

1.3 研究平台 Research Platform

本研究平台是由冯奥特及其同事在伯兹霍尔海军研究所海洋系统实验室开发的 REMUS 系列。REMUS 是一种低成本、模块化的车辆, 用于远程海洋调查和浅水水雷反制。REMUS 车辆的规格请参见第 2 章。

This research platform was developed by Fong et al. at the Naval Research Laboratory's Marine Systems Laboratory. REMUS is a low-cost, modular vehicle with applications in autonomous docking, long-range oceanographic survey, and shallow-water mine reconnaissance [30]. See Chapter 2 for the specifications of the REMUS vehicle.

REMUS 目前使用基于现场调节的 PID 控制器。之前尝试将更先进的控制器应用于 REMUS 遭遇了困难，因为缺乏用于描述车辆动力学的数学模型。of a mathematical model to describe the vehicle dynamics.

1.4 模型代码 Code

作者使用 MATLAB 开发了模拟器代码。尽管与其他编译器相比，MATLAB 运行较慢，但该程序极大地方便了数据可视化。在开发代码时，作者没有使用任何 MATLAB 特定的函数，因此将模型代码导出到另一种更快速的语言进行控制器开发将很容易。author did not use any MATLAB-specific functions, so exporting the model code to another, faster language for controller development will be easy.

1.5 建模假设 Modeling Assumptions

为了简化自主水下航行器建模的挑战，有必要对模型开发的基础做出一些假设，it is necessary to make some assumptions on which to base the model development.

1.5.1 环境假设 Environmental Assumptions

作者对该车辆及其环境作出了以下假设。The author made the following assumptions about the vehicle with respect to its environment:

- * *The vehicle is deeply submerged in a homogeneous, unbounded fluid.* 换句话说，车辆位于远离自由表面（无表面效应，即无波浪或车辆造波载荷）i.e. 墙壁和底部的位置。loads), walls and bottom.
- * *The vehicle does not experience memory effects.* 模拟器忽略了车辆穿越自身尾流的影响。
- *The vehicle does not experience memory effects.* The simulator neglects the effects of the vehicle passing through its own wake.
- * *The vehicle does not experience underwater currents.*

1.5.2 车辆动力学假设 Vehicle Dynamics Assumptions

作者对车辆本身作出了以下假设。The author made the following assumptions about the vehicle itself:

- * *The vehicle is a rigid body of constant mass.* 换句话说，车辆的质量和质心在操作过程中不会变化。我们假设控制鳍无论迎角如何都不会失速。我们还假设鳍的响应是瞬时的。distribution do not change during operation.
- *Control surface assumptions.* We assume that the control fins do not stall regardless of angle of attack. 这意味着车辆执行器的响应时间相比车辆姿态的响应时间很短。我们将使用一个极其简单的推进模型，该模型将车辆的螺旋桨视为恒定推力和扭矩的来源。that that vehicle actuator time response is small in comparison with the vehicle attitude time response.
- *Thruster assumptions:* We will be using an extremely simple propulsion model, which treats the vehicle propeller as a source of constant thrust and torque.
- *There exist no significant vehicle dynamics faster than 45 Hz (the modeling time step).*

Chapter 2

REMUS Autonomous Underwater Vehicle

In order to calculate the vehicle parameters, we must first define the vehicle profile, determine the mass and mass distribution, and finally identify the necessary control fin parameters.

2.1 Vehicle Profile

The REMUS vehicle body profile is defined by the following equations. These equations describe the vehicle profile in terms of the maximum diameter and drag coefficient. These equations have been modified so as to be defined in terms of the following parameters:

- $a, b,$ and c are the nose-section, the constant center-section, and tail-section radius center-section, respectively
- m , an exponential parameter to give different body shapes.
- 2θ , tail section angle at the tip of the tail
- d , maximum body diameter

These equations assume the origin at the nose of the vehicle.

Nose shape is given by the semi-elliptical radius distribution

$$r(\Xi) = \frac{1}{2}d \left[1 - z \left(\frac{\text{offset} - a}{a} \right)^2 \right]^{\frac{1}{m}} \quad (2.1)$$

where r is the radius measured perpendicular to the centerline, z is the axial position, a is the nose radius, b is the constant radius, c is the tail radius, θ is the tail angle, and m is the exponential parameter. The vehicle coordinate system is shown in Figure 3-1 for a diagram of the vehicle coordinate system.

The tail shape is given by the equation

$$r(E) = \frac{1}{2}d - \left[\frac{F_0}{2c} - \frac{\tan\theta}{c} \right] (\Xi - l)^2 + \left[\frac{d}{c^3} - \frac{\tan\theta}{c^2} \right] (\Xi - l_f)^3 \quad (2.2)$$

where l_f is the forward body length

$$l_f = a + b - \text{offset} \quad (2.3)$$

where r is the vehicle radius, z is the axial position, a is the nose radius, b is the constant radius, c is the tail radius, θ is the tail angle, and m is the exponential parameter. The vehicle coordinate system is shown in Figure 3-1 for a diagram of the vehicle coordinate system.

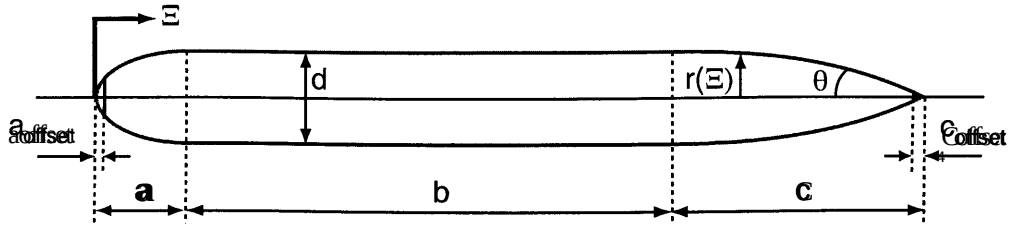


Figure 2.2 Myring Profile. Vehicle hull radius as a function of axial position

作为参考, Myring [22] 第 189 页] 假设总体长度为 100 单位, 并通形式为 $r(x) = \frac{d}{2} \left(1 - \frac{x}{l} \right)^n$ 的 $r(x)$ 来分类类型, 其中 d 为最大直径, n 为指数, l 为总长度。REMUS 基于 Myring B 型轮廓, 该轮廓由代码 15/55/1.25/0.4363/5 给出。Table 2.1 gives the dimensionalized Myring parameters.

Table 2.1 Myring Parameters for STD REMUS

Parameter	Value	Units	Description
a	+1.91e-001	m	Nose Length
a_{offset}	+1.65e-002	m	Nose Offset
b	+6.54e-001	m	Midbody Length
c	+5.41e-001	m	Tail Length
c_{offset}	+3.68e-002	m	Tail Offset
n	+2.00	n/a	Exponential Coefficient
θ	+4.36e-001	radians	Included Tail Angle
d	+1.91e-001	m	Maximum Hull Diameter
l_f	+8.28e-001	m	Vehicle Forward Length
l	+1.33e+000	m	Vehicle Total Length

2.2 声纳换能器 Sonar Transducer

REMUS 车辆配备有一个前置声纳换能器, 该换能器为直径为 10 厘米 (4 英寸) 的圆柱体。其余换能器的尺寸见图 2.2。Transducer dimensions are given in Figure 2-2.

2.3 控制翼 Control Fins

REMUS 载具配备了四个相同的控制翼, 安装在船体尾部附近, 呈矩形排列。这些鳍具有 π 弧度 (180 度) 的横截面, 其余尺寸见图 2.3。相关的鳍参数见表 2.2。

2.4 车辆重量与浮力 Weight and Buoyancy

REMUS 车辆的重量可能会因任务而异, 这取决于所使用的电池类型和添加的压载物数量。REMUS 通常配重约 1.5 磅的浮力, 以便在计算机或电源故障的情况下最终浮到水面。车辆重量和浮力的典型值列于表 2.3。

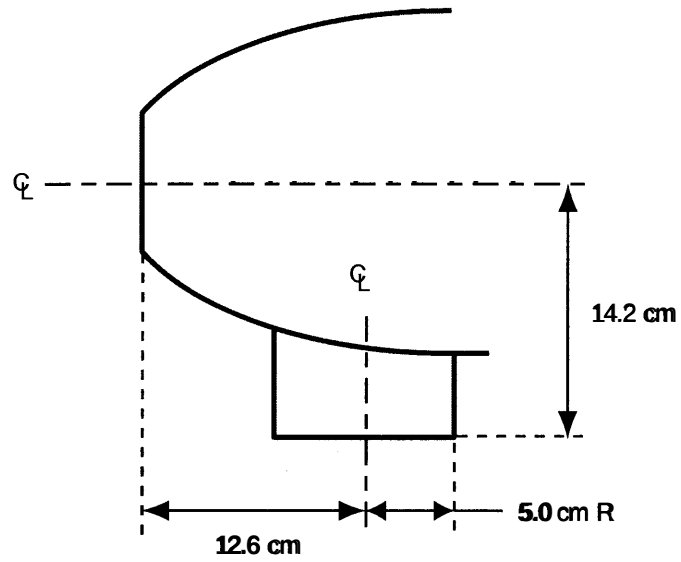


图2-2:2-REMUS 低频声呐换能器 (XZ平面)
Fig.2-2:2-REMUS Low Frequency Sonar Transducer (XZ-plane)

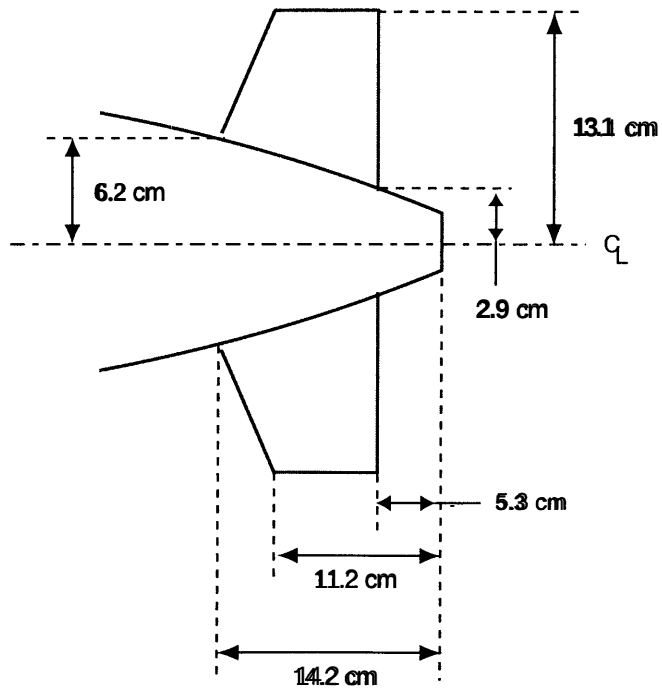


图2-3:2-REMUS 尾鳍 (XY平面和 XZ平面)
Fig.2-3:2-REMUS Tail (XY and XZ plane)

Table 2. REMUS 翼鳍参数

Parameter	Value	Units	Description
S_{fin}	+6.65e-003	m ²	Planform Area
b_{fin}	+8.57e-002	m	Span
$x_{finpost}$	-6.38e-001	m	Moment Arm wrt Vehicle Origin at CB
δ_{max}	+1.36e+001	deg	Maximum Fin Angle
a_{fin}	+5.14e+000	m	Max Fin Height Above Centerline
C_{mean}	+7.47e-002	m	Mean Chord Length
t	+6.54e-001	n/a	Fin Taper Ratio (Whicker-Felner)
c_{df}	+5.58e-001	n/a	Fin Crossflow Drag Coefficient
AR_e	+2.21e+000	n/a	Effective Aspect Ratio
$\bar{a}$	+9.00e-001	n/a	Lift Slope Parameter
CL_a	+3.12e+000	n/a	Fin Lift Slope

Table 2.3 车辆重量和浮力

Parameter	Value	Units
W	+2.99e+002	N
B	+3.06e+002	N

2.5 浮心和重心 Buoyancy and Gravity

在现场操作 REMUS 时对于特定的 REMUS 车辆, 由于很少对船体外部进行更改, 其浮心位置大致保持不变, 另一方面, 当车辆重量可能有所变动时, 由于并不清楚任何变动, 通常只需要更换电池组并重新配重车辆。另一方面, 在任务之间, 经常需要更换电池组并重新配重车辆。

The average values are given in Tables 2.4 and 2.5.

Table 2.4 浮心相对于车辆前端的原点

Parameter	Value	Units
x_{cb}	-6.11e-001	m
y_{cb}	+0.00e+000	m
z_{cb}	+0.00e+000	m

2.6 惯性张量 Inertia Tensor

车辆惯性张量是相对于位于车辆浮心的车体固定原点的, 固定于惯性系 z_{cb} 和车辆中心浮心 z_{cb} 和 z_{cb} 的乘积, 我们假设它们为零, 实际上假设车辆具有两个轴对称平面。我们假设它们为零, 实际上假设车辆具有两个轴对称平面。

这些数值是根据车辆重量清单, 列出各个车辆内部组件的位置和重量的表格估算的。虽然重量变化显然会影响车辆的惯性矩, 但我们假设这些变化足够小, 可以忽略不计。估算值见表 2.6。

2.7 最终车辆概况 Final Vehicle Profile

图 2.4 显示了完整车辆剖面, 并绘制在圆柱体上以供参考。表 2.7 给出了一些额外的船体参数, 大多为船体几何的函数, 是船体几何的函数, 是船体几何的函数。

Table 2: 重心Center of Gravity的位置 Origin at CB

Parameter	Value	Units
x_{cg}	+0.00e+000	m
y_{cg}	+0.00e+000	m
z_{cg}	+1.96e-002	m

Table 2.3: 关于Moments的转动惯量wrt Origin at CB

Parameter	Value	Units
I_{xx}	+1.77e001	$\text{kg} \cdot \text{m}^2$
I_{yy}	+3.45e+000	$\text{kg} \cdot \text{m}^2$
I_{zz}	+3.45e+000	$\text{kg} \cdot \text{m}^2$

注意, 车辆浮力和总浮力位置的估算仅基于裸船体轮廓, 并未考虑车辆的鳍和容器, 也未考虑车辆首盖中进水的部分。表 2.4 中给出的 x_{cb} 值以及表 2.3 中给出的总浮力 B 是基于实验测量得出的。The flooded sections in the vehicle nose cap. The x_{cb} value given in Table 2.4 and the total buoyancy B given in Table 2.3 are based on experimental measurements.

Table 2.7: STD REMUS 艇体参数

Parameter	Value	Units	Description
ρ	+1.03e+003	kg/m^3	Seawater Density
A_f	+2.85e-002	m^2	Hull Frontal Area
A_p	+2.26e-001	m^2	Hull Projected Area (xz plane)
S_w	+7.09e-001	m^2	Hull Wetted Surface Area
V	+3.15e-002	m^3	Estimated Hull Volume
B_{est}	+3.17e+002	N	Estimated Hull Buoyancy
$x_{cb}(est)$	+5.54e-003	m	Est. Long. Center of Buoyancy

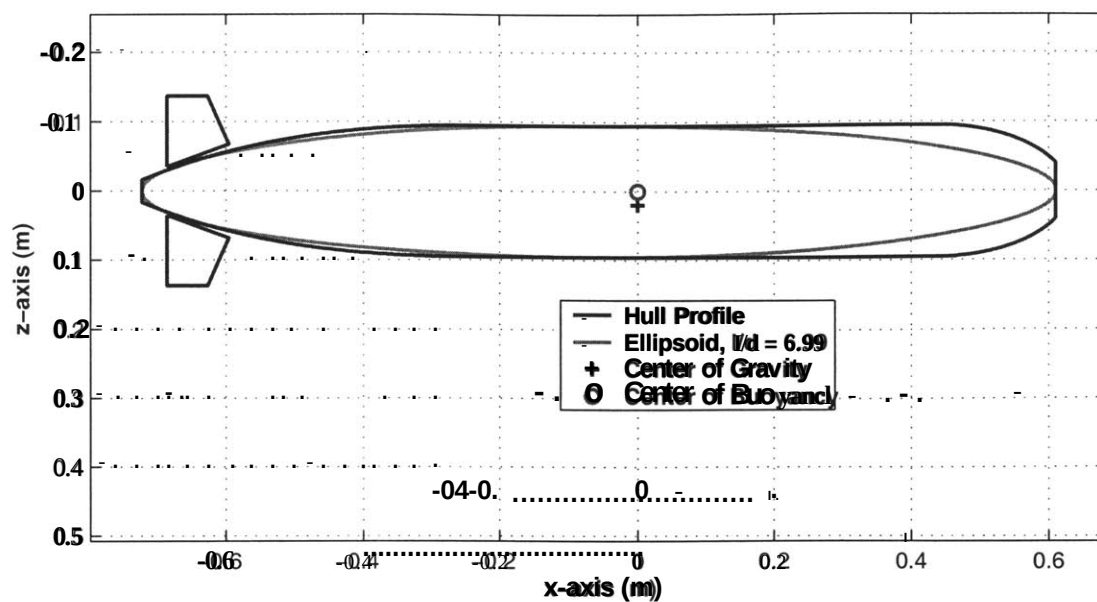


图2-4:2 STD REMUS轮廓 (x-z 平面) plane)

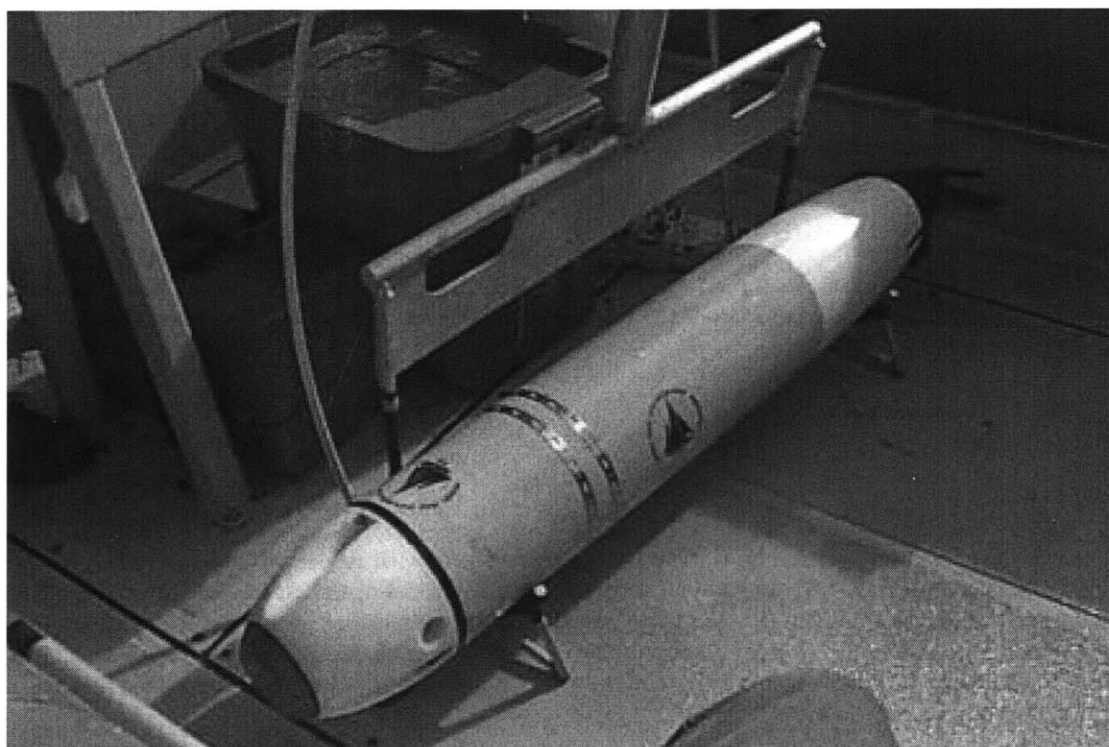


图2-5:2 REMUS自主水下航行器 Autonomous Underwater Vehicle

Chapter 3

控制方程的要素 The Governing Equations

在本章中，我们定义了控制车辆运动的方程。这些方程包括以下要素。 These equations consist of the following elements:

- **Kinematics:** 运动的几何方面 aspects of motion
- **Rigid-body Dynamics:** 车辆惯性矩阵 the vehicle inertia matrix
- **Mechanics:** 引起运动的力和力矩 forces and moments causing motion

这些元素将在以下章节中进行讨论。 These elements are addressed in the following sections.

3.1 车身固定坐标系原点 Vehicle Coordinate System Origin

请注意，在所有后续计算中，车辆车身固定坐标系的原点位于车辆的浮心，如第2.5节所定义，并在图2-4中所示。 Please note that in all subsequent calculations, the origin of the vehicle body-fixed coordinate system is located at the vehicle center of buoyancy, as defined in Section 2.5 and illustrated in Figure 2-4.

3.2 车辆运动学 Vehicle Kinematics

在惯性参考系或地球固定参考系中，描述刚体固定参考系的运动。车辆在六自由度的一般运动可以通过以下向量来描述。 The general motion of the vehicle in six degrees of freedom can be described by the following vectors:

$$\begin{aligned} \eta_1 &= [\mathbf{x} \ \mathbf{y} \ \mathbf{z}]^T; & \eta_2 &= [\phi \ \theta \ \psi]^T \\ \mathbf{v}_1 &= [\mathbf{u} \ v \ w]^T; & \mathbf{v}_2 &= [p \ q \ r]^T \\ \boldsymbol{\tau}_1 &= [\mathbf{X} \ \mathbf{Y} \ \mathbf{Z}]^T; & \boldsymbol{\tau}_2 &= [\mathbf{K} \ \mathbf{M} \ \mathbf{N}]^T \end{aligned}$$

其中 η_1 描述了车辆相对于惯性或地固定参考系的位置和姿态， η_2 描述了车辆相对于地固定参考系的平移和旋转速度。描述了作用在车辆上的总力和力矩相对于地固定参考系。车辆坐标系的示意图见图3-1。 See Figure 3-1 for a diagram of the vehicle coordinate system.

以下变换将机体固定坐标系和惯性或地固定坐标系之间的平移速度联系起来。 The following coordinate transformation relates the translation velocities between body-fixed and inertial or earth-fixed coordinates:

$$\begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\mathbf{y}} \\ \dot{\mathbf{z}} \end{bmatrix} = \mathbf{J}_1(\eta_2) \begin{bmatrix} u \\ v \\ w \end{bmatrix} \quad (3.1)$$

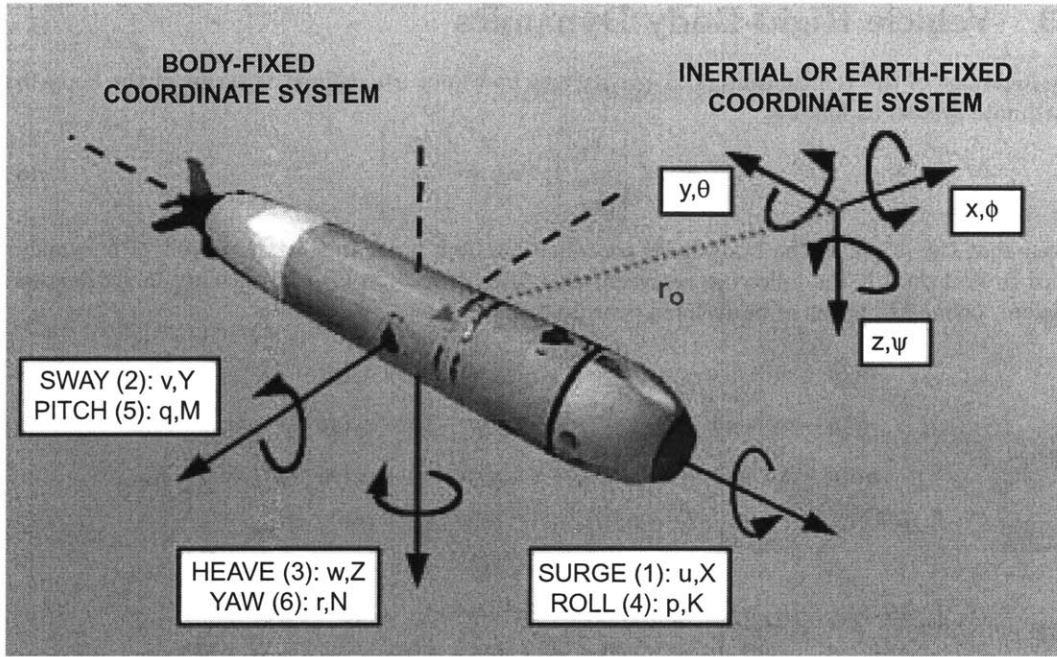


Figure 3: REMUS Body-Fixed and Inertial Coordinate Systems

哪里

$$\mathbf{J}_1(\eta) = \begin{bmatrix} \cos \psi \cos \theta & -\sin \psi \cos \theta & \sin \psi \sin \theta & \sin \theta \sin \phi & \cos \psi \sin \theta \cos \phi & \sin \psi \sin \theta \cos \phi \\ \sin \psi \cos \theta & \cos \psi \cos \theta & \cos \psi \sin \theta & \sin \theta \cos \phi & -\cos \psi \sin \theta \cos \phi & \sin \psi \sin \theta \cos \phi \\ -\sin \theta & \cos \theta & \sin \theta & \cos \theta \sin \phi & \cos \theta \cos \phi & \sin \theta \cos \phi \end{bmatrix} \quad (3.2)$$

注意 $\mathbf{J}_1(\eta)$ 是正交的 (orthogonal):

$$(\mathbf{J}_1(\eta))^{-1} = (\mathbf{J}_1(\eta))^T \quad (3.3)$$

第二个坐标变换将机体固定坐标系与地球固定坐标系之间的旋转速度联系起来: and earth-fixed coordinates:

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \mathbf{J}_2(\eta) \begin{bmatrix} \mathbf{p} \\ \mathbf{q} \\ \mathbf{r} \end{bmatrix} \quad (3.4)$$

哪里

$$\mathbf{J}_2(\eta) = \begin{bmatrix} 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi / \cos \theta & \cos \phi / \cos \theta \end{bmatrix} \quad (3.5)$$

注意, $\mathbf{J}_2(\eta)$ 在俯仰角为 $\pm 90^\circ$ 时未定义。这不是问题, 因为车辆运动通常不会接近此奇异点。如果我们处于必须模拟车辆通过极端俯仰角运动的情况, 我们可以采用其他运动学表示方法, 例如四元数或罗德里格斯参数 [17]。

3.3 车辆刚体运动 Rigid-Body Dynamics

车辆重心和浮力中心的位置在体固定坐标系中定义如下, buoyancy are defined in terms of the body-fixed coordinate system as follows:

$$\mathbf{r}_G = \begin{bmatrix} x_g \\ y_g \\ z_g \end{bmatrix} \quad \mathbf{r}_B = \begin{bmatrix} x_b \\ y_b \\ z_b \end{bmatrix} \quad (3.6)$$

鉴于第2.1节所述, 固定于物体的坐标系的原点平动系统, 以物体固定坐标系运动的刚体 as a rigid body in six degrees of freedom, the following represent equations of motion for a rigid body in six degrees of freedom, defined in terms of body-fixed coordinates:

$$\begin{aligned} m [\ddot{x} - v\dot{r} + w\dot{q} - x_g(\dot{q}^2 + r^2) + y_g(\dot{p}\dot{q} - \dot{r}) + z_g(\dot{p}\dot{r} + \dot{q})] &= \sum X_{\text{ext}} \\ m [\ddot{y} - w\dot{p} + ur - y_g(\dot{r}^2 + \dot{p}^2) + z_g(\dot{q}\dot{r} - \dot{p}) + x_g(\dot{q}\dot{p} + \dot{r})] &= \sum Y_{\text{ext}} \\ m [\ddot{z} - u\dot{q} + vp - z_g(\dot{p}^2 + \dot{q}^2) + x_g(\dot{r}\dot{p} - \dot{q}) + y_g(\dot{r}\dot{q} + \dot{p})] &= \sum Z_{\text{ext}} \\ I_{xx}\dot{p} + (I_{zz} - I_{yy})qr - (\dot{r} + p\dot{q})I_{xz} + (\dot{r}^2 - \dot{q}^2)I_{yz} + (p\dot{r} - \dot{q})I_{xy} \\ &+ m[y_g(\dot{u} - u\dot{q} + v\dot{p}) - z_g(\dot{v} - w\dot{p} + ur)] = \sum K_{\text{ext}} \\ I_{yy}\dot{q} + (I_{xx} - I_{zz})rp - (\dot{p} + q\dot{r})I_{xy} + (\dot{p}^2 - \dot{r}^2)I_{xz} + (\dot{q}\dot{p} - \dot{r})I_{yz} \\ &+ m[z_g(\dot{u} - u\dot{q} + v\dot{p}) - x_g(\dot{v} - w\dot{p} + ur)] = \sum M_{\text{ext}} \\ I_{zz}\dot{r} + (I_{yy} - I_{xx})pq - (\dot{q} + r\dot{p})I_{yz} + (\dot{q}^2 - \dot{p}^2)I_{xy} + (\dot{r}\dot{q} - \dot{p})I_{xz} \\ &+ m[x_g(\dot{v} - w\dot{p} + ur) - y_g(\dot{u} - u\dot{q} + v\dot{p})] = \sum N_{\text{ext}} \end{aligned} \quad (3.7)$$

其中 m 是车辆质量, 前五个方程表示平动运动, 最后一个方程表示旋转运动。请注意, 这些方程忽略零值的浮心项。three rotational motion. Note that these equations neglect the zero-valued center of buoyancy terms.

给定以车辆重心为原点的机体固定坐标系, 我们得到如下对角惯性张量。center of buoyancy, we have the following, diagonal inertia tensor.

$$\mathbf{I} = \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix}$$

这是基于第2.6节中提出的假设, 即车辆的惯性积很小。2.6, that the vehicle products of inertia of inertia are small.

这简化了运动方程为 equations of motion to the following:

$$\begin{aligned} m [\ddot{x} - v\dot{r} + w\dot{q} - x_g(\dot{q}^2 + r^2) + y_g(\dot{p}\dot{q} - \dot{r}) + z_g(\dot{p}\dot{r} + \dot{q})] &= \sum X_{\text{ext}} \\ m [\ddot{y} - w\dot{p} + ur - y_g(\dot{r}^2 + \dot{p}^2) + z_g(\dot{q}\dot{r} - \dot{p}) + x_g(\dot{q}\dot{p} + \dot{r})] &= \sum Y_{\text{ext}} \\ m [\ddot{z} - u\dot{q} + vp - z_g(\dot{p}^2 + \dot{q}^2) + x_g(\dot{r}\dot{p} - \dot{q}) + y_g(\dot{r}\dot{q} + \dot{p})] &= \sum Z_{\text{ext}} \\ I_{xx}\dot{p} + (I_{zz} - I_{yy})qr + m[y_g(\dot{u} - u\dot{q} + v\dot{p}) - z_g(\dot{v} - w\dot{p} + ur)] &= \sum K_{\text{ext}} \\ I_{yy}\dot{q} + (I_{xx} - I_{zz})rp + m[z_g(\dot{u} - u\dot{q} + v\dot{p}) - x_g(\dot{v} - w\dot{p} + ur)] &= \sum M_{\text{ext}} \\ I_{zz}\dot{r} + (I_{yy} - I_{xx})pq + m[x_g(\dot{v} - w\dot{p} + ur) - y_g(\dot{u} - u\dot{q} + v\dot{p})] &= \sum N_{\text{ext}} \end{aligned} \quad (3.8)$$

通过假设 y_g 相对于其他项较小, 我们可以进一步简化这些方程, 考虑到 y_g 是车辆内部组件的。考虑车辆进行特殊配重, m 则 y_g 项可以忽略不计。US 导致车辆刚体的 y_g 是 specially ballasted y_g is in fact negligible. This results in the following equations for the vehicle rigid body

ynamics:

$$\begin{aligned}
 m [\dot{u} - v r + w p q - x y (q^2 + r^2) + z g (p r + \dot{q})] &= \sum X_{\text{ext}} \\
 m [\dot{v} - w p p + u r + z g (q r - p \dot{q}) + x g (q p + \dot{r})] &= \sum Y_{\text{ext}} \\
 m [\dot{w} - u q + v p - x g (p^2 + q^2) + x y (r p - \dot{q})] &= \sum Z_{\text{ext}} \\
 I_{xx} \dot{p} + (I_{zz} - I_{yy}) q r + m [-z g (\dot{v} - w p + u r)] &= \sum K_{\text{ext}} \\
 I_{yy} \dot{q} + (I_{xx} - I_{zz}) r p + m [z g (\dot{u} - v r + w p q) - x g (\dot{w} - u q + v p)] &= \sum M_{\text{ext}} \\
 I_{zz} \dot{r} + (I_{yy} - I_{xx}) p q + m [x g (\dot{v} - w p + u r)] &= \sum N_{\text{ext}}
 \end{aligned} \tag{3.9}$$

3.4 车辆力学 Vehicle Mechanics

在车辆运动方程中, 外力和力矩, external forces and moments

$$\sum \mathbf{F}_{\text{ext}} = \mathbf{F}_{\text{hydrostatic}} + \mathbf{F}_{\text{lift}} + \mathbf{F}_{\text{drag}} + \mathbf{F}_{\text{control}}$$

以车辆系数来描述, 例如, 轴向阻力 coefficients. For example, axial drag

$$F_d = - \left(\frac{d}{2} \rho C_d A_f \right) \frac{u}{|u|} = X_{u|u|} \frac{u}{|u|} \implies X_{u|u|} = \frac{\partial F_d}{\partial (u|u|)} = - \frac{\rho h}{2} C_d A_f$$

These coefficients are based on a combination of theoretical equations and empirical formulae. The actual values of these coefficients are derived Chapter 4.

Chapter 4

Coefficient Derivation

在本章中,我们推导了定义车辆上力和力矩的系数。计算每个系数所需的车辆和流体参数值包含在描述该系数的章节中,要么在附录A中列出。The vehicle weight and buoyancy are expressed as $W = mg$ and $B = \rho \nabla g$, where ρ is the density of the surrounding fluid and ∇ the total volume displaced by the vehicle.

4.1 Hydrostatics

由于车辆重量和浮力的共同作用,车辆会受到静水力和力矩的作用。假设车辆的质量为 m , 重量 $W = mg$ 。车辆的浮力表示为 $B = \rho \nabla g$, 其中 ρ 是周围流体的密度, ∇ 是车辆排开的总体积。weight $W = mg$. The vehicle buoyancy is expressed as $B = \rho \nabla g$, where ρ is the density of the surrounding fluid and ∇ the total volume displaced by the vehicle.

有必要将这些力和力矩用体坐标系表示,这可以通过使用方程3.2中给出的变换矩阵来完成。This is accomplished using the transformation matrix given in Equation 3.2:

$$\mathbf{f}_G(\mathbf{1}_2) = J_1^{-1} \begin{bmatrix} 0 \\ 0 \\ W \end{bmatrix} \quad \mathbf{f}_B(\mathbf{1}_2) = J_h^{-1} \begin{bmatrix} 0 \\ 0 \\ B \end{bmatrix} \quad (4.1)$$

The hydrostatic forces and moments on the vehicle can be expressed as:

$$\begin{aligned} \mathbf{F}_{HS} &= \mathbf{f}_G - \mathbf{f}_B \\ \mathbf{M}_{HS} &= \mathbf{r}_G \times \mathbf{f}_G - \mathbf{r}_B \times \mathbf{f}_B \end{aligned} \quad (4.2)$$

These equations can be expanded to give linear equations for hydrostatic forces and moments:

$$\begin{aligned} X_{HS} &= (W - B) \sin \theta \\ Y_{HS} &= (W - B) \cos \theta \sin \phi \\ Z_{HS} &= (W - B) \cos \theta \cos \phi \\ K_{HS} &= -(y_g W - y_b B) \cos \theta \cos \phi - (z_g W - z_b B) \cos \theta \sin \phi \\ M_{HS} &= -(z_g W - z_b B) \sin \theta - (x_g W - x_b B) \cos \theta \cos \phi \\ N_{HS} &= -(x_g W - x_b B) \cos \theta \sin \phi - (y_g W - y_b B) \sin \theta \end{aligned} \quad (4.3)$$

注意,静水力矩在外倾和横倾方向上是稳定的,这意味着静水力矩会抵抗这些角度方向的偏转。Note, the hydrostatic roll and yaw moments are stabilizing, meaning that the hydrostatic moment opposes deflections in those angular directions.

4.2 水动力阻尼 Hydrodynamic Damping

众所周知, 在六自由度下高速运动的水下航行器的阻尼是耦合且高度非线性的。为了简化对航行器的建模, 我们将作出以下假设:

- We will neglect linear and angular coupled terms. 我们将假设诸如 $\dot{U}Y$ 之类的项相对较小, 计算这些项超出了本工作的范围。
- We will assume the vehicle is torus (or cylinder) and not starboard (or starboard) symmetric. 我们将忽略由换能器引起的车辆不对称性, 这使我们能够忽略由阻力引起的力矩。这允许我们忽略诸如 $\dot{U}Y$ 之类的高阶项。
- We will neglect any damping terms greater than second-order. This will allow us to drop such higher-order terms as Y_{vv} .

水动力阻尼的主要组成部分是边界层产生的表面摩擦, 这些边界层部分为层流部分, 湍流, 以及由涡流脱落引起的阻力。无量纲分析帮助我们预测车辆周围的流动类型。雷诺数表示惯性力与粘性力的比值, 其表达式为

$$Re = \frac{U l}{\nu} \quad (4.4)$$

其中 U 是车辆运行速度, 通常为 105 KTS (节), 通常是特征长度 (3 knots = 1.67 m/s), 流体粘度, 海水在 15°C 时的值为 $1.190 \times 10^{-6} \text{ m}^2/\text{s}$ 。Newman [24] 给出的值为 $1.190 \times 10^{-6} \text{ m}^2/\text{s}$ 。

这产生了一个雷诺数 $Re \approx 1.3 \times 10^6$ 。对于表面光滑的物体, 这落在层流和湍流之间的过渡区。然而, 由于车辆上的许多接缝、凹痕和凸起, 这很可能使车辆周围的流动进入湍流状态。我们可以利用这些信息来估算车辆的阻力系数。我们可以使用这些信息来估计车辆的阻力系数。

请注意, 粘性阻力总是与车辆运动相反, 为了得到正确的符号, 在所有阻力方程中都必须考虑 $v|v|$, 而不是 v^2 。

4.2.1 轴向阻力 Axial Drag

车辆轴向阻力可以用以下经验关系来表示 following empirical relationship:

$$X = - \left(\frac{1}{2} \rho c_d A_f \right) u |u| \quad (4.5)$$

该方程得出以下非线性轴向阻力系数: linear axial drag coefficient:

$$X_{u|u} = - \frac{1}{2} \rho c_d A_f \quad (4.6)$$

其中 ρ 是周围流体的密度, A_f 是车辆正面面积, c_d 是车辆的轴向阻力系数, and c_d the axial drag coefficient of the vehicle.

Bottaccini [7, 第 206 页], Horrocks [15, 第 3-12 页] 和 Triantafyllou [29] 提供了计算轴向阻力系数的经验公式。例如, Triantafyllou:

$$c_d = \frac{c_{ss} \pi A_p}{c_{ss} \pi A_p} \left[1 + 60 \left(\frac{d}{l} \right)^3 + 0.0025 \left(\frac{l}{d} \right) \right] \quad (4.7)$$

其中 c_{ss} 是 Schlichting 对平板表面摩擦的估计值, A_p 是车辆的平面面积, A_f 是车辆的正面面积 (frontal area)。From *Principles of Naval Architecture* [26], 我们得到 c_{ss} 的估计值为 3.397×10^{-3} 。

这些经验方程得出的 c_{df} 值在0.1到0.2的范围内。在海洋系统实验室进行的水池测量实验 conducted 具有推进效率0.4的 c_{df} 的估计值为0.2。Lab measuring the propulsion efficiency of the vehicle resulted in an estimate for c_{df} of 0.2.

作者在罗德岛大学进行的车辆轴向阻力水池测量, 如第5章中描述, 得到的轴向阻力系数为0.27。这个较高的数值反映了车辆船体的阻力以及经验估算中忽略的阻力来源, 例如车辆的鳍片和声纳发射器, 以及车辆前端区域的空腔。我们将在车辆仿真中使用这个更高的、实验测得的数值。 This higher value reflects the drag of the vehicle hull plus the drag of sources neglected in the empirical estimate, such as the vehicle fins and sonar transponder, and the pockets in the vehicle nose section. We will use this higher, experimentally-measured value in the vehicle simulation.

请参见表C.1 获取 c_{df} 的最终值

4.2.2 横流阻力 Crossflow Drag

车辆横流阻力被定义为船体横流阻力与鳍横流阻力的总和。横流阻力 c_{df} 的计算方法类似于计算船体附加质量的方法: 总横流阻力近似为船体横截面积上的横流阻力的总和。 The method used to calculate the hull added mass: the total hull drag is approximated as the sum of the drags on the two-dimensional vehicle cross-sections.

横流阻力是计算附加质量的相当准确的方法, 但仅限于低速流动, 即 $Re < 10^5$ 。然而, 这种方法允许我们在运动方程中包含所有项, 这在低速流动中, 我们试图与实验数据和海上车辆观测的比较, 会识别出横流阻力项中的任何误差。 In the vehicle simulation, we will attempt to correct any errors in the crossflow drag terms through comparison with experimental data and observations of the vehicle at sea.

非线性横流阻力系数表示如下: coefficients are expressed as follows:

$$\begin{aligned} Y_{v|v} &= Z_{w|w} = -\frac{1}{2} \rho c_{dc} \int_{x_t}^{x_{b2}} 2R(x) dx - 2 \cdot \left(\frac{1}{2} \rho S_{fin} c_{df} \right) \\ M_{w|w} &= -N_{v|v} = \frac{1}{2} \rho c_{dc} \int_{x_t}^{x_{b2}} x^2 2R(x) dx - \frac{2x_{fin}}{t} \left(\frac{1}{2} \rho S_{fin} c_{df} \right) \\ Y_{r|r} &= -Z_{q|q} = -\frac{1}{2} \rho c_{dc} \int_{x_t}^{x_{b2}} 2x|x| R(x) dx - \frac{2x_{fin}}{t} \left(\frac{1}{2} \rho S_{fin} c_{df} \right) \\ M_{q|q} &= N_{r|r} = -\frac{1}{2} \rho c_{dc} \int_{x_t}^{x_{b2}} 2x^3 R(x) dx - \frac{2x_{fin}^3}{t} \left(\frac{1}{2} \rho S_{fin} c_{df} \right) \end{aligned} \quad (4.8)$$

其中 ρ 是海水密度, c_{dc} 是船体的阻力系数, $R(x)$ 是船体半径 a 作为轴向位置的函数, h_{fin} 是鳍片的高度, S_{fin} 是鳍片的表面积, c_{df} 是鳍片的横流阻力系数。2. 积分的限制请参考表A.2。 c_{df} is the crossflow drag coefficient of the control fins. See Table A.2 for the limits of integration.

Hoerner [15] 给出了横流阻力系数 c_{df} 的经验公式, 它是使用 Whicker 和 Fehner [32] 开发的公式推导出的: is derived using the formula developed by Whicker and Fehner [32]:

$$c_{df} = 0.1 + 0.7t \quad (4.9)$$

其中 t 是鳍片高度 h_{fin} 与沿车辆纵向轴线鳍片顶端和底端宽度的比值。根据这个公式, 我们得到 c_{df} 的估计值为 0.56。 t is the ratio of the fin height to the width of the fin at the top and bottom of the fin along the vehicle centerline. From this formula, we get an estimate for c_{df} of 0.56.

请参见表C.2 获取最终系数值。 efficient values.

4.2.3 滚动阻力 Rolling Drag

我们将假设主要部分来自鳍片的横流阻力。假设车辆滚动阻力主要来自鳍片的横流阻力。 Assuming that the principle component comes from the crossflow drag of the fins.

$$F = (Y_{vv} f_{meas}) r_{mean}^2 |p| \quad (4.10)$$

其中 Y_{vv} 是车辆横流阻力系数的鳍片, r_{mean} 是车辆中心线以上的平均鳍片高度, p 是车辆滚动角。 This yields the following equation for the vehicle rolling drag height above the vehicle centerline. This yields the following equation for the vehicle rolling drag

系数:

$$K_{ppj} = Y_{vyn}^3 m_{\text{can}} \quad (4.11)$$

这在其量上是实际值的粗略近似。最好使用实验数据。It would be better to use experimental data.

根据这个粗略估算, 请参见表 4.3 中的系数值。this rough approximation.

4.3 添加的质量 Mass

附加质量是车辆加速时移动水质量的量度。理想液体的力和力矩可以用以下方程表示: Ideal fluid forces and moments can be expressed by the equations:

$$\begin{aligned} F_j &= -\dot{m}_{ji} \dot{x}_i - \epsilon_{jkl} \omega_l \Omega_k m_{ji} \\ M_j &= -\dot{m}_{ji} \dot{x}_i s_{ji} - \epsilon_{jkl} \omega_l \Omega_k m_{ji} s_{ji} - \epsilon_{jkl} \omega_l \omega_k m_{ji} \\ \text{where } i &= 1, 2, 3, 4, 5, 6 \\ \text{and } j, k, l &= 1, 2, 3 \end{aligned} \quad (4.12)$$

其中, where 交替张量 ϵ_{jkl} 在指标为循环顺序 (123, the 31 indices) 时等于 +1, 在指标 31, 32 非循环顺序 (132, are cyclic (321)) 2 时等于 -1, 如果任意 pair 对指标相等则为零。See Newman 方程的展开 [10] for the expansion of these equations [10].

Due to body top-bottom and front-back symmetry, the vehicle added mass matrix reduces 收件人:

$$\begin{bmatrix} m_{11} & 0 & 0 & 0 & 0 & 0 \\ 0 & m_{22} & 0 & 0 & 0 & m_{26} \\ 0 & 0 & m_{33} & 0 & m_{35} & 0 \\ 0 & 0 & 0 & m_{44} & 0 & 0 \\ 0 & 0 & m_{53} & 0 & m_{55} & 0 \\ 0 & m_{62} & 0 & 0 & 0 & m_{66} \end{bmatrix} \quad (4.13)$$

相当于是 equivalent to:

$$\begin{bmatrix} X_u & 0 & 0 & 0 & 0 & 0 \\ 0 & Y_v & 0 & 0 & 0 & N_v \\ 0 & 0 & Z_w & 0 & M_w & 0 \\ 0 & 0 & 0 & K_p & 0 & 0 \\ 0 & 0 & Z_q & 0 & M_q & 0 \\ 0 & Y_r & 0 & 0 & 0 & N_r \end{bmatrix} \quad (4.14)$$

将这些剩余项代入方程 4.12 中, 流体力和力矩的展开方程, 得到以下方程: forces and moments from Equation 4.12 yields the following equations:

$$\begin{aligned} \dot{X}_A &= X_u \dot{u} + Z_{\dot{w}} \dot{w} + Z_{\dot{q}} \dot{q} - Y_{\dot{w}} \dot{r} - Y_{\dot{r}} \dot{r}^2 \\ Y_A &= Y_v \dot{v} + Y_{\dot{r}} \dot{r} + X_{\dot{u}} \dot{u} - Z_{\dot{w}} \dot{w} - Z_{\dot{q}} \dot{q} \\ Z_A &= Z_w \dot{w} + Z_{\dot{q}} \dot{q} - X_{\dot{u}} \dot{u} + Y_{\dot{v}} \dot{v} + Y_{\dot{r}} \dot{r} \\ K_A &= K_{pp} \dot{p} \\ M_A &= M_{\dot{w}} \dot{w} + M_{\dot{q}} \dot{q} - (Z_{\dot{u}} - X_{\dot{u}}) \dot{u} \dot{w} - Y_{\dot{v}} \dot{v} + (K_p - N_r) \dot{r} \dot{p} - Z_{\dot{q}} \dot{q} \\ N_A &= N_{\dot{v}} \dot{v} + N_{\dot{r}} \dot{r} - (X_{\dot{u}} - Y_{\dot{v}}) \dot{u} \dot{v} + Z_{\dot{q}} \dot{q} \dot{p} - (K_p - M_q) \dot{p} \dot{q} + Y_{\dot{w}} \dot{w} \end{aligned} \quad (4.15)$$

4.3.1 轴向附加质量 Axial Added Mass

为了估算轴向附加质量, 我们将车辆船体形状近似为椭球体, 其长轴为车辆长度的 $\frac{1}{2}$, 短轴为车辆直径的 $\frac{1}{2}$ 。The major axis is the vehicle length, and the minor axis is the vehicle diameter. For a comparison of the two shapes. Blevins [6, p.407] gives the following empirical formula for the

椭球的轴附加质量 of an ellipsoid:

$$X_{\dot{u}} = -m_{11} = -\frac{4\rho\pi}{3} \left(\frac{l}{2}\right) \left(\frac{d}{2}\right)^2 \quad (4.16)$$

or

$$X_{\dot{u}} = -m_{11} = -\frac{4\beta\rho\pi}{3} \left(\frac{d}{2}\right)^3 \quad (4.17)$$

其中 ρ 是周围流体的密度, β 和 β_3 是Blevins测量的经验参数, 并由车辆长度与直径的比值确定, as measured by Blevins and determined by the ratio of the vehicle length to diameter as shown in Table 4.1.

Table 4.1: 轴附加质量参数 α and β

l/d	α	β_3
0.01	-	0.6348
0.1	6.148	0.6148
0.2	3.008	0.6016
0.4	1.428	0.5712
0.6	0.9078	0.5447
0.8	0.6514	0.5211
1.0	0.5000	0.5000
1.5	0.3038	0.4557
2.0	0.2100	0.4200
2.5	0.1563	0.3908
3.0	0.1220	0.3660
5.0	0.05912	0.2956
7.0	0.03585	0.2510
10.0	0.02071	0.2071

请参阅表 C.3 获取最终系数值 coefficient values.

4.3.2 横流附加质量 Added Mass

车辆附加质量是通过将圆柱形和十字形船体横截面使用条带理论计算的, 根据 Newman [24], 单个圆柱切片 z 片的单位长度附加质量给出如下, the added mass per unit length of a single cylindrical slice is given as:

$$m_a(x) = \pi\rho R(x)^2 \quad (4.18)$$

其中 ρ 是周围流体的密度, $R(x)$ 是轴面位置的船体半径函数, 如方程 4.18 和 4.22 所示, 带鳍圆的附加质量在 Blevins [6] 中给出如下: 2.1 and 2.2. The added mass of a circle with fins is given in Blevins [6] as:

$$m_{af}(x) = \pi\rho \left(r_{fm}^2 - R(x)^2 + \frac{R(x)^4}{a_{fm}^2} \right) \quad (4.19)$$

其中 r_{fm} , 如表 2.2 所定义, 是车辆鳍片 z 轴上的最大高度 above 车辆长度 l 的方程 4.18 和 vehicle 进行积分. Integrating Equations 4.18 and 4.19 over the length of the vehicle, we arrive at the following

横流附加质量的方程w added mass:

$$\begin{aligned}
 Y_{\dot{v}} &= -M_{22} = - \int_{x_t}^{x_f} m_a(x) dx - \int_{x_{fin2}}^{x_{f2}} m_{af}(x) dx - \int_{x_{f2}}^{x_{b2}} m_a(x) dx \\
 Z_{\dot{w}} &= -M_{33} = -M_{22} = Y_{\dot{v}} \\
 M_{\dot{w}} &= -M_{33} = \int_{x_t}^{x_f} x m_a(x) dx - \int_{x_f}^{x_{f2}} x m_{af}(x) dx - \int_{x_{f2}}^{x_{b2}} x m_a(x) dx \\
 N_{\dot{r}} &= -m_{53} = -M_{\dot{w}} \\
 Y_{\dot{r}} &= -m_{26} = -m_{62} = N_{\dot{r}} \\
 Z_{\dot{q}} &= -m_{35} = -m_{53} = M_{\dot{w}} \\
 M_{\dot{q}} &= -m_{55} = - \int_{x_{tail}}^{x_{fin2}} x^2 m_a(x) dx - \int_{x_{fin}}^{x_{fin2}} x x^2 m_{af}(x) dx - \int_{x_{fin2}}^{x_{bow2}} x^2 m_a(x) dx \\
 N_{\dot{r}} &= -m_{66} = -M_{\dot{q}}
 \end{aligned} \tag{4.20}$$

参见表 A.2 解积分的界限 of integration.

请参见表 C.5 获取最终系数值 efficient values.

4.3.3 滚转附加质量 Added Mass

为了估算横滚附加质量, 我们将假设车辆船体的相对光滑部分不会在横滚中产生任何附加质量, 我们还将忽略声纳应答器和任何其他小型突起。我们将忽略附加质量 generated by the sonar transponder and any other small protuberances. Given those assumptions, we need only consider the hull section containing the vehicle control fins.

Blevins [6] 提出了带鳍滚动圆的附加质量的以下经验公式 added mass of a rolling circle with fins:

$$K_{\dot{p}} = - \int_{x_{fin}}^{x_{fin2}} \frac{2}{\pi} p a^4 dx \tag{4.21}$$

其中 a 是鳍片相对于车辆中心线的高度, 在本例中平均为 0.175 米。积分的限制见表 A.2. See Table A.2 for the limits of integration.

请参见表 C.5 以获取最终系数值 efficient value.

4.3.4 附加质量交叉项 Mass Cross-terms

剩余的 remaining cross-terms 是由附加质量耦合产生的, 并且包括 and are listed below:

$$\begin{aligned}
 X_{\dot{q}} &= Z_{\dot{w}} & X_{q\dot{q}} &= Z_{\dot{q}} & X_{vr} &= -Y_{\dot{v}} & X_{rr} &= -Y_{\dot{r}}
 \end{aligned} \tag{4.22}$$

$$Y_{\dot{u}} = X_{\dot{u}} \quad Y_{\dot{w}p} = -Z_{\dot{w}} \quad Y_{\dot{p}q} = -Z_{\dot{q}} \tag{4.23}$$

$$Z_{u\dot{q}} = -X_{\dot{u}} \quad Z_{\dot{u}p} = Y_{\dot{v}} \quad Z_{\dot{r}p} = Y_{\dot{r}} \tag{4.24}$$

$$M_{u\dot{w}a} = -(Z_{\dot{w}} - X_{\dot{q}})M_r = M_{vp} = -Y_{\dot{r}}M_{rp} = (K_{\dot{p}} - N_{\dot{r}})M_{uq} = -Z_{\dot{q}} \tag{4.25}$$

$$N_{u\dot{v}a} = -(X_{\dot{u}} - Y_{\dot{v}}) \quad N_{\dot{p}p} = Z_{\dot{q}} \quad N_{\dot{p}q} = -(K_{\dot{p}} - M_{\dot{q}}) \quad N_{\dot{u}r} = Y_{\dot{r}} \tag{4.26}$$

附加质量交叉项 cross-terms 和 $M_{u\dot{w}a}$ 被称为达朗贝尔力矩, 它与理想无粘流中物体在迎角下经历的力矩有关 the moment experienced by a body at an angle of attack in ideal, inviscid flow.

请参见表 C.6, C.7 和 C.8 以获取最终系数值 efficient values.

4.4 车体抬升 Body Lift

车辆车体 lift 源自车辆以迎角穿过水面移动, 导致流体分离, 并在车辆船体的尾部区域产生随之而来的压和下降。这种压力下降被建模为作用于压力中心点力。随着这个压力中心的 This pressure drop is modeled as a point force applied at the center of pressure. As this center of

pressure on the hull fixed coordinate system's origin. This force also leads to a pitching moment about the origin.

In determining the best method for comparing the three empirical methods, the four methods ranged over an order of magnitude. Unfortunately, the lack of agreement between the empirical methods, it would be preferable to base the body lift estimates on actual REMUS data, from perhaps tow tank tests or measurements of the vehicle mounted on a rotating arm.

在那之前, 作者决定使用 author 的估算 [16], 因为它看起来最可靠 [16], which appeared the most reliable.

4.4.1 车体升力 Lift Force

To calculate body lift, we will use the empirical formula developed by Hoerner [16], which states:

$$L_{body} = -\frac{1}{2}\rho d^2 c_{ydw}^2 \quad (4.27)$$

其中 ρ 为周围流体的密度, A 为船体的投影面积, u 为车辆的前进速度, c_{yd} 为车体升力系数. 根据 Hoerner 的表示法表示为 c_{yd} , and c_{yd} the body lift coefficient, which by Hoerner's notation is expressed as:

$$c_{ydw} = c_{yd}(\beta) = \frac{dc_{yd}}{d\beta} \beta \quad (4.28)$$

哪里 β is the vehicle angle of attack in radians and is given by the relationship:

$$\tan \beta = \frac{w}{u} \implies \beta \approx \frac{w}{u} \quad (4.29)$$

Hoerner gives the relationship for lift slope:

$$\frac{dc_{yd}}{d\beta^\circ} = c_{y\beta}^\circ = \left(\frac{l}{d}\right) c_{y\beta}^\circ \quad (4.30)$$

where l is vehicle length and d is diameter. Hoerner [16, pg. 13-3] states that

$$\text{for } 6.7 \leq \frac{l}{d} \leq 10, \quad c_{y\beta}^\circ = 0.003 \quad (4.31)$$

Note, the equation has to be converted to degrees. Hoerner lift slope coefficient $c_{y\beta}^\circ$ is defined in terms of radians as follows:

$$c_{y\beta}^\circ = c_{y\beta} \left(\frac{180}{\pi}\right) \quad (4.32)$$

将上述给出的关系代入方程 4.27 后, 我们得到车辆升力的下方方程:

$$L_{body} = -\frac{1}{2}\rho d^2 c_{ydw}^2 \quad (4.33)$$

这导致以下车体升力系数:

$$Y_{LM} = Z_{LM} = -\frac{1}{2}\rho d^2 c_{ydw}^2 \quad (4.34)$$

请参阅表 C4 以获取最终系数值.

4.4.2 车体提升力矩 Body Lift Moment

Hoerner 估计, 对于一个有攻角的旋转体, 粘性力的作用点位于从鼻端起到整个机体长度的 0.6 到 0.7 之间。他的实验结果表明, 流体在船体前端能够顺畅流动, 并且横向力只在船体后半部分的风洞侧产生。The flow goes smoothly around the forward end of the hull, and that the lateral force only develops on the leeward side of the rear half of the hull.

根据这些发现, 我们将假设在身体固定坐标, in body-fixed coordinates:

$$x_{cp} = -0.65l - x_{\xi \ell \rho} \quad (4.35)$$

这就导致了以下公式用于车体升力力矩: body lift moment:

$$M_{uwl} = -N_{uwl} = -\frac{1}{2} \rho c_L^2 a_{cp} x_{cp} \quad (4.36)$$

请参阅表 C.4 获取最终系数值。

4.5 鳍提升力矩

REMUS 车辆的姿态由两个水平鳍翼 (或尾舵面) 和两个垂直鳍翼 (或方向舵) 控制, 这些鳍翼会一起移动; 换句话说, 尾舵面不会互相独立移动, 舵面也不会独立移动。The pairs of fins moves together in other words, the stern planes do not move independently of each other, nor do the rudder planes.

对于车辆控制鳍, 鳍提升力的经验公式给出如下:

$$L_{fin} \approx \frac{1}{2} \rho c_L S_{fin} \delta_e^2 \quad (4.37)$$

$$M_{fin} = x_{fin} L_{fin}$$

其中 c_L 是鳍升力系数, S_{fin} 是鳍的平面面积, δ_e 是以弧度表示的有效鳍角, v_e 是有效鳍速度, x_{fin} 是以机身参考坐标表示的鳍柱轴向位置。

鳍升力系数 c_L 是有效鳍攻角 α 的函数, Hoerner [16] 和 [B.21] 给出了鳍升力作为 α (以弧度) 度值的以下经验公式:

$$c_{L\alpha} = \frac{dc_L}{d\alpha} = \left[\frac{1}{2\alpha\pi} + \frac{1}{\pi(AR_e)} \right]^{-1} \quad (4.38)$$

其中因子 f 被 Hoerner 发现大约为 0.9 (AR) 是有效鳍展长比 (AR) 以下公式给出: the effective fin aspect ratio, which is given by the formula:

$$AR_e = 2(AR) = 2 \left(\frac{b_{fin}^2}{S_{fin}} \right) \quad (4.39)$$

由于鳍位于车辆坐标系原点的某个偏移位置, 它会根据以下有效速度, it experiences the following effective velocities:

$$\begin{aligned} v_{fin} &= u + z_{fin} q - y_{fin} r \\ v_{fin} &= v + x_{fin} p - z_{fin} p \\ w_{fin} &= w + y_{fin} p - x_{fin} q \end{aligned} \quad (4.40)$$

其中 x_{fin} , y_{fin} 和 z_{fin} 是鱼鳍支柱的车辆参考坐标。对于 REMUS 车辆, 我们将忽略涉及 y_{fin} 和 z_{fin} 的项, 因为它们相比车辆平移速度很小。as they are small compared to the vehicle translational velocities.

有效鳍角 δ_e 和 δ_{re} 可以表示为 δ_{re} can be expressed as

$$\begin{aligned} \delta_{re} &= \delta_r - \beta_{re} \\ \delta_{se} &= \delta_s + \beta_{se} \end{aligned} \quad (4.41)$$

where δ_r and δ_s are the rudder angles in the rudder plane and the yaw plane, respectively, and β_e is the effective angle of attack. As shown in Figures 4-1 and 4-2, if the angles are small, these effective angles can be expressed as:

$$\begin{aligned}\beta_{oe} &= \frac{v_{fin}}{u_{fin}} \approx \frac{1}{u} (\psi + x_{fin} r) \\ \beta_{se} &= \frac{w_{fin}}{u_{fin}} \approx \frac{1}{u} (w - x_{fin} q)\end{aligned}\quad (4.42)$$

Based on Equation 4.40

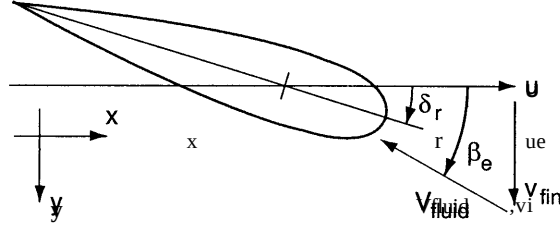


Figure 4-1 Rudder Angle of Attack

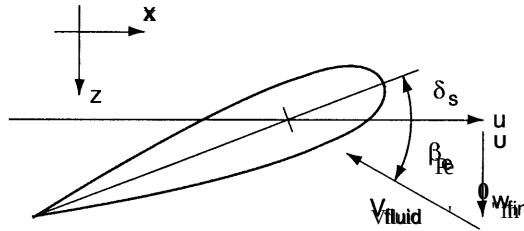


Figure 4-2 Yaw Plane Angle of Attack

Substituting the results of Equations 4.41 and 4.42 into Equation 4.40 results in the following equations for fin lift and moment:

$$\begin{aligned}Y_s &= \frac{1}{2} \rho C_L S_{fin} [u^2 \delta_r - uv - x_{fin} (ur)] \\ Z_s &= -\frac{1}{2} \rho C_L S_{fin} [u^2 \delta_s + uw - x_{fin} (uq)] \\ M_s &= \frac{1}{2} \rho C_L S_{fin} x_{fin} [u^2 \delta_s + uw - x_{fin} (uq)] \\ N_r &= \frac{1}{2} \rho C_L S_{fin} x_{fin} [u^2 \delta_r - uv - x_{fin} (ur)]\end{aligned}\quad (4.43)$$

Finally, we can separate the equation into the following sets of fin lift coefficients:

$$\begin{aligned}Y_{u\delta_r} &= Y_{uv} = \rho C_L S_{fin} \\ Z_{u\delta_s} &= Z_{uw} = -\rho C_L S_{fin} \\ Y_{ur} &= -Z_{uq} = -\rho C_L S_{fin} x_{fin}\end{aligned}\quad (4.44)$$

and rudder moment coefficients:

$$\begin{aligned}M_{u\delta_s} &= M_{uw} = \rho C_L S_{fin} x_{fin} \\ N_{u\delta_r} &= N_{ur} = \rho C_L S_{fin} x_{fin} \\ M_{uq} &= N_{ur} = -\rho C_L S_{fin} x_{fin}^2\end{aligned}\quad (4.45)$$

See Table C.10 for the final coefficient values.

4.6 推进模型 Propulsion Model

我们将为 REMUS 推进系统使用一个非常简单的模型。该模型将螺旋桨视为恒定推力和扭矩的源。这些系数的数值来自 Ben Allen 的海洋系统实验室进行的车辆设计阶段螺旋桨试验，以及作者在海上进行的实验。

这个简单的模型对于车辆稳态下的小幅扰动是可接受的。如果对模拟输出的检查表明需要更复杂的模型，我们可以尝试用螺旋桨模型替代，比如 Yoerger 和 Slotine 的模型 [35]，或者使用实验获得的数值。

4.6.1 螺旋桨推力 Propeller Thrust

在海上测试中，我们观察到 REMUS 车辆在螺旋桨转速为 1500 RPM 时能够保持 3.0 m/s (3 kn) 的前进速度。我们将假设在这种稳定速度下，螺旋桨推力与车辆的轴向阻力相匹配。the propeller thrust matches the vehicle axial drag.

$$\begin{aligned} X_{prop} &= -X_{fu} u |u| \\ &= -22.28 X_{fu} |u| \end{aligned} \quad (4.46)$$

为了模拟的目的，我们将假设车辆仅在此前进速度的偏差范围内运动。最终系数值见表 C.9。from this forward speed. See Table C.9 for the final coefficient value.

4.6.2 螺旋桨扭矩 Propeller Torque

在海上试验中，REMUS 车辆在稳态条件下以 1500 转/分钟运行且俯仰角为零时，被观察到保持平均横滚偏移 4

of -5.3 degrees (-9.3 x 10-2 radians). We will assume that under these conditions, the propeller torque matches the hydrostatic roll moment.

$$\begin{aligned} K_{prop} &= -K_{RGS} = (y_g W - y_b B) \cos \theta \cos \phi + (z_g W - z_b B) \cos \theta \sin \phi \\ &= 0.995(y_g W - y_b B) - 0.093(z_g W - z_b B) \end{aligned} \quad (4.47)$$

请参见表 C.9 获取最终系数值。

4.7 组合条款 Combined Terms

将方程 4.21 到 4.47 的同类项组合起来，我们得到如下结果：

$$\begin{aligned} Y_{uv} &= Y_{uvl} + Y_{uvf} \\ Y_{ur} &= Y_{ura} + Y_{urf} \\ Z_{uw} &= Z_{uwl} + Z_{uwf} \\ Z_{uq} &= Z_{uqa} + Z_{uqf} \\ M_{uv} &= M_{uva} + M_{uvl} + M_{uvf} \\ M_{uq} &= M_{uqa} + M_{uqf} \\ N_{uv} &= N_{uva} + N_{uvl} + N_{uvf} \\ N_{ur} &= N_{ura} + N_{urf} \end{aligned} \quad (4.48)$$

4.8 总车辆力和力矩 Total Vehicle Forces and Moments

结合车辆的系数方程

• *Hydrostatics*: Equation 4.6

• *Hydrodynamic Damping*: Equation 4.8

• *Added Mass*: Equations 4.16, 4.20, 4.21 and 4.22

• *Body Lift and Moment*: Equations 4.34 and 4.36

• *Propeller Thrust and Torque*: Equations 4.46 and 4.47

• *Fin Lift and Moment*: Equations 4.44 and 4.45

• *Propeller Thrust and Torque*: Equations 4.46 and 4.47

The sum of the depth and pitch moments on the vehicle can be expressed as:

$$\begin{aligned}
 \sum X_{\text{ext}} &\equiv X_{HS} + X_{u|u}|u| + X_{\dot{\theta}} + X_{w|w}|w| + X_{q|q}|q| + X_{v|v}|v| + X_{r|r}|r| \\
 &\quad + X_{\text{prop}} \\
 \sum Y_{\text{ext}} &\equiv Y_{HS} + Y_{v|v}|v| + Y_{r|r}|r| + Y_{\dot{\psi}} + Y_{\dot{\theta}} \\
 &\quad + Y_{u|u}|u| + Y_{w|w}|w| + Y_{p|p}|p| + Y_{uv|uv}|uv| + Y_{us|us}|u|^2 \dot{\theta}_r \\
 \sum Z_{\text{ext}} &\equiv Z_{HS} + Z_{w|w}|w| + Z_{q|q}|q| + Z_{\dot{\psi}} + Z_{\dot{\theta}} \\
 &\quad + Z_{u|u}|u| + Z_{v|v}|v| + Z_{r|r}|r| + Z_{uw|uw}|u|w| + Z_{us|us}|u|^2 \dot{\theta}_s \\
 \sum K_{\text{ext}} &\equiv K_{HS} + K_{p|p}|p| + K_{\dot{\psi}} + K_{\text{prop}} \\
 \sum S_{\text{ext}} &\equiv M_{HS} + M_{u|u}|u| + M_{q|q}|q| + M_{u\dot{\psi}} + M_{q4} \\
 &\quad + M_{uq|uq}|u|q| + M_{vp|vp}|v|p| + M_{rp|rp}|r|p| + M_{uw|uw}|u|w| + M_{us|us}|u|^2 \dot{\theta}_s \\
 \sum N_{\text{ext}} &\equiv N_{HS} + N_{u|u}|u| + N_{r|r}|r| + N_{\dot{\psi}} + N_{\dot{\theta}} \\
 &\quad + N_{uv|uv}|u|v| + N_{vp|vp}|v|p| + N_{pq|pq}|p|q| + N_{uv|uv}|u|v| + N_{us|us}|u|^2 \dot{\theta}_r
 \end{aligned} \tag{4.49}$$

See Tables 4.1 and 4.3 for non-zero vehicle coefficients.

表 4.2 STDSREHENS 非线性 机动系数: 力量

Parameter	Value	Units	Description
X_{du}	-1.622e+0000	kg/m	Cross-flow Drag
X_u	-9.30e+0001	kg	Added Mass
X_{vq}	-3.55e+0001	kg/rad	Added Mass Cross-term
X_{qh}	-1.93e+000	kg - m/rad	Added Mass Cross-term
X_{ur}	+3.55e+0001	kg/rad	Added Mass Cross-term
X_{rr}	-1.93e+0000	kg - m/rad	Added Mass Cross-term
X_{prop}	+3.86e+0000	N	Propeller Thrust
Y_{vu}	-1.331e+0002	kg/m	Cross-flow Drag
Y_{rr}	+6.32e+0001	kg - m/rad ²	Cross-flow Drag
Y_{uv}	-2.88e+0001	kg/m	Body Lift Force and Fin Lift
Y_u	-3.55e+0001	kg	Added Mass
Y_r	+1.93e+000	kg - m/rad	Added Mass
Y_{ur}	+5.22e+0000	kg/rad	Added Mass Cross Term and Fin Lift
Y_{vp}	+3.55e+001	kg/rad	Added Mass Cross-term
Y_{pq}	+1.93e+000	kg - m/rad	Added Mass Cross-term
Y_{uudr}	+9.64e+000	kg/(m - rad)	Fin Lift Force
Z_{du}	-1.331e+0002	kg/m	Cross-flow Drag
Z_{rr}	-6.32e+0001	kg - m/rad ²	Cross-flow Drag
Z_{uv}	-2.88e+0001	kg/m	Body Lift Force and Fin Lift
Z_{vb}	-3.55e+0001	kg	Added Mass
Z_q	-1.93e+000	kg - m/rad	Added Mass
Z_{uq}	-5.22e+0000	kg/rad	Added Mass Cross-term and Fin Lift
Z_{rp}	-3.55e+0001	kg/rad	Added Mass Cross-term
Z_{rp}	+1.93e+000	kg/rad	Added Mass Cross-term
Z_{uuds}	-9.64e+0000	kg/(m - rad)	Fin Lift Force

表 4.3 STDSREHENS 非线性 机动系数: Moments

Parameter	Value	Units	Description
K_{pp}	-1.30e+003	kg - m ² /rad ²	Rolling Resistance
K_p	-1.41e+002	kg - m ² /rad	Added Mass
K_{prop}	-5.43e+0001	N.m	Propeller Torque
M_{uu}	+3.18e+000	kg	Cross-flow Drag
M_{qh}	-9.40e+0000	kg - m ² /rad ²	Cross-flow Drag
M_{uv}	+2.40e+0001	kg	Body and Fin Lift and Munk Moment
M_u	-1.93e+000	kg - m	Added Mass
M_{rp}	-4.88e+0000	kg - m ² /rad	Added Mass
M_{uq}	-2.00e+0000	kg - m/rad	Added Mass Cross Term and Fin Lift
M_{yp}	-1.93e+000	kg - m/rad	Added Mass Cross Term
M_{rp}	+4.88e+0000	kg - m ² /rad ²	Added Mass Cross-term
M_{uuds}	-6.15e+0000	kg/rad	Fin Lift Moment
N_{uv}	-3.18e+0000	kg	Cross-flow Drag
N_{rr}	-9.40e+0000	kg - m ² /rad ²	Cross-flow Drag
N_{uv}	-2.40e+0001	kg	Body and Fin Lift and Munk Moment
N_u	+1.93e+000	kg - m	Added Mass
N_{rp}	-4.88e+0000	kg - m ² /rad	Added Mass
N_{ur}	-2.00e+000	kg - m/rad	Added Mass Cross Term and Fin Lift
N_{vp}	-1.93e+000	kg - m/rad	Added Mass Cross Term
N_{pq}	-4.88e+0000	kg - m ² /rad ²	Added Mass Cross-term
N_{uuds}	-6.15e+0000	kg/rad	Fin Lift Moment

Chapter 5

Vehicle Flow Experiments

In 1999, during the summer period, of 1999, while at the naval systems laboratory of the British Royal Naval School of Maritime Systems, the author conducted a series of experiments to measure the vehicle axial drag coefficient and hull resistance. The experiments were intended to measure the vehicle axial drag coefficient and hull resistance, and to assist in estimating the overall efficiency of the vehicle propulsion system. The experiments involved recording axial and lateral drag data for a range of vehicle speeds and hull configurations, as well as thrust data from bollard pull tests for a range of propeller speeds. These experiments provided the author with the opportunity to experimentally measure the vehicle axial drag coefficient.

5.1 动机 Motivation

然而自主水下航行器 important 更重要的特性之一是其续航能力, 即该车辆完成任务所需的航程和速度 at the 推进系统效率的提升意味着 accomplish 下航程更长, 或能够在更短时间内覆盖相同距离。system 提升整体效率的努力都将使 vehicle 更适用。for a given speed, or the ability to cover the same distance in a reduced time. More efforts to improve the overall efficiency will result in a more useful vehicle.

REMUS是一种低成本、基于微处理器的自主水下航行器,with approximately 100台设备在数百次任务中累计为underwater的 underwater 使用,时间 3 years. 该车辆设计在the vehicle 在海中的使用性方面非常成功,do in demonstration 然而其航程和速度usefulness of 有限。现有AUVs 对the ocean 飞机螺旋桨配合直流有限电机、螺旋桨轴和轴封。最近的一次设计工作对该设计进行了修改,以提供显著更高的推进性能。to this design to provide significantly greater propulsion performance.

由于既不知道实物的车辆阻力系数也不知道螺旋桨效率, 因此对于域的车辆测试确定船体阻力系数和螺旋桨效率的影响差异, 因此测试过程的第一步是量化现在车辆在拖曳水槽中的阻力来源, 然后确定可能的改进措施。Therefore the first step in the design process entailed quantifying the sources of drag in a tow-tank on an existing vehicle, and then determining what improvements were possible.

5.2 Laboratory Facilities and Equipment

实验在罗德岛大学纳拉甘西特湾校区的Steensky楼内的牵引槽进行。罗德岛大学的牵引槽位于Sheets Building内为30米x18米x3米。3米深Campus 2牵引槽的牵引车的停放行程为12米。水(water)是 appropriate 水槽周围的示意图 long 参见在 meters 槽的照片请参见图 5-2, deep (100 by 12 by 5 feet). The tow tank carriage had a useful run of almost 21 meters (70 feet). See Figure 5-1 for a diagram of the tow tank layout, and Figure 5-2 for a picture of the tank.

考虑到水箱和悬架在车辆做bank, roll 我们能够在测试中使用实际的REMUS车辆, 而不是仅仅模拟REMUS 车辆通过一个流线性支撑架悬浮在水上, 而该支撑架通过弹性底座与底板固定到拖车上。有关拖车设置和车辆安装的示意图, 请参见图5-3; 有关车辆支撑架的照片, 请参见图5-4。 See Figure 5-3 for a diagram of the carriage setup and vehicle mounting, and Figure 5-4 for a photo of the vehicle on the strut.

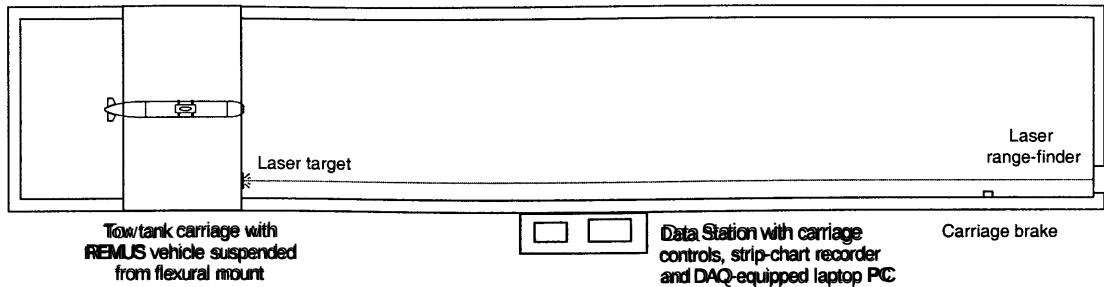


Figure 5.1: 5UR1拖车油箱布局 Layout

5.2.1 弯曲 Flexural Mount

弹性底座是 mount 由两个水平板 consisting of two parallel horizontal plates connected by vertical springs. 下板在水平面内移动。板相对板的运动通过两个垂直安装的线性可变差动变压器 (LVDT) 进行测量。这种机电换能器将板沿每个水平轴的直线运动转换为相应的电信号。

LVDT 输出信号被放大并经过电子滤波, 然后传输到数据站, 在那里它们被绘制在条带图记录器上, 并通通过连接到笔记本电脑的模拟到数字转换器板进行采样。

我们校准了轴向安装的 LVDT 校准到比横向安装仪器显著更高的精度。因此, 我们仅将横向安装的 instrument 用于粗略测量 (gross measurements) of the lateral drag, as an indicator of strut, vehicle or fin misalignment.

5.2.2 拖罐 Tow Tank Carriage

拖罐车是 tank carriage 是一种带有硬橡胶轮子的大 flat 平板车 with 电动机驱动。车辆不是沿着铁轨行驶, 而是在油罐墙的平顶上滚动。

参见图 5.1 了解拖车水箱的图 the tow tank carriage.

所需的车辆速度由数据站上的一个电位器设置, 并由一个电动机控制器使用安装在电机轴上的编码器和光传感器来测量车速。在前进行程中, 当行程中的触发开关被安装在罐壁上时, 车辆会停止。在倒车时, 车辆仅由警觉的操作员在数据站刺按电机停止开关来停止。

我们操作车的速度更多地受限于 carriage 的强度, 而不是滑车电机的能力。我们的最大速度大约是每秒 1.5 米或 3 节, 这是车辆的操作速度。在那个速度下, 由瞬时启动产生的支撑杆振动需要几秒钟才能衰减, 使我们在滑车开始减速之前只有几秒钟的有效数据。

实际的慢速速度是使用安装在坦克远端的激光测距仪测量的。测距仪 (mounted at the far end of the tank) 测量飞行时间, 相反, Nova Range-finder NR-100 did not measure time-of-flight; instead, it was calibrated to measure distance based on the location of the projected dot. For a given distance, the instrument output a corresponding voltage.

模拟测距仪信号被传输到数据站, 在那里它被输入到笔记本电脑的模数转换器板采样。两个数字信号都使用数据收集软件记录, 然后用 MATLAB 处理。

5.3 阻力 Drag 试验程序 Experimental Procedure

阻力测试实验程序包括以下步骤:

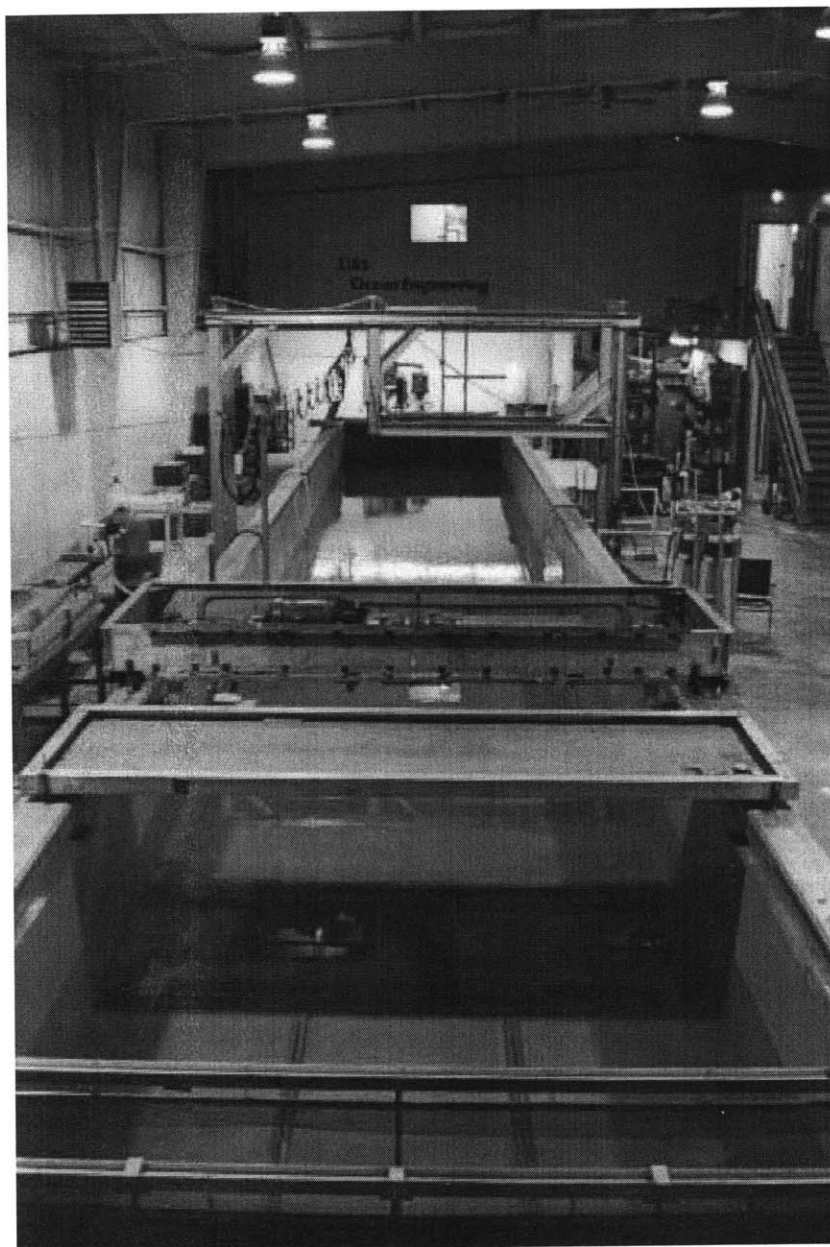


图5-2c UR1 托槽罐 (Photo courtesy of UR1's Ocean Engineering Department)

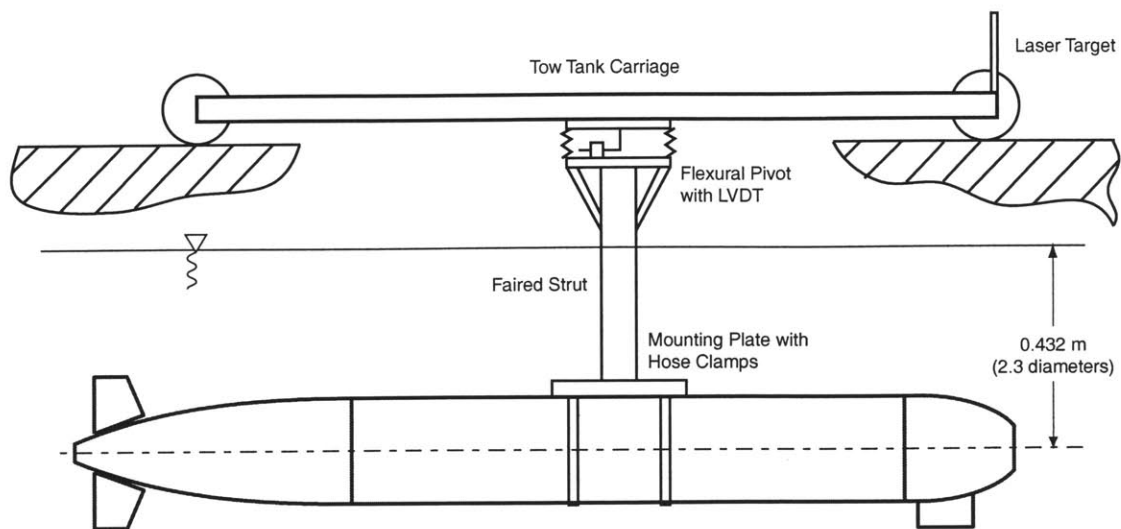


图5-3: 5运输装置设置和车辆安装 Vehicle Mounting

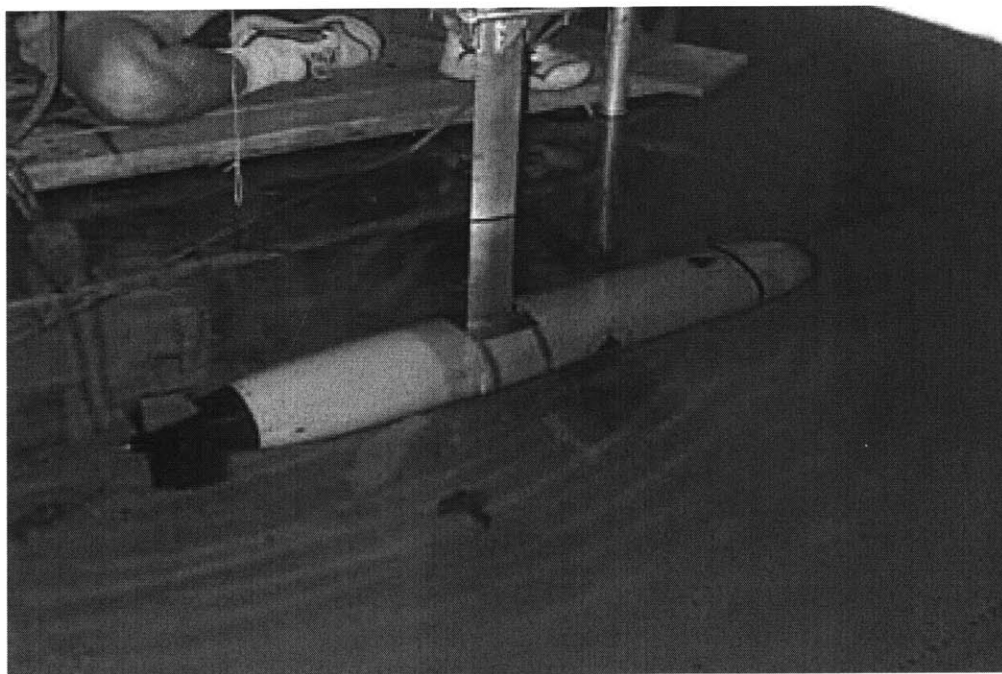


图5-4: 5-URI拖罐车底盘 Tank Carriage

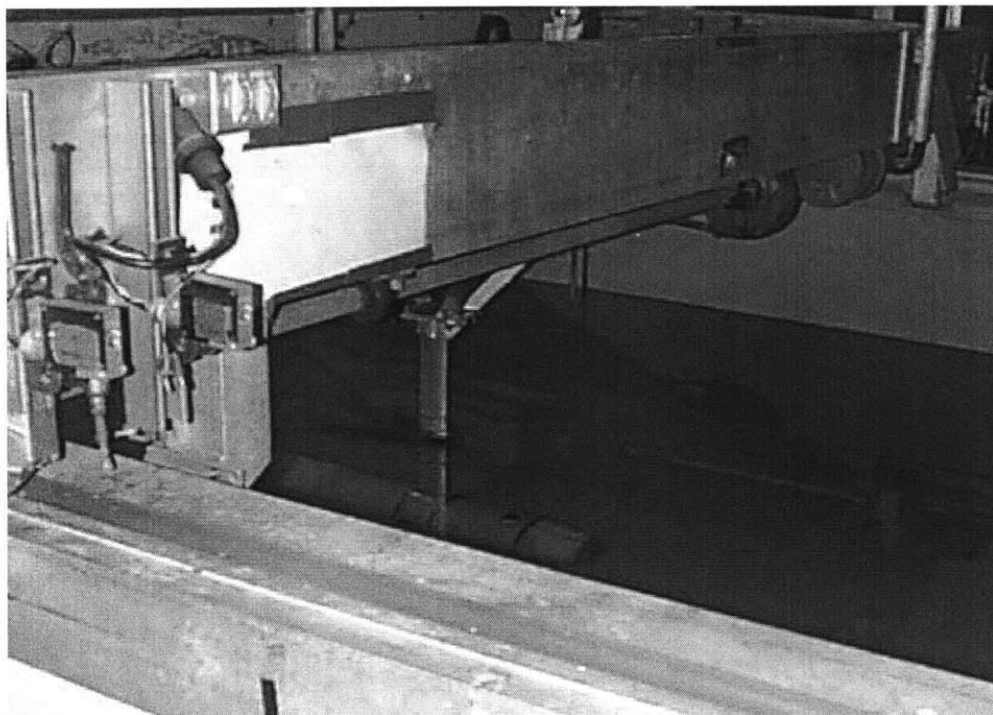


图5-5: 5-URL缓冲罐小车 Tank Carriage

- LVDT和支柱预校准 * 车辆安装和
对齐检查 "鳍片对齐检查" 车辆阻力
测试 * LVDT 和支柱后校准
- fin alignment check
- vehicle drag runs
- LVDT and strut post-calibration

5.3.1 仪器校准 Instrument Calibration

在校准LVDT时, 我们会对挠性滑架施加一系列已知载荷, 并记录输出电压。这是通过将重量挂在系在底部挠性板后端的一根绳子上, 并通过滑轮实现的。鉴于LVDT轴上有少量摩擦, 在挂上重量后, 我们会敲击挠性支架, 让振动衰减, 并在敲击几次后记录平均的稳定值。

在油箱运行期间, 我们会定期检查LVDT电源的输出电压, 因为老化仪器的输出似乎在加热时略有变化。

在校准支柱时, 我们会让小车以不同的速度运行, 让支柱浸入水中, 同时记录轴向和横向阻力。如有必要, 我们会重新调整支柱的对齐, 并再次进行测试。随后测得的支柱阻力作为小车速度的函数, 将从每次车辆运行的总阻力中扣除, 从而分离出车辆的阻力。

在完成仪器校准和安装车辆后, 我们会用铅垂线检查车辆的偏航对准, 并通过观察油箱侧窗来检查车辆的俯仰对准。

在最初的实验中, 我们将以类似的方式检查车辆鳍的对齐。不幸的是, 在WHOH尾部的鳍驱动链条都很松, 因此很难保持鳍正确对齐。我们尝试更换到鳍更紧的尾部, 但它

仍然很难观察到下鳍的力齐情况。最后，我们发现每天收集一次拆下鳍片的数据集是很方便的，each day at the end of the day with the fins removed, in order to verify the alignment of the vehicle.

5.3.2 拖曳 Runs

拖曳槽实验在五种不同的速度下进行，速度范围在0到1.5米每秒之间。实验之前，我们会开始处理阻力数据，wait for the drag data while we waited for the waves in the tank to damp out.

After spending several sessions preparing the lab equipment and developing our calibration protocol, we gave out four days of vehicle tests. Table 5-1 gives the dates and details of these experimental runs.

Table 5. REMUS拖曳实验 Runs

Date	Filename	Vehicle (notes)
09 Jun 1999	remxfps7	WHOI1
16 Jun 1999	remdxfps8	WHOI1 (DOCK2 tail)
16 Jun 1999	remdxfps8b	WHOI1 (DOCK2 tail)
16 Jun 1999	rmfdxfps8	WHOI1 (DOCK2 tail, no fins)
16 Jun 1999	rmfdxfps8b	WHOI1 (DOCK2 tail, no fins)

5.3.3 信号处理 Signal Processing

For a given run, we would collect data from three channels simultaneously: vehicle axial drag, vehicle lateral drag, and carriage speed at a frequency of 400 Hz per channel. To remove sensor noise and the high frequency strut and carriage vibrations, we filtered the data using a zero-phase forwards and reverse digital filter of order 250, and with a cut-off frequency of 2.5 Hz. Figure 5-6 shows a comparison of the filtered and unfiltered data for a single channel.

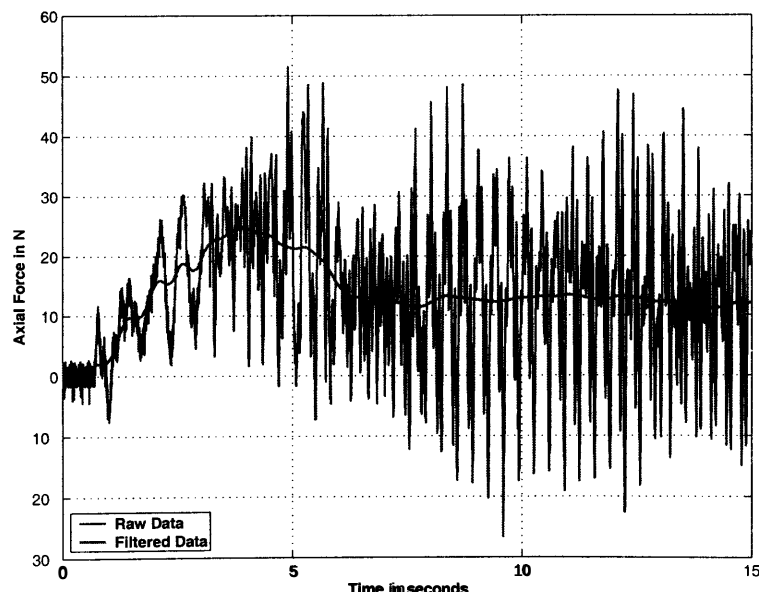


Figure 5-6: Vehicle Axial Drag, Carriage Speed, 1.5 m/s, 16 June 1999, remdxfps8b, 16 June 1999

5.4 实验结果 Experimental Results

图5-7显示了在不同配置下前进速度与车辆轴向阻力的关系图。这些数据被平均以找到前进速度与轴向阻力之间的关系，基于以下公式：

$$c_d = \frac{2E_d}{\rho A_f v^2} \quad (5.1)$$

其中 E_d 是测得的阻力（在减去支撑杆阻力后）， ρ 是流体密度 (999.1 kg/m^3)， A_f 是车辆正面面积 (0.029 m^2)， v 是测得的车辆前进速度， c_d 是车辆阻力系数。由此得到的实验平均阻力系数为 0.267。The resulting parabolic fit is also plotted in Figure 5-7. Again, Table 5.1 gives the dates and details of these experimental runs.

尽管该车辆在 2.3 个车体直径的深度被牵引，但在车速超过每秒一米时，水槽中仍观察到明显的波浪产生。在图 5-7 中，这种额外的波浪阻力可以看到为在较高车速下实验数据偏离抛物线拟合曲线的现象。

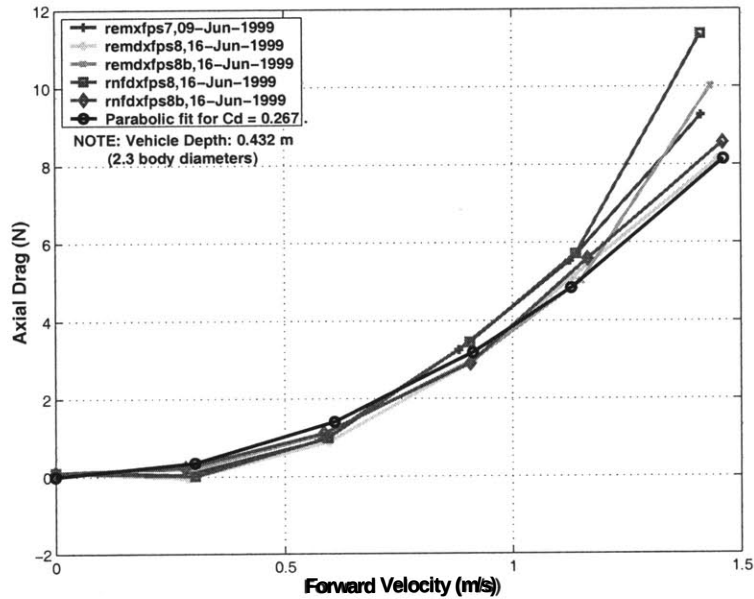


图5-7: 5前进速度与车辆轴向和横向阻力 (实验详情见表5.1) (See Table 5.1 for experiment details)

5.5 基于组件的阻力模型 Component-based Drag Model

Bottaccini [7] 和 Hoerner [15] 建议，对于类似 REMUS 的鱼雷形状，即长宽与最大直径之比为 6 到 11 的细长体，阻力系数为 0.08 到 0.1。鉴于实验测得的阻力系数为 0.267，很明显，各种船体突出部分对总车辆阻力贡献显著。

表 5.2 列出了不同的车辆组件及其对车辆总阻力的估计贡献。车辆船体的阻力系数值取自 Myring [22] 对 B 船体轮廓的研究。车辆组件的阻力系数估计取自 Hoerner [15]。所有估计均假设车辆运行速度为 1.54 米每秒 (3 节)。由此得到的估计值为

根据方程5.1,由总阻力得到的整体阻力系数为0.26, coefficient of 0.26, which compares well with the experimental results.

在表5.3中,进行了类似的基于组件的分析,以预测配备侧扫声纳的REMUS车辆的总阻力 the sidescan sonar-equipped REMUS vehicle.

Table 5.2. REMUS 基于组件的阻力分析 Drag Analysis - Standard Vehicle

	Qty	Cd	Length m	Width m	Diam. m	Area m ²	Drag N
Myring Hull	1	0.10			0.19	2.E-04	3.39
Fins	4	0.02	0.09	0.08		5.E-05	0.62
LBL Transducer	1	1.20	0.03	0.05		1.E-05	2.07
Nose Pockets	3	1.17			0.03	4.E-06	2.68
Blunt Nose	1	?					
Total Vehicle Drag:							8.77
Effective Cd:							0.26

Table 5.3. 基REMUS的REMUS阻力分析 Drag Analysis - Sonar Vehicle

	Qty	Cd	Length m	Width m	Diam. m	Area m ²	Drag N
Myring Hull	1	0.10			0.19	2.E-04	3.39
Fins	4	0.02	0.09	0.08		5.E-05	0.62
LBL Transducer	1	1.20	0.03	0.05		1.E-05	2.07
Nose Pockets	3	1.17			0.03	4.E-06	2.68
Blunt Nose	1	?					0.00
SSS Transducers	2	0.40	0.04	0.04		1.E-05	1.47
ADCP Transducers	8	0.20			0.05	1.E-05	3.86
Total Vehicle Drag:							14.09
Effective Cd:							0.42

Chapter 6

Vehicle Simulation

In this chapter, we first derive the equations of motion for the governing vehicle motion. We then derive a kinematic equation relating motion in the vehicle coordinate frame to that of the inertial or Earth-fixed reference frame. Finally, we use that numerical approximation to write a computer simulation of the vehicle motion.

6.1 Combined Nonlinear Equations of Motion

For the vehicle dynamics equations (for the vehicle rigid body) and dynamic equations (Equation 3.8) with the equations for the vehicle in six degrees of freedom (Equation 4.49), we arrive at the combined nonlinear equations of motion for the REMUS vehicle in six degrees of freedom.

Surge motion along the x-axis:

$$m [\ddot{x} - v\dot{r} + w\dot{q} - xg(q^2 + r^2) + yg(\dot{p} - \dot{r}) + zg(\dot{p} + \dot{r})] = X_{HS} + X_{u|u|} |u| + X_{\dot{u}} \dot{u} + X_{uq} u\dot{q} + X_{qqq} q^3 + X_{vrvr} + X_{rrrr} + X_{prop} \quad (6.1)$$

Sway motion along the y-axis:

$$m [\ddot{y} - w\dot{p} + u\dot{r} - yg(\dot{r}^2 + \dot{p}^2) + zg(q\dot{r} - 3\dot{p}) + xg(q\dot{p} + \dot{r})] = Y_{HS} + Y_{y|y|} |y| + Y_{\dot{y}} \dot{y} + Y_{r|r|} |r| + Y_{\dot{q}} \dot{q} + Y_{\dot{r}} \dot{r} + Y_{ur} u\dot{r} + Y_{wp} w\dot{p} + Y_{pq} p\dot{q} + Y_{uuv} u^2 \dot{v} + Y_{uus} u^2 \dot{s} \quad (6.2)$$

Heave motion along the z-axis:

$$m [\ddot{z} - u\dot{q} + v\dot{p} - zg(\dot{p}^2 + \dot{q}^2) + xg(r\dot{p} - \dot{q}) + yg(r\dot{q} + \dot{p})] = Z_{HS} + Z_{w|w|} |w| + Z_{q|q|} |q| + Z_{j\dot{j}} \dot{j} + Z_{\dot{q}} \dot{q} + Z_{uq} u\dot{q} + Z_{vp} v\dot{p} + Z_{rpr} r\dot{p} + Z_{uuv} u^2 \dot{v} + Z_{uus} u^2 \dot{s} \quad (6.3)$$

Roll, or rotation about the x-axis:

$$I_{xx} \ddot{\phi} + (I_{zz} - I_{yy})q\dot{r} + m[yg(\dot{r} - u\dot{q} + v\dot{p}) - zg(\dot{r} - w\dot{p} + u\dot{r})] = K_{HS} + K_{p|p|} |p| + K_{\dot{p}} \dot{p} + K_{prop} \quad (6.4)$$

Pitch, or rotation about the y-axis:

$$I_{yy} \ddot{\theta} + (I_{xx} - I_{zz})r\dot{p} + m[zg(\dot{u} - v\dot{r} + w\dot{q}) - xg(\dot{u} - u\dot{q} + v\dot{p})] = M_{HS} + M_{w|w|} |w| + M_{q|q|} |q| + M_{j\dot{j}} \dot{j} + M_{\dot{q}} \dot{q} + M_{uq} u\dot{q} + M_{vp} v\dot{p} + M_{rpr} r\dot{p} + M_{uuv} u^2 \dot{v} + M_{uus} u^2 \dot{s} \quad (6.5)$$

偏航, 或绕 z-axis 的旋转:

$$\begin{aligned} I_{zz}\dot{r} + (I_{yy} - I_{xx})pq + m[xg(\delta - wp + ur) - yg(t\delta - vr + wq)] = \\ N_{HS} + N_{v|v}|v| + N_{r|r}|r| + N_{\dot{v}}\dot{v} + N_{\dot{r}}\dot{r} \\ + N_{ur}ur + N_{wp}wp + N_{pq}pq + N_{uv}uv + N_{u\delta, u\dot{\delta}}\delta^2\dot{\delta}r \end{aligned} \quad (6.6)$$

我们将发现将车辆运动方程中的速度项与加速度项分离开会更方便。因此, 这些方程可以重新写为 vehicle equations of motion. The equations can thus be re-written as:

$$\begin{aligned} (m - X_{\dot{v}})\dot{v} + mz_g\dot{q} - my_g\dot{r} &= X_{HS} + X_{u|u}|u| \\ &+ (X_{uq} - m)wq + (X_{qg} + mx_g)\dot{q}^2 + (X_{vr} + m)vr + (X_{rr} + mx_g)r^2 \\ &- m\dot{u}q\dot{p}q - mz_g\dot{p}r + X_{p\dot{p}}\dot{p} \\ (m - Y_{\dot{v}})\dot{v} - mz_g\dot{p} + (mx_g - Y_{\dot{q}})\dot{r} &= Y_{HS} + Y_{v|v}|v| + Y_{r|r}|r| \\ &+ my_g r^2 + (Y_{wr} - m)ur + (Y_{wp} + m)u\dot{p} + (Y_{qg} - mx_g)pq \\ &+ Y_{uv}uv + m\dot{u}p^2 + mz_gqr + Y_{u\delta, u\dot{\delta}}\delta^2\dot{\delta}r \\ (m - Z_{\dot{b}})\dot{b} + my_g\dot{v} - (mx_g + Z_{\dot{q}})\dot{q} &= Z_{HS} + Z_{u|u}|u| + Z_{q|q}|q| \\ &+ (Z_{uq} + m)uq + (Z_{vp} - m)v\dot{p} + (Z_{rp} - mx_g)r\dot{p} + Z_{u\dot{u}}u\dot{u} \\ &+ mz_g(p^2 + q^2) - my_g r\dot{q} + Z_{u\delta, u\dot{\delta}}u^2\dot{\delta}_s \\ - mz_g\dot{v} + my_g\dot{u} + (I_{xx} - K_{\dot{p}})\dot{p} &= K_{HS} + K_{p|p}|p| \\ &- (I_{zz} - I_{yy})qr + m(uq - vp) - mz_g(wp - ur) + K_{p\dot{p}}\dot{p} \\ mz_g\ddot{u} - (mx_g + M_{\dot{b}})\dot{b} + (I_{yy} - M_{\dot{q}})\dot{q} &= M_{HS} + M_{u|u}|u| + M_{q|q}|q| \\ &+ (M_{uq} - mx_g)uq + (M_{vp} + mx_g)v\dot{p} + [M_{rp} - (I_{xx} - I_{zz})]r\dot{p} \\ &+ mz_g(vr - wq) + M_{uv}uv + M_{u\dot{u}}u^2\dot{\delta}_s \\ - my_g\dot{q} + (mx_g - N_{\dot{v}})\dot{v} + (I_{zz} - N_{\dot{r}})\dot{r} &= N_{HS} + N_{v|v}|v| + vN_{r|r}|r| \\ &+ (N_{ur} - mx_g)ur + (N_{vp} + mx_g)v\dot{p} + [N_{rq} - (I_{yy} - I_{xx})]p\dot{q} \\ &- my_g(vr - wq) + N_{uv}uv + N_{u\dot{u}}u^2\dot{\delta}_r \end{aligned} \quad (6.7)$$

最后, 这些方程可以总结为如下矩阵形式 matrix form as follows:

$$\begin{bmatrix} X_{\dot{v}} & 0 & 0 & 0 & mz_g & -my_g \\ 0 & m - Y_{\dot{v}} & 0 & -mz_g & 0 & mx_g - Y_{\dot{q}}K \\ 0 & 0 & m - Z_{\dot{b}} & my_g & -mx_g - Z_{\dot{q}} & 0 \\ 0 & -mz_g & my_g & I_{xx} - K_{\dot{p}} & 0 & 0 \\ mz_g & 0 & -mx_g - M_{\dot{b}} & 0 & I_{yy} - M_{\dot{q}} & 0 \\ -my_g & mx_g - N_{\dot{v}} & 0 & 0 & 0 & I_{zz} - N_{\dot{r}}J \end{bmatrix} \begin{bmatrix} \dot{u} \\ \dot{v} \\ \dot{w} \\ \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} \sum X \\ \sum Y \\ \sum Z \\ \sum K \\ \sum M \\ \sum N \end{bmatrix} \quad (6.8)$$

或者

$$\begin{bmatrix} \dot{u} \\ \dot{v} \\ \dot{w} \\ \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} m - X_{\dot{v}} & 0 & 0 & 0 & mz_g & -my_g \\ 0 & m - Y_{\dot{v}} & 0 & -mz_g & 0 & mx_g - Y_{\dot{q}}K \\ 0 & 0 & m - Z_{\dot{b}} & my_g & -mx_g - Z_{\dot{q}} & 0 \\ 0 & -mz_g & my_g & I_{xx} - K_{\dot{p}} & 0 & 0 \\ mz_g & 0 & -mx_g - M_{\dot{b}} & 0 & I_{yy} - M_{\dot{q}} & 0 \\ -my_g & mx_g - N_{\dot{v}} & 0 & 0 & 0 & I_{zz} - N_{\dot{r}}J \end{bmatrix}^{-1} \begin{bmatrix} \sum X \\ \sum Y \\ \sum Z \\ \sum K \\ \sum M \\ \sum N \end{bmatrix} \quad (6.9)$$

6.2 运动方程的数值积分 Numerical Integration of the Equations of Motion

定义车辆加速度的非线性微分方程 (using the vehicle accelerations (Equation 6.9) and 参考系中的加速度 (reference frame acceleration) (Equation 6.4) gives us the vehicle accelerations 在不同时间参考系中的速度位置姿态。Given the complex and highly non-linear nature of these equations, we will use numerical integration to solve for the vehicle speed, position, and attitude in time.

考虑在每个时间步 each 我们可以将方程 6.9 表示为: Equation 6.9 as follows:

$$\dot{x}_n = f(x_n, u_n) \quad (6.10)$$

其中 x 是车辆状态向量: state vector:

$$x = [u \ v \ w \ p \ q \ r \ x \ y \ z \ \phi \ \theta \ \psi]^T \quad (6.11)$$

并且 u_n 是输入向量 input vector:

$$u_n = [\delta_s \ \delta_r \ X_{prop} \ K_{prop}]^T \quad (6.12)$$

请参阅第 3.2 节以获取车辆状态和输入的定义, 并参阅 Vehicle 和 States 以了解符号的约定。Figures 4-1 and 4-2 for the fin angle sign conventions.

The following sections summarize 三种数值积分方法 of numerical integration in order of increasing accuracy.

6.2.1 欧拉方法's Method

我们将首先考虑欧拉方法 Euler's method, 这是一种简单的数值近似方法, 其包括应用迭代公式 which consists of applying the iterative formula:

$$x_{n+1} = x_n + f(x_n, u_n) \cdot \Delta t \quad (6.13)$$

其中 Δt 是建模时间步长 modeling 欧拉方法虽然简单, 但不精确, 因为它可能在较大的时间步长下导致解发散, is unacceptable as it can lead to divergent solutions for large time steps.

6.2.2 改进的欧拉方法 Improved Euler's Method

The following method improves the accuracy of the Euler's method by averaging the slopes for two points along the line. We first calculate the following:

$$\begin{aligned} k_1 &= x_n + f(x_n, u_n) \cdot \Delta t \\ k_2 &= f(k_1, u_{n+1}) \end{aligned} \quad (6.14)$$

然后将它们组合起来以计算新的状态向量 the new state vector:

$$x_{n+1} = x_n + \frac{\Delta t}{2} (f(x_n, u_n) + k_2) \quad (6.15)$$

这种方法比欧拉方法更准确 more accurate than Euler's method.

6.2.3 龙格库塔法 Runge-Kutta Method

该方法通过对四个点的斜率进行平均, 进一步提高了近似的准确性。我们首先计算以下内容 slope at four points. We first calculate the following:

$$\begin{aligned} k_1 &= x_n + f(x_n, u_n) \\ k_2 &= f\left(x + \frac{\Delta t}{2} k_1, u_{n+\frac{1}{2}}\right) \\ k_3 &= f\left(x + \frac{\Delta t}{2} k_2, u_{n+\frac{1}{2}}\right) \\ k_4 &= f(x + \Delta t k_3, u_{n+1}) \end{aligned} \quad (6.16)$$

插值后的输入向量在哪里 input vector

$$u_{n+\frac{1}{2}} = \frac{1}{2} (u_n + u_{n+1}) \quad (6.17)$$

我们将上述方程结合起来得到

$$x_{n+1} = x_n + \frac{\Delta t}{6} (k_1 + 2k_2 + 2k_3 + k_4) \quad (6.18)$$

这种方法是目前三种方法中最准确的，这就是我们将在车辆模型代码中使用的方法。the vehicle model code.

6.3 计算机模拟 Computer Simulation

如引言所述，作者使用MATLAB实现了这一数值近似。模型代码见附录A。该模型代码的MATLAB原理是，模型代码在每个时间步，计算车辆的力和力矩。The model code works by calculating for each time step the forces and moments on the vehicle as a function of vehicle speed and attitude. These forces determine the vehicle body-fixed accelerations and earth-relative rates of change. These accelerations are then used to approximate the new vehicle velocities, which become the inputs for the next modeling time step.

该车辆模型需要两个输入：

- “Initial conditions, or the starting vehicle state vector inputs, or vehicle俯仰鳍和尾翼角度，这些角度可作为预定矢量。用于比较模型输出与现场数据时，或在控制系统设计中每个时间步计算。
- Control inputs, or the vehicle pitch fin and stern plane angles, either given as a pre-determined vector, when comparing the model output with field data, or calculated at each time step, in the case of control system design.

Chapter 7

Field Experiments

7.1 Motivation

为了验证车辆模型的准确性，作者进行了一系列实验，the author conducted a series of experiments in order to verify the accuracy of the vehicle model. 这些实验测量了车辆对舵角和尾平面角阶跃变化的响应。这些实验在海洋系统实验室工作人员的协助下进行。地点包括新泽西州科尔顿的罗格斯海洋田野站。the changes in rudder and stern plane angle. These experiments were conducted with the help of the staff at both the Rutgers University Marine Field Station in

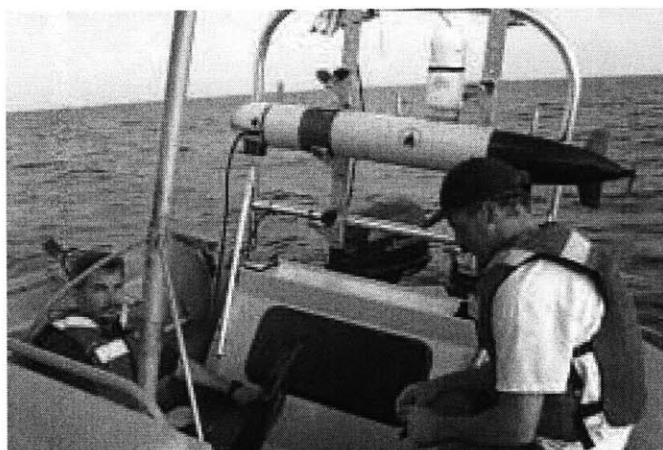


Figure 7-1: The author (left) and Mike Purcell (right) conducting vehicle experiments at the Rutgers University Marine Field Station in Colonia, New Jersey. Photo courtesy of Prof. Nuno Cruz, of Porto University.

7.2 Measured States

在每次实验中，我们测量了车辆的深度和姿态。这些都在车辆中表示出来。in each experiment, we measured the depth and attitude of the vehicle. These are represented in the vehicle model through the following vehicle-referenced vehicle states:

$$x = [z \ 0 \ \phi \ \theta]^\top \psi \quad (7.1)$$

在这些实验中，我们还记录了车辆的鳍角，这在车辆模型中表示为 the vehicle model by the following control inputs:

$$u = [u_s \ u_r]^\top \quad (7.2)$$

请参见图4-1了解车辆坐标系的示意图。The 参见图4-1和图4-2了解控制舵符号约定的示意图。4-2 for diagrams of the control fin sign conventions.

请注意, 对于本章中描述的所有现场测试, 车辆螺旋桨未被用作控制输入, 而是保持在恒定的1500转/分钟。由于不同螺旋桨转速下的螺旋桨推力和扭矩难以估算, 坚持使用恒定值使我们能够从车辆模型比较中消除一个不确定性来源。

7.3 车辆传感器Sensors

在作者的实地实验中, 可用的导航传感器如表1所示。对于每个传感器, 我们将介绍传感器的功能及其已知的限制。

请注意, 传感器的精度通常与成本相关。例如, 像RMU155这样的车辆设计相对低成本, 但精度相对较低。高精度的陀螺罗盘可能会使车辆成本翻倍。车辆设计面临的挑战是确定能够满足车辆导航需求的最廉价的传感器组合。

7.3.1 标题 Heading: 磁性罗盘Magnetic Compass

车辆航向是通过三轴磁通门磁力计测量的。该磁力计感知车辆相对于地球磁场的方向。由于磁场的敏感性, 例如车辆内部各种磁性物体产生的磁场, 磁罗盘需要进行校准。校准过程通常要求磁罗盘在车辆周围进行校准, 同时通过俯仰和滚转。通过积分偏航角速度并将其与测得的航向进行比较, 可以制作一张航向偏差随车辆航向变化的表格。

这种磁罗盘可以校正恒定的磁噪声来源, 例如车辆电池, 但不能校正间歇性噪声, 例如螺旋桨电机。因此, 航向测量可能会偏差多达五度。

7.3.2 偏航率 yaw rate 陀螺 Tuning Fork Gyro

车辆偏航率使用音叉陀螺仪测量。通过对传感器输出进行积分, 可以获得航向。容易受到漂移的影响, 因此在测量高频车辆运动时更准确。因此, 在测量高频车辆运动时更准确。通过将低通滤波的指南针数据与高通滤波的高频车辆偏航率陀螺仪数据结合, 我们可以得出车辆航向的更准确估计。

7.3.3 姿态 Attitude 传感器 Sensor

车辆的俯仰和滚转是通过电解倾斜和横滚传感器测量的。该传感器测量杯中的导电液体团的位置。例如, 当车辆向前倾斜时, 导电液体团向前移动, 指示车辆俯仰向下。

该传感器对于低频运动非常准确, 但由于导电液体的惯性, 在捕捉高频运动时仍然会有问题。此外, 液体的运动是耦合的, 因此高车辆偏航率或纵向加速度会产生错误的俯仰测量。

7.3.4 深度 Depth 压力 Pressure Sensor

车辆深度由压力传感器测量。该仪器对周围海水温度的变化有一定的敏感性, 但其误差相对于指南针和姿态传感器的误差来说, 幅度较小。

7.4 实验步骤 Experimental Procedure

在本节中, 我们描述了表1中列出的现场实验所使用的程序。这些实验在大约1米深的水中进行。实验地点包括在兹霍尔附近的哈德利湾。

海洋生物地理学及在新泽西州罗格斯斯的罗格斯海洋实地站附近的大西洋沿岸 Rutgers, New Jersey.

Table: 7. 车辆侧地实验 Field Experiments

Date	Filename	Vehicle	Location
29 Jun 1998	d980729a	STD REMUS (Dock1)	RUMFS
29 Jun 1998	d980729b	STD REMUS (Dock1)	RUMFS
27 Oct 1998	d981027	STD REMUS (Dock1)	Hadley's
28 Oct 1998	d981028	STD REMUS (Dock1)	Hadley's
26 Jun 1999	a990726	NSW REMUS (NSW)	RUMFS
27 Jun 1999	a990727	NSW REMUS (NSW)	RUMFS

7.4.1 发射前检查清单 Check List

在每次通信之前, 作者都会按照图 2 所示的流程进行数据检查, 以检查车辆是否加密, 2 to check the communication status and to verify operation of the vehicle sensors and communications.

7.4.2 修边Trim and Ballast Check

Following the pre-launch weight check, the vehicle's buoyancy was measured in a sea water tank, and the vehicle ballast adjusted to achieve the 1.5 pounds of positive buoyancy as described in Section 2.4.

7.4.3 车辆任务编程Mission Programming

The USCM4 使用基于任务的编程架构, 任务中的每个元素都被命名, 而元实验使用 Early element 的目标: mission is called an objective. The following types of objectives were used in the thesis experiments:

- **SETPOINT**: This command tells the vehicle to move to the given latitude and longitude, using the given transponder beacons. The command specifies the vehicle's starting position, the given latitude and longitude, the distance and heading range, and the propeller RPM. The command tells the vehicle to start its mission by reaching into the water and spinning the vehicle propeller. The mission program starts, the propeller starts spinning on its own, we push the vehicle underwater and it is on its way.
- **LONG BASELINE**: This command tells the vehicle to navigate to the given latitude and longitude, using the given transponder beacons. In this mode, the vehicle uses long baseline navigation, dead-reckoning its position between acoustic fixes. See Roger Stokey's paper [28] for details on REMUS navigation.
- **TIMER**: This command tells the vehicle to maintain the given depth or heading using feedback control, or to maintain a given, fixed fin angle or propeller RPM. The timer commands represent the experimental sections of each mission.

这些目标被编辑并作为文本文件发送到车辆。有关示例任务文件，请参见附录 F 中的附录 F 中的示例任务文件。

为了测量车辆对起升角度阶跃变化的响应,车辆被下边changes下指令和
following commands:

- 对 *Tim* “抬头”命令, *ed* 车辆被指令潜入 2 米深,, 以避免被水面 as 对 *tim* “低头”命令, *six* 车辆被指令潜入 2 米深, *oid* breaking the surface. For “pitch down” commands, the vehicle was commanded to 2 meters depth.

THE SIS EXPERIMENT PRE-LAUNCH CHECKLIST

DATE: _____
 MISSION: _____
 OPERATOR: _____
 MISSION OBJECTIVE: _____

- 1 Check vehicle dock. Reset if necessary.
- 2 Check that the OHS is working.
- 3 Check pressure gauge: should have vehicle altitude < 40381 m.
- 4 Check that CTD data makes sense in air:

Cond: _____ Temp: _____ Press: _____

- 5 Record compass pitch and roll offsets:

Pitch: _____ Roll: _____

- 6 Check ADC operation.
Listen for transmission pings.
Ensembles being collected in text window.
- 7 Check acoustics on bench.
Uncomment ATS diagnostics in mission file.
Set remus ranger (ranger.jff) to TRANSDUC mode.
Place transducers 1 to 2 inches apart.
Check numbers:
SIN > 75000
Attenuation < 45 (Drop < 26 when removed)
BBN < 6
Peak > 28000
Turn RANGEROFF. Verify attenuator drops below 26.
Comment ATS diagnostics backout of mission file.
- 8 Verify REMUS ranger operation on 5 seconds.
- 9 Verify REMUS ranger abort.
Make sure the prop is clear of then sensitive pings.
- 10 Set data rate to maximum in remus_vini.
Set "Log every nth status msg" = 1.
- 11 Check pitch and roll in alignments.
Record offsets from remus_vini.

Rich: _____ Rider: _____

- 12 Program LOCATION file.
Triple check all buoy LAT/LONG, be aware of offsets and always use the true lat/long locations indicated in the LOCATION INFO pull down.
- 13 Program MISSION file.
- 14 Check mission using ROUTE INFO.
Verify DEPTH, Distance and RPM for each leg.
Verify total time of mission.
- 15 Clear all fault messages.
- 16 RESTART vehicle.
- 17 Check that all lights are GREEN.
Read any fault messages that appear.
- 18 Check battery voltage by turning off external power supply.
Should be 22.46V.
Record battery voltage: _____

REMUS: HOUSING OPEN

- 1 Check vehicle onshore power.
- 2 Check IMU alignment in mounting bracket.
- 3 Align battery slide to marks on chassis screw holes.
- 4 Check screws on chassis blind-mate connector mounting plate.
- 5 Check IMU serial and power connections, tie-wrap compass serial connector.
- 6 Check IMU compass calibration.
ASU commands through serial debug:
'h' - clears hard iron calibration
'b' - stops polling
'c' - clears compass calibration
'r' - restarts compass
- 7 Check IMU operation in heading, pitch, and roll.
Compass light GREEN?
- 8 Check 9V sonar back-up battery voltage. Replace if necessary.
- 9 Check forward and aft O-rings for ricks, lubricate if necessary.
- 10 Note ADI Mail section: _____

REMUS: HOUSING CLOSED

- 1 Check midbody housing forward and aft for pinched O-rings.
- 2 Be sure that all files have been uploaded and archived.
Delete
C:\remus\ntf
C:\adp.dat
C:\gps.dat
Archive
C:\shadow\mission\mission.ntf
C:\shadow\mission\location.ntf
C:\shadow\remus_vini
C:\adp.dat
- 3 Set battery type in remus_vini file.
Clear battery if new or fresh charge.

Fig 7-2: 7-2 REMUS 发射前检查表 (Checklist) (Page One)

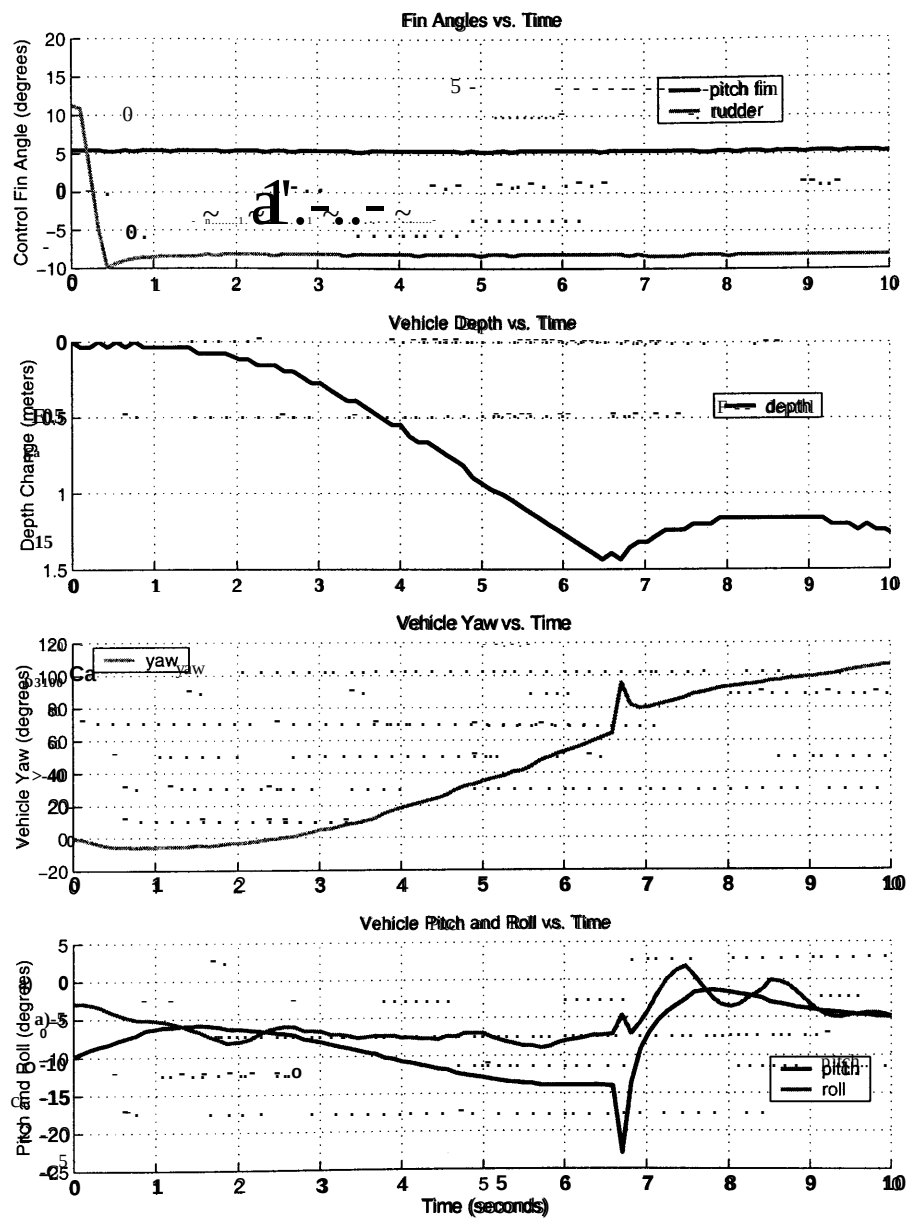


Figure 7: REMUS Mission Data: Vehicle Dynamics of the bottom [d980729a, Obj. 6]

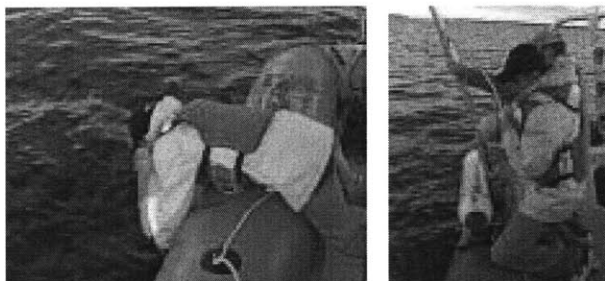


图7-4: 7发射和回收REMUS车辆 [Photo courtesy of Rob Colclough, WHOI OSL] Goldsborough, WHOI OSL]

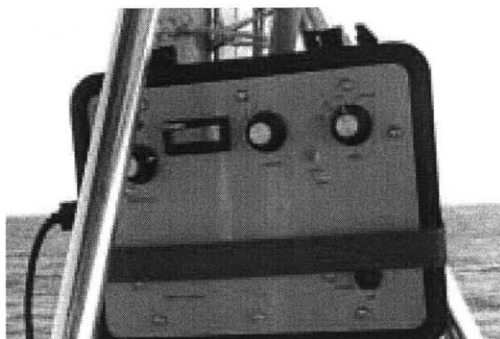


图7-5: 7REMUS游侠US Ranger

在图7-9中,guide还显示车辆在前两秒内,为了保持恒定深度,需要调整鳍角为五十四度。这表明车辆的配重略微向前倾。这可能是由于车辆发射过程中内部配重的移动造成的。vehicle was ballasted slightly nose-down. This may be due to internal ballast weights shifting during the launch of the vehicle.

尽管该车辆在整個过程中都处于闭环航向控制下,但您会在数据中注意到一些航向漂移。目前尚不清楚这是否反映了实际车辆运动,还是仪器误差。clear whether this reflects actual vehicle motion, or instrument error.

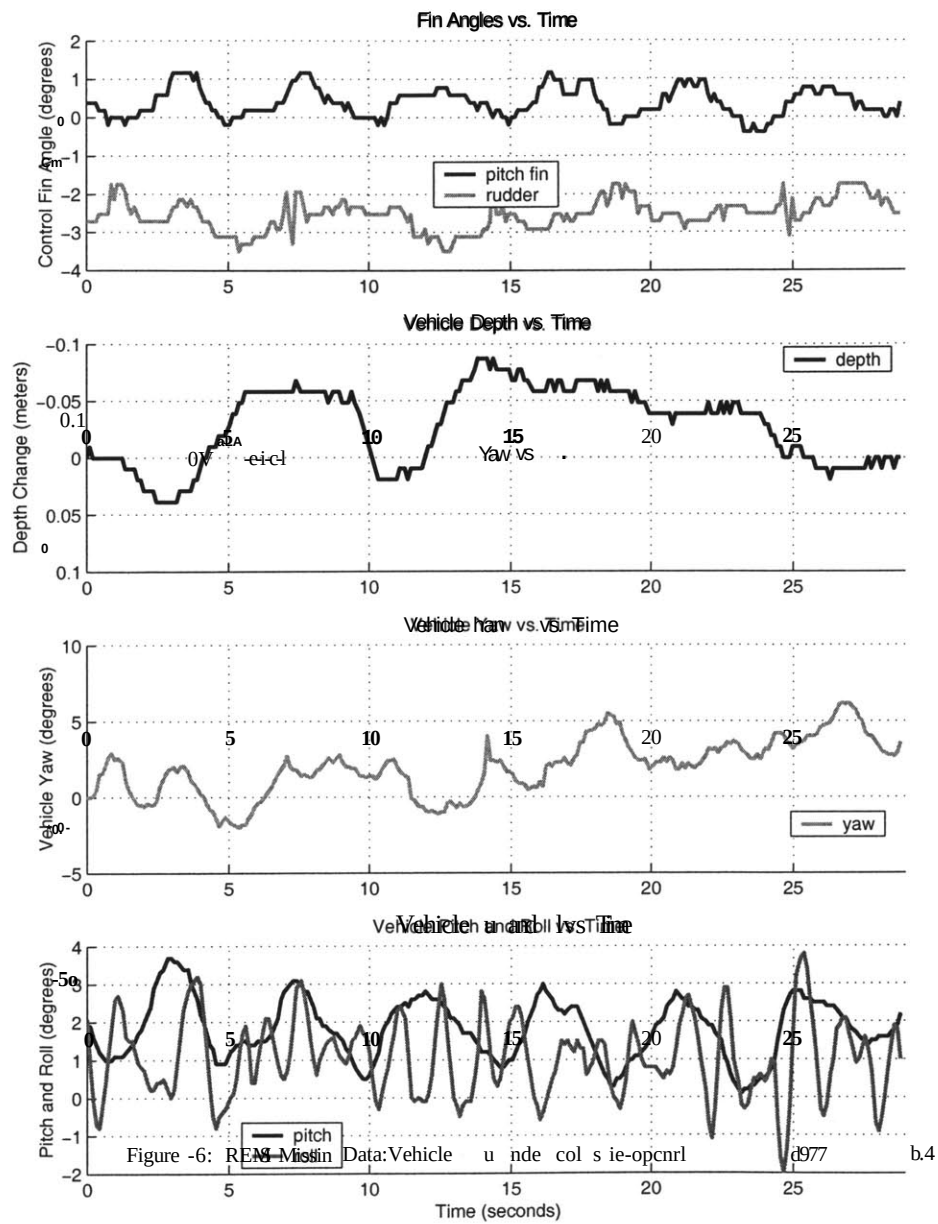


Figure 7-6: REMUS Mission Data: Vehicle under closed-loop control. [d990727, Obj. 4]

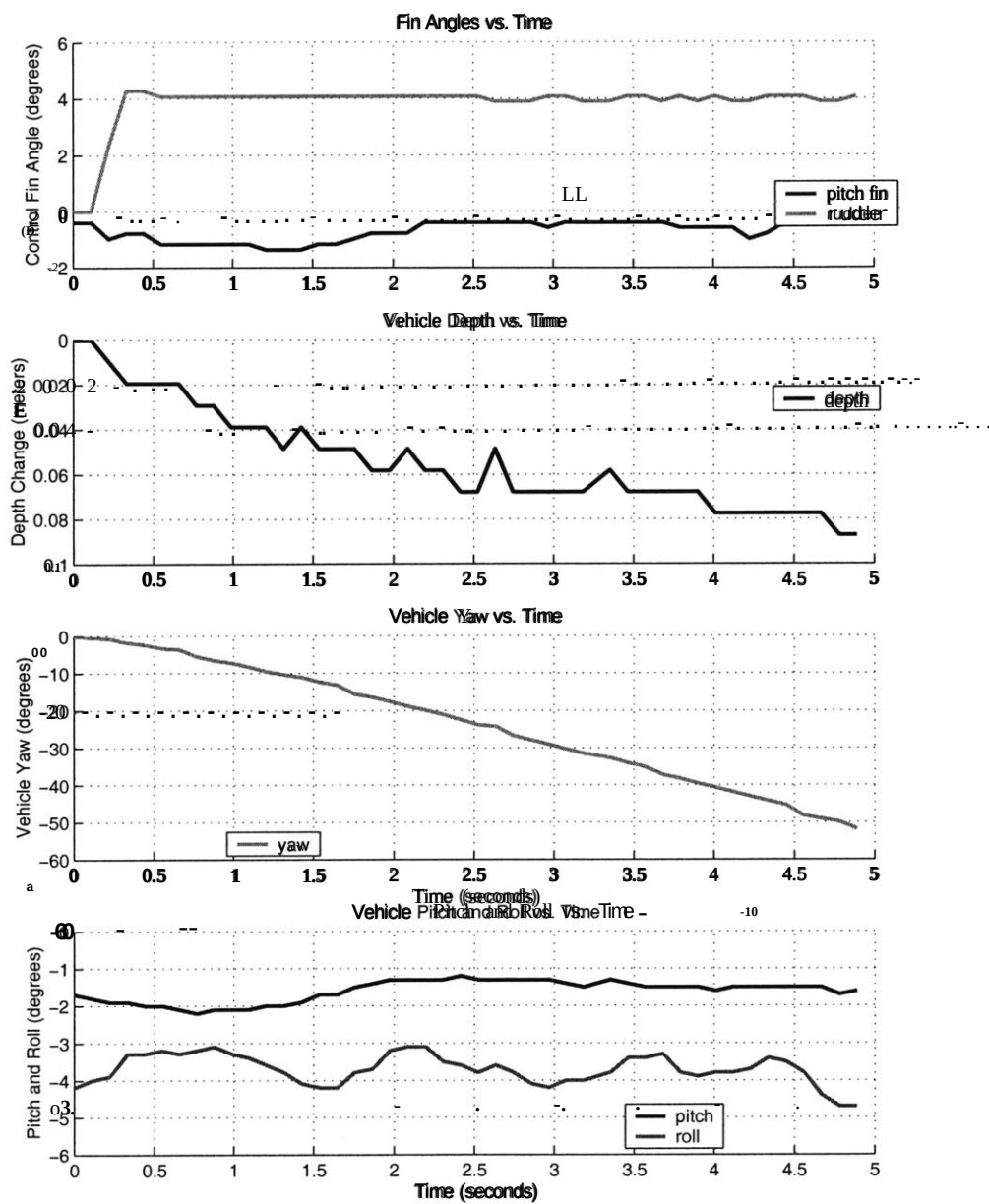


Figure 2. RMU3000 Mission Data (Step 1028, Obj. 25)

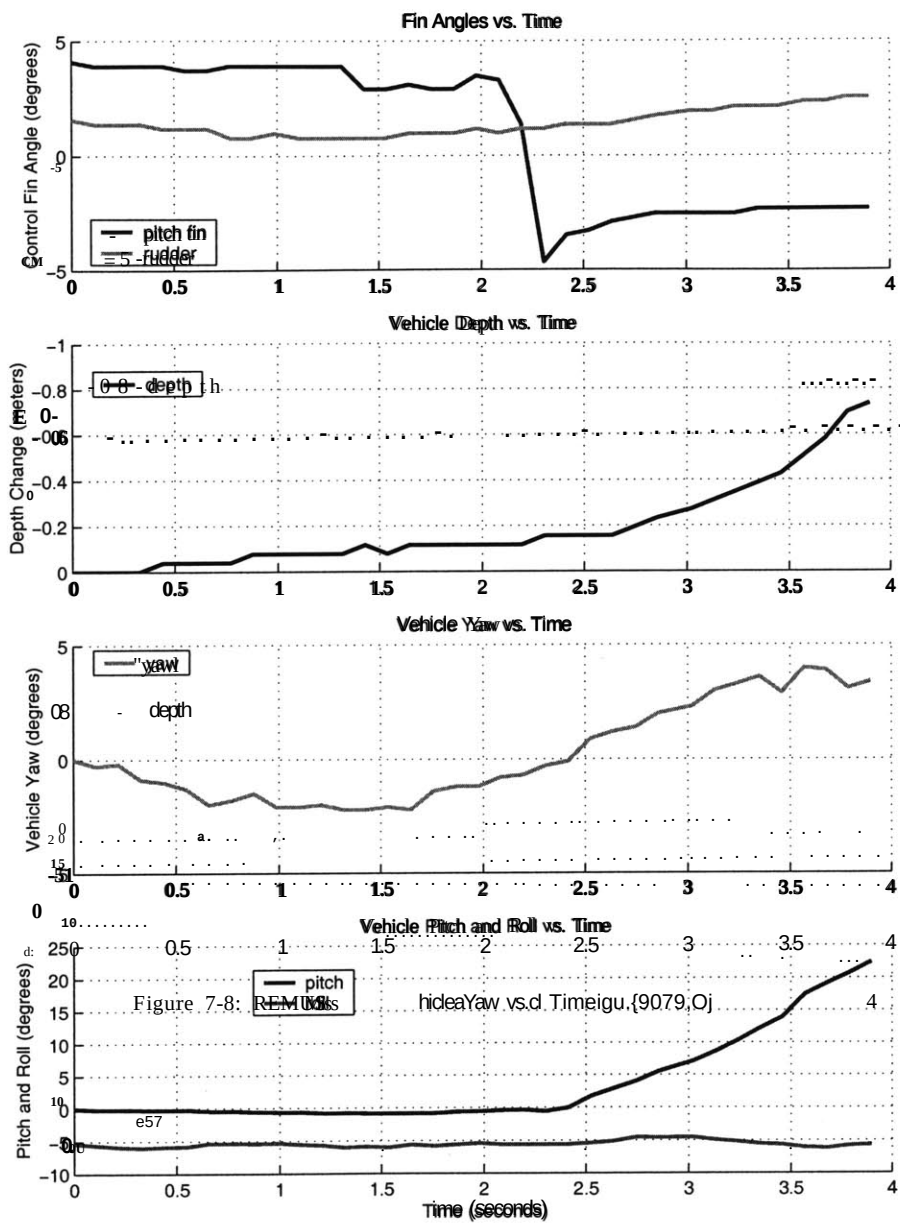


Figure 7-8: REMUS US A data. Vehicle Yaw vs. Time (9079, Obj. 14)

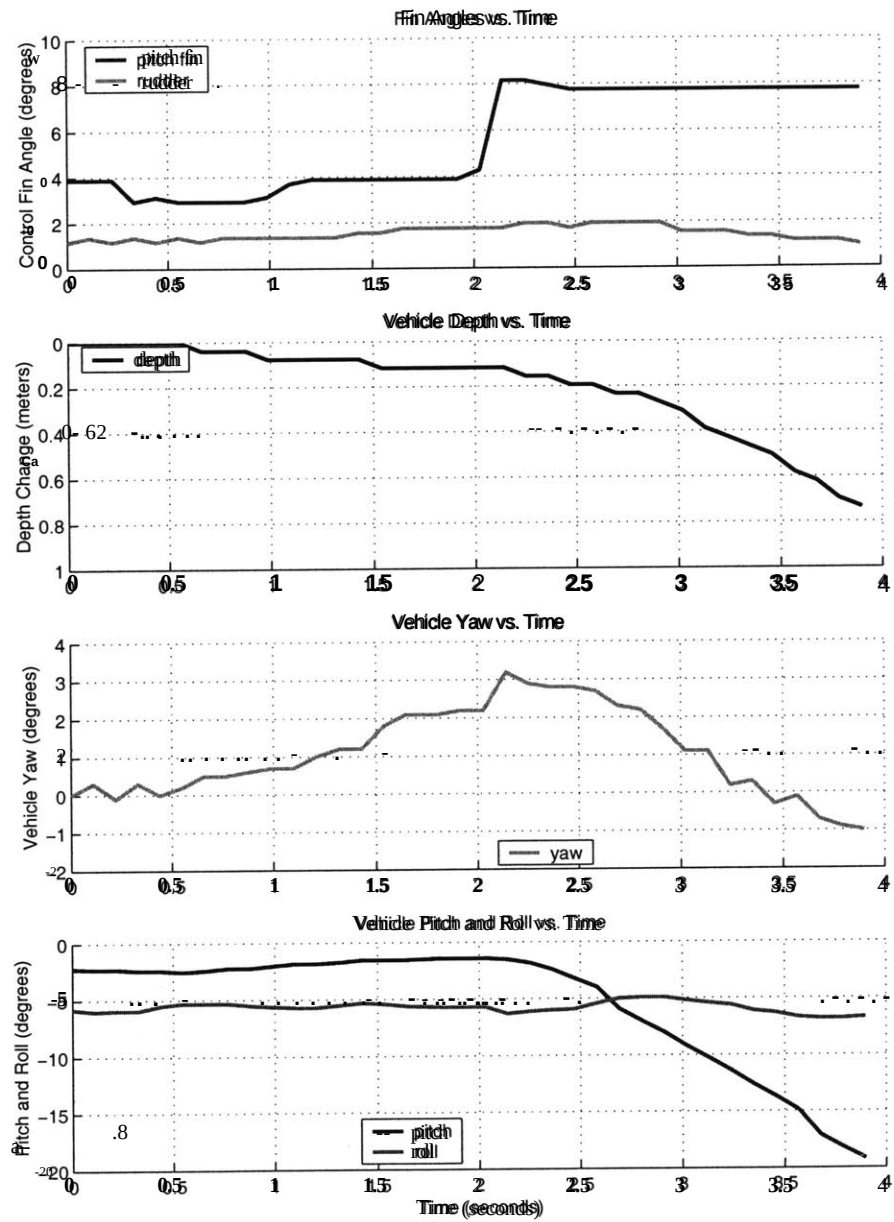


Figure 7-9. REMUS Mission Data as Vehicle pitches down. [d980729b, Obj. 22]

Chapter 8

Comparisons of Output and Experimental Data

在本章中，我们将把模拟器输出与第4章中描述的实验数据进行比较。我们将讨论两组数据之间的差异，以及用于修正这些差异的系数调整。

8.1 Model Preparation

模型被给予与实验数据相匹配的初始条件和最终输入。早期模型的比较促使作者调整一些车辆系数。

8.1.1 Initial Conditions

模型的每次运行都给出了以下初始条件：

Table 8.1: REMUS Simulator Initial Conditions			
Parameter	Value	Units	Description
z_g	+11.96e-002	m	vertical center of gravity
u	+1.54e+000	m/s	Forward velocity
ϕ	-5.00e+000	deg	Roll Angle

推进器转速为1500RPM时，车辆的前进速度为1.54 m/s (3 ft/s)。初始横滚角是由于推进器扭矩引起的实验测量的稳态横滚偏移。

其余的角速度和速度被输入为零。虽然在实验中没有直接测量车辆的角速度和速度，但假设它们都很小。

8.1.2 Coefficient Adjustments

作者认为有必要根据下表中的系数，对第4章中得出的部分车辆系数进行调整。

这些调整是基于与实验数据的比较，并且并不完全符合。在第4.2.3节和第4.3.3节中，用于计算滚动阻力的方法具有高度的不确定性。当探索更准确的方法来计算这种滚动阻力和附加质量时，第4.2.2节中用于估算横流阻力的条带积分法也被认为是不准确的。

请注意，在Table 4.2.3列出了未调整的车辆系数，而在附录中列出了已调整的车辆系数。C list the adjusted vehicle coefficients.

表B12:8.车辆系数调整因子 Coefficient Adjustment Factors

Coefficient	Adjustment Factor	Description
K_{pp}	100	Rolling Resistance Moment
$K_{\dot{p}}$	5	Roll Added Mass Moment
Y_{qv}	10	Sway Resistance Force
Z_{Ww}	10	Heave Resistance Force
M_{qm}	12.5	Pitch Rate Resistance Moment
N_{rr}	10	Yaw Rate Resistance Moment

8.2 模型比较中的不确定性 Model Comparison

以下不确定性影响了模型比较的准确性 the accuracy of the model comparison:

- **Vehicle Initial Conditions** 最大的不确定性是在每个实验目标开始时的车辆状态。我们无法测量中流波浪影响以及非轴车辆速度，这些都会影响车辆在环境操作过程中的运动。non-axial vehicle velocities, which would have all affected the vehicle motion during open-loop maneuvers.
- **Control Fin Alignment** 尽管在每次实验前各成员都检查了车辆鳍片的状态，但在车辆运输和发射过程中很难防止车辆控制鳍片被撞击。这可能导致鳍片的错位达到五度，多 getting knocked during vehicle transportation and launch. This could have resulted in fin misalignments as great as five degrees.
- **Attitude Sensor Dynamics** 车辆姿态传感器对由于车辆加速度引起的耦合很敏感，虽然最可能是 vehicle 小影响，但作者没有机会刻画这些传感器的动态特性。the author did not have the opportunity to characterize these sensor dynamics.

8.3 水平面动力学 Horizontal Plane Dynamics

图 8-18 显示了 8 车辆对舵角阶跃变化的响应。在图 8-18 中，在车辆进入水池前，舵角为 0 度，持续 15 秒，然后舵角为 4 度，持续 25 秒，然后舵角为 -4 度，持续 30 秒。

图 8-19 显示了车辆偏航角速率的阶跃变化。这符合图 8-18 中的实验数据，比较符合，which compares well with the experimental data in Figure 7-7, Section 7.5.1.

图 8-20 显示了实验数据与模拟数据的直接比较。模拟车辆的偏航角速率与实验非常接近。车辆的偏航角速率以及车辆俯仰角和横滚角的差异可能与模拟器初始条件的不同有关。Discrepancies between the vehicle depth rates, and vehicle pitch and roll angles likely have to do with differences in the simulator initial conditions.

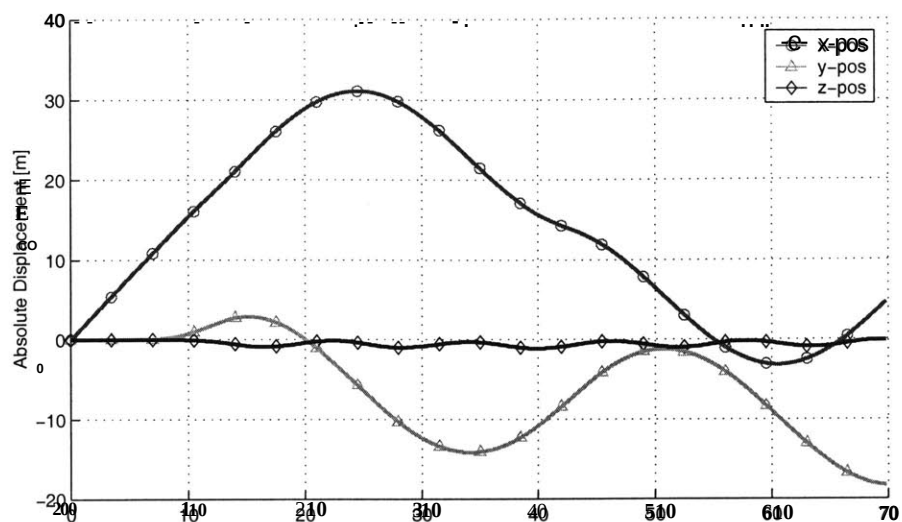
8.4 垂直面动力学 Vertical Plane Dynamics

垂直面运动不同于平面运动，因为它受到重心分离的影响，必须仔细考虑车辆对正负俯仰鳍角的响应 to both positive and negative pitch fin angles, due to the effect of the vehicle center of gravity-center of buoyancy separation.

8.4.1 车辆抬头 Pitching Up

图 8-21 显示了 8 车辆对俯仰角阶跃变化的响应。在图 8-21 中，在车辆进入水池前，俯仰角为 0 度，持续 15 秒，然后俯仰角为 10 度，持续 2 秒，然后俯仰角为 -10 度，持续 2 秒。

图 8-22 显示了车辆深度约为 0.5 米的俯仰角速率的阶跃变化。这两者与图 8-21 中的实验数据比较符合，which compares well with the experimental data in Figure 7-8, Section 7.5.2.



aL)

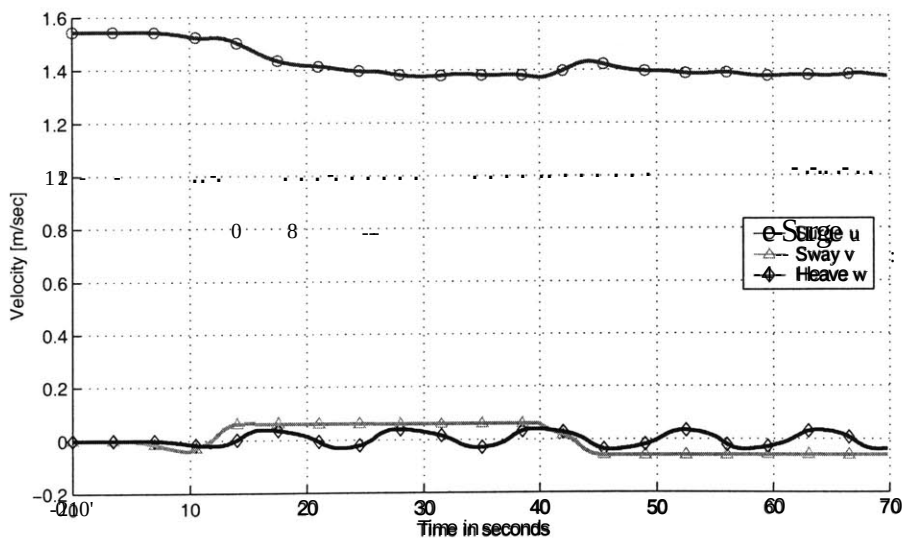


Figure 8-1: REMUS 仿真器数据: 车辆在水平面响应的线性位移和速度
 Figure 8-1: REMUS simulator data: Linear displacements and velocities for vehicle response in the horizontal plane

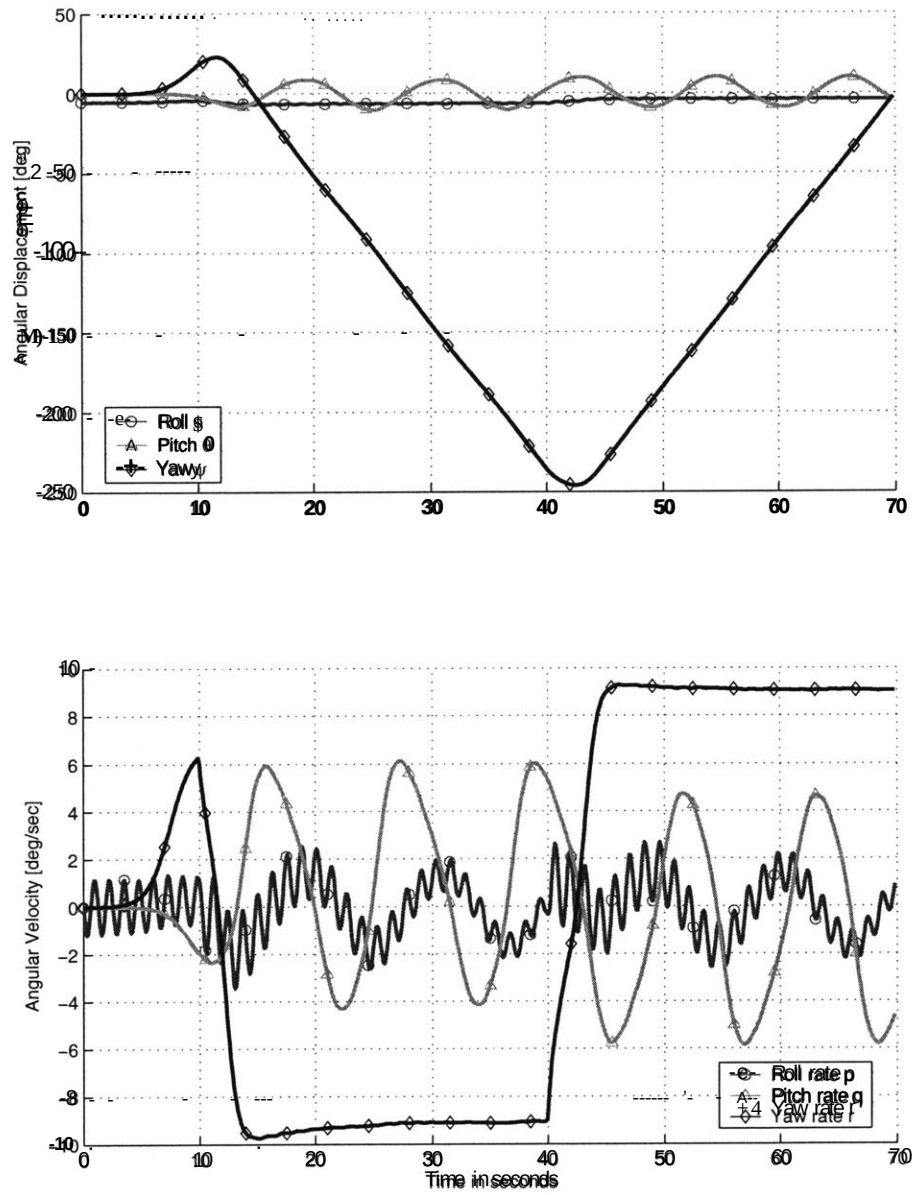


图8-2:8-REMUS仿真器数据 车辆在水面响应的角位移和角速度
 Fig.8-2:8-REMUS simulator data for vehicle response in the horizontal plane

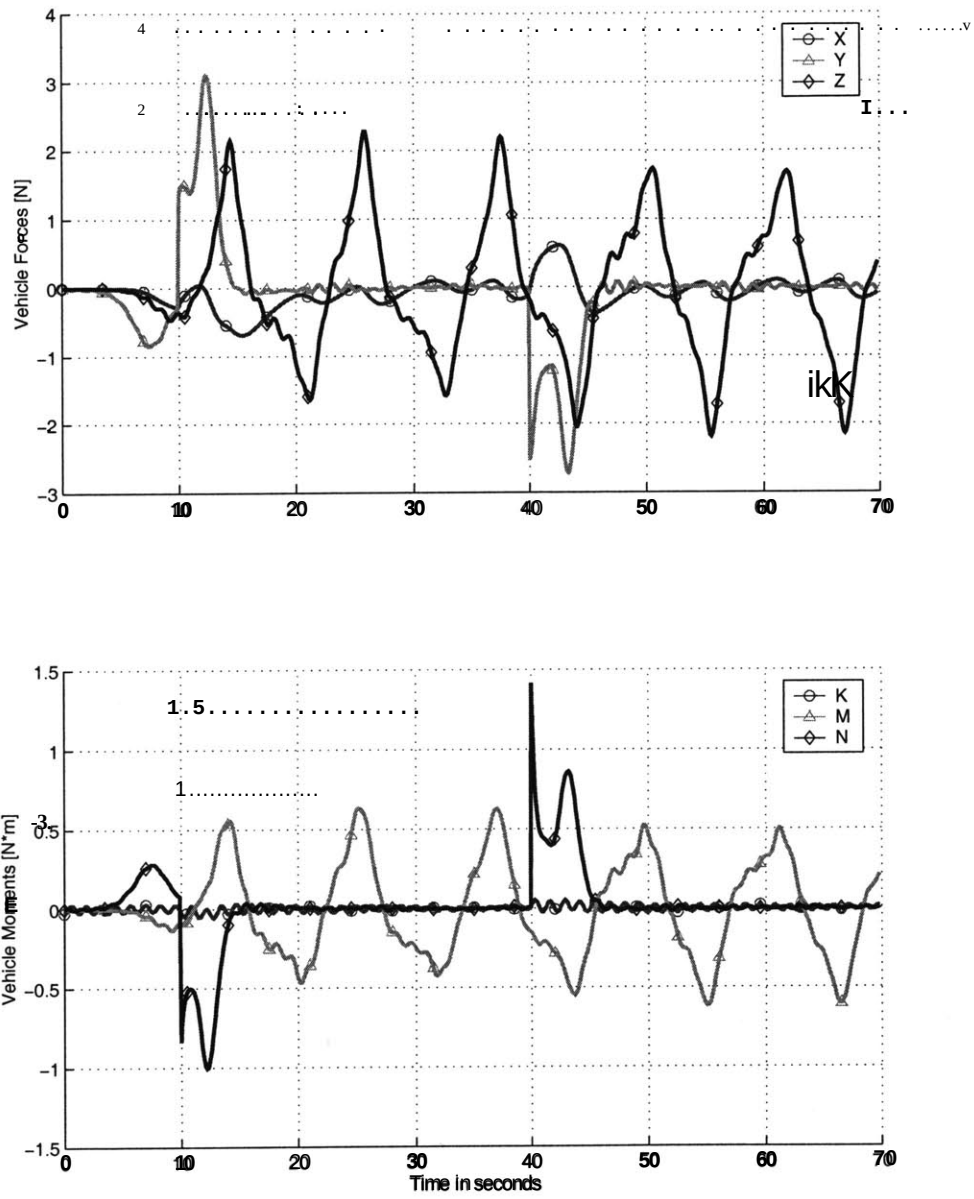


图 8-3: 8BEMES 模拟器数据 or 车辆在水平面响应的力和力矩 for vehicle response in the horizontal plane

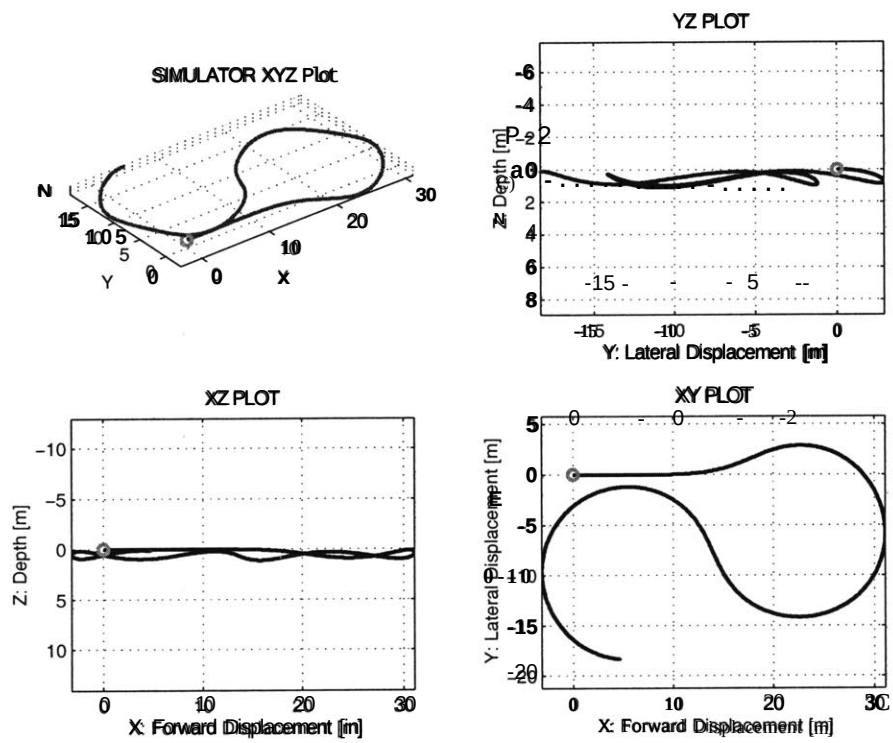


图8-4:8-REDCS 模拟器数据: 车辆在水平面上的车辆响应轨迹 the response in the horizontal plane

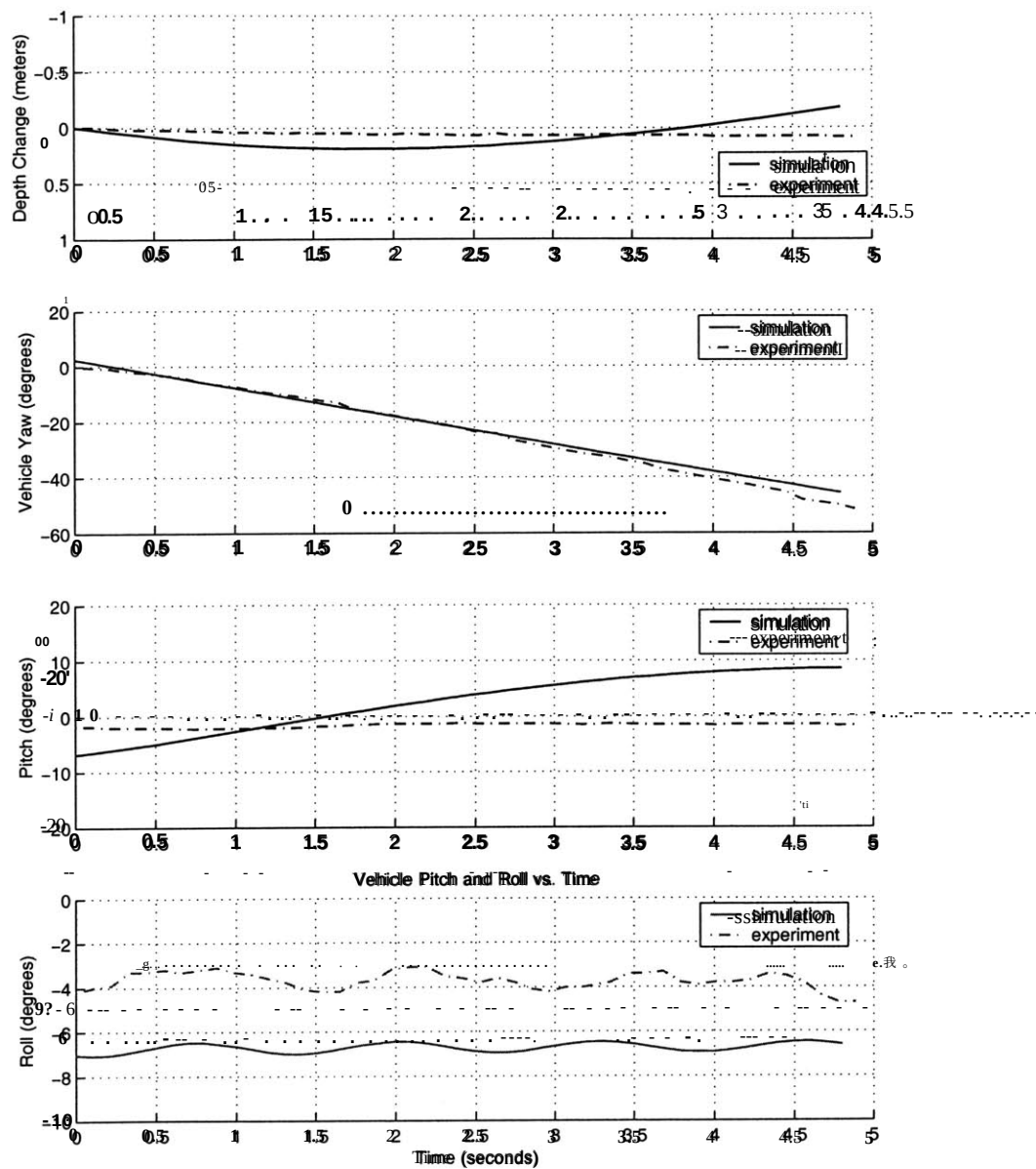


图 8.5: REMUS 模拟器数据。车辆在偏航平面响应的比较图
Figure 8.5: REMUS Simulator Data Comparison plots for vehicle response in the yaw plane

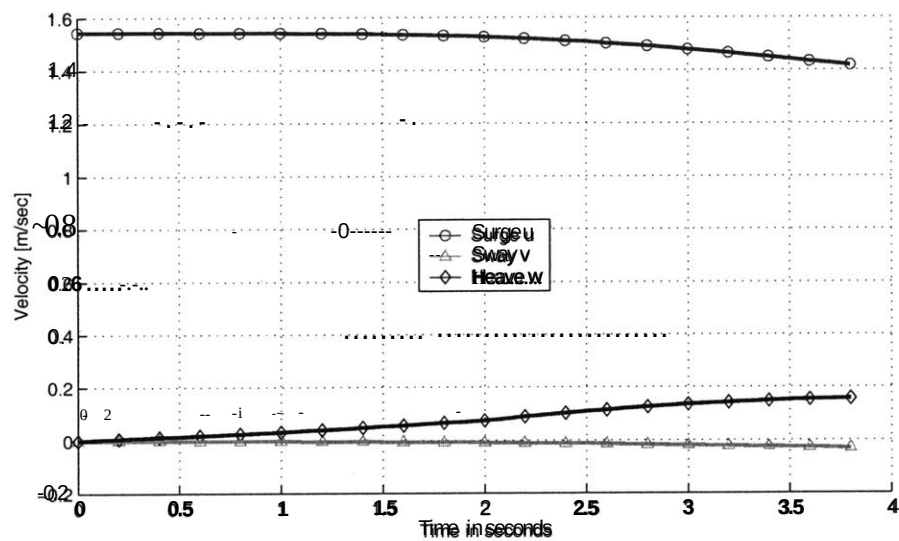
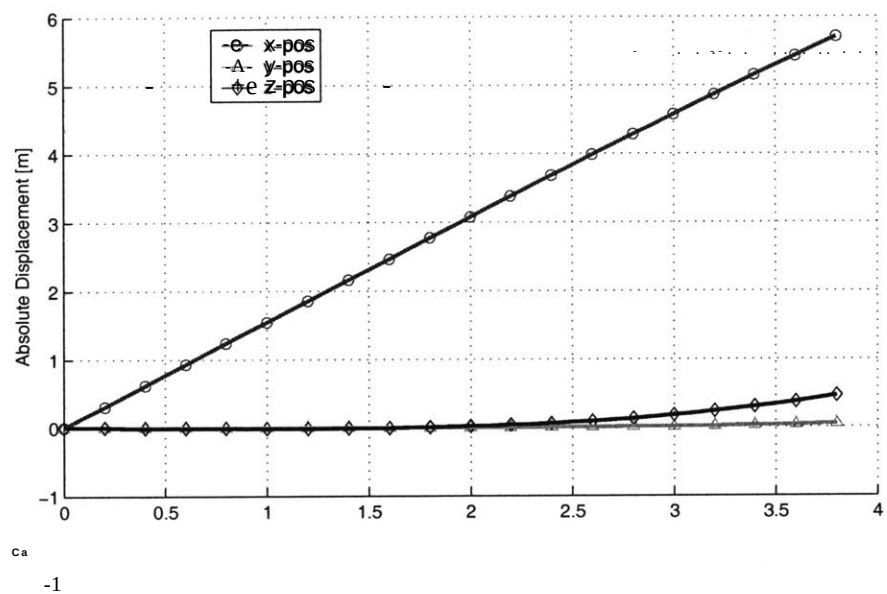


图8-6:8-REDSU模拟器数据 车辆俯仰向上时的线性位移和速度

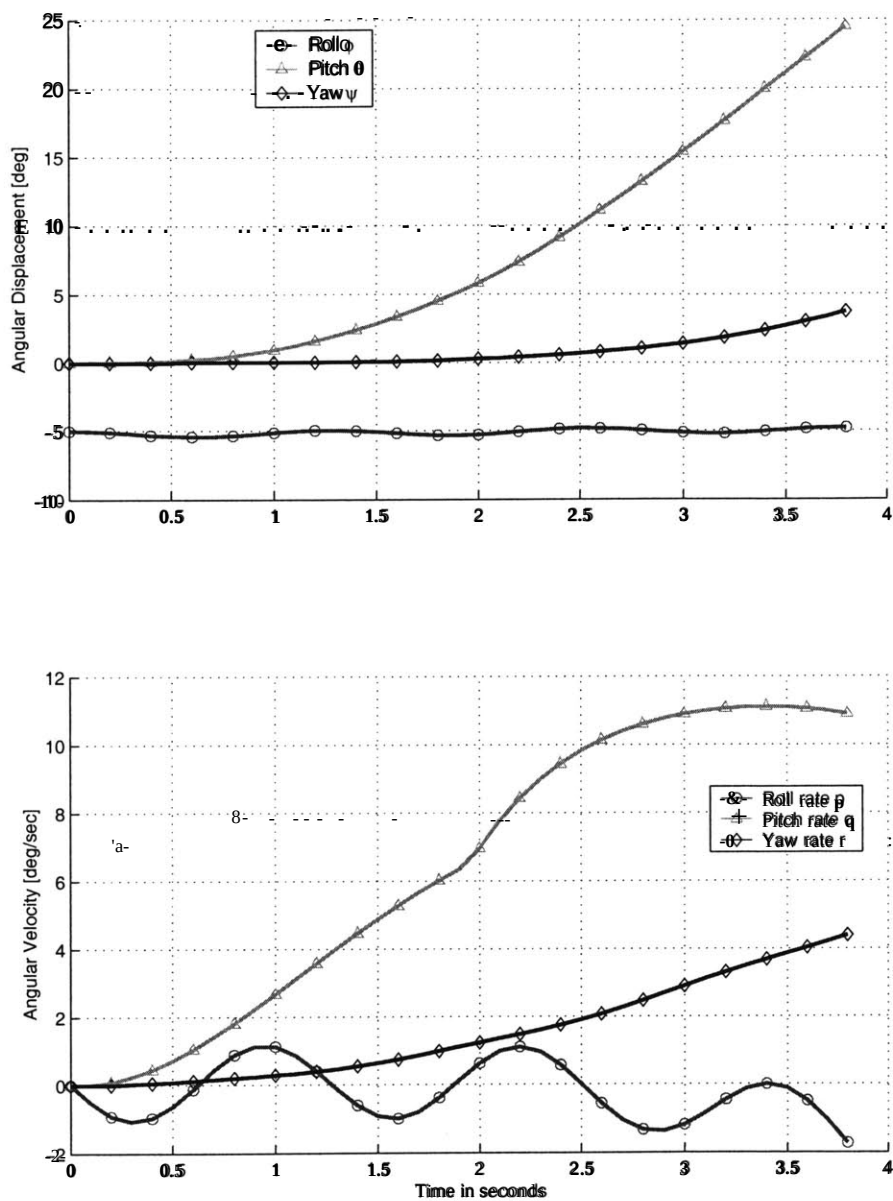


Figure 8-7 REMUS Simulator Data: Angular displacements and velocities for vehicle pitching up

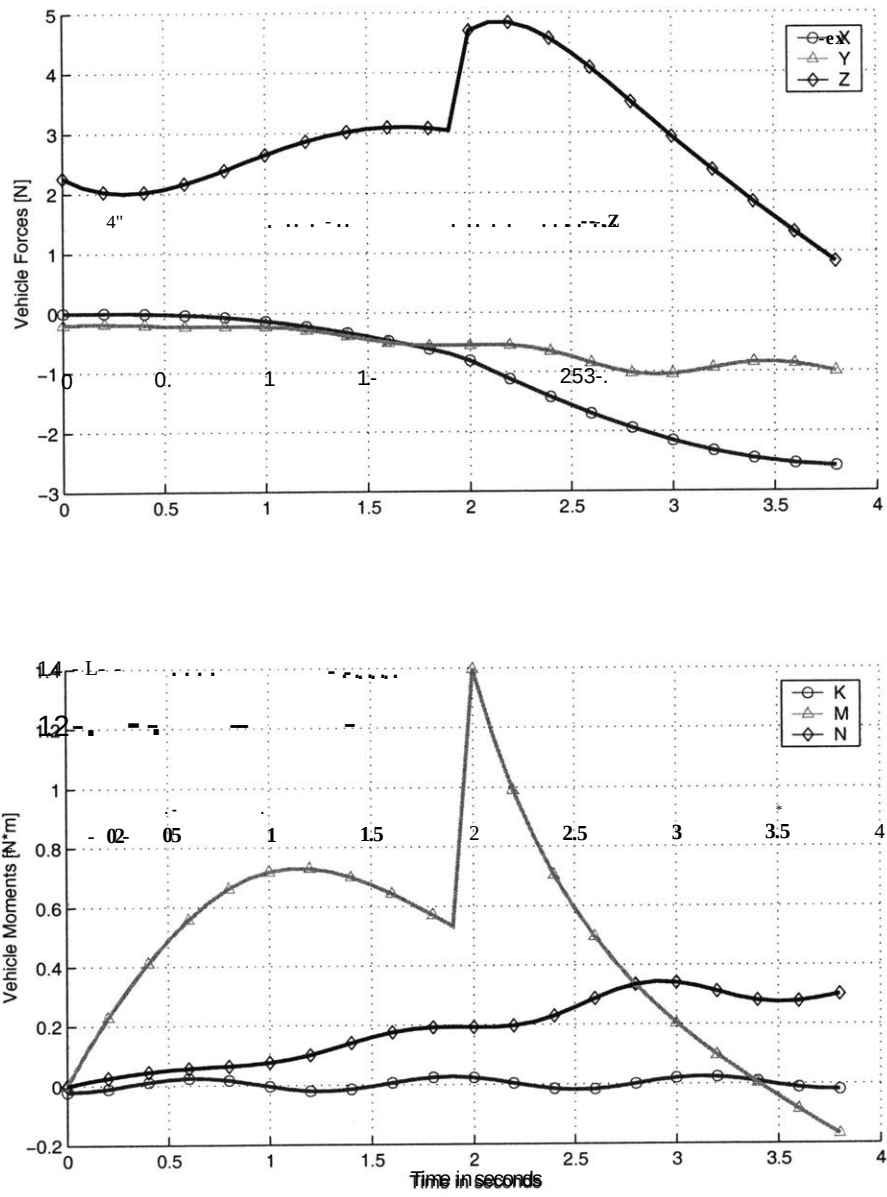


Figure 8: REMUS 模拟器数据: 车辆俯仰向上时的力和力矩 for vehicle pitching up

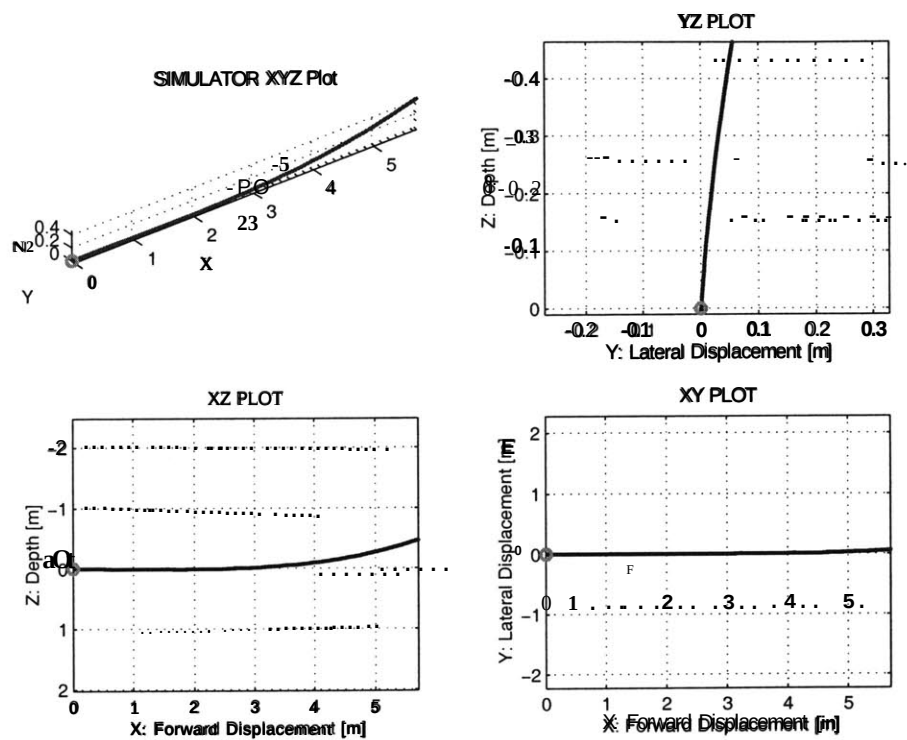


图 8-9:8-REIMSUS 仿真器数据 for 车辆Pitching up 的车辆轨迹

啦

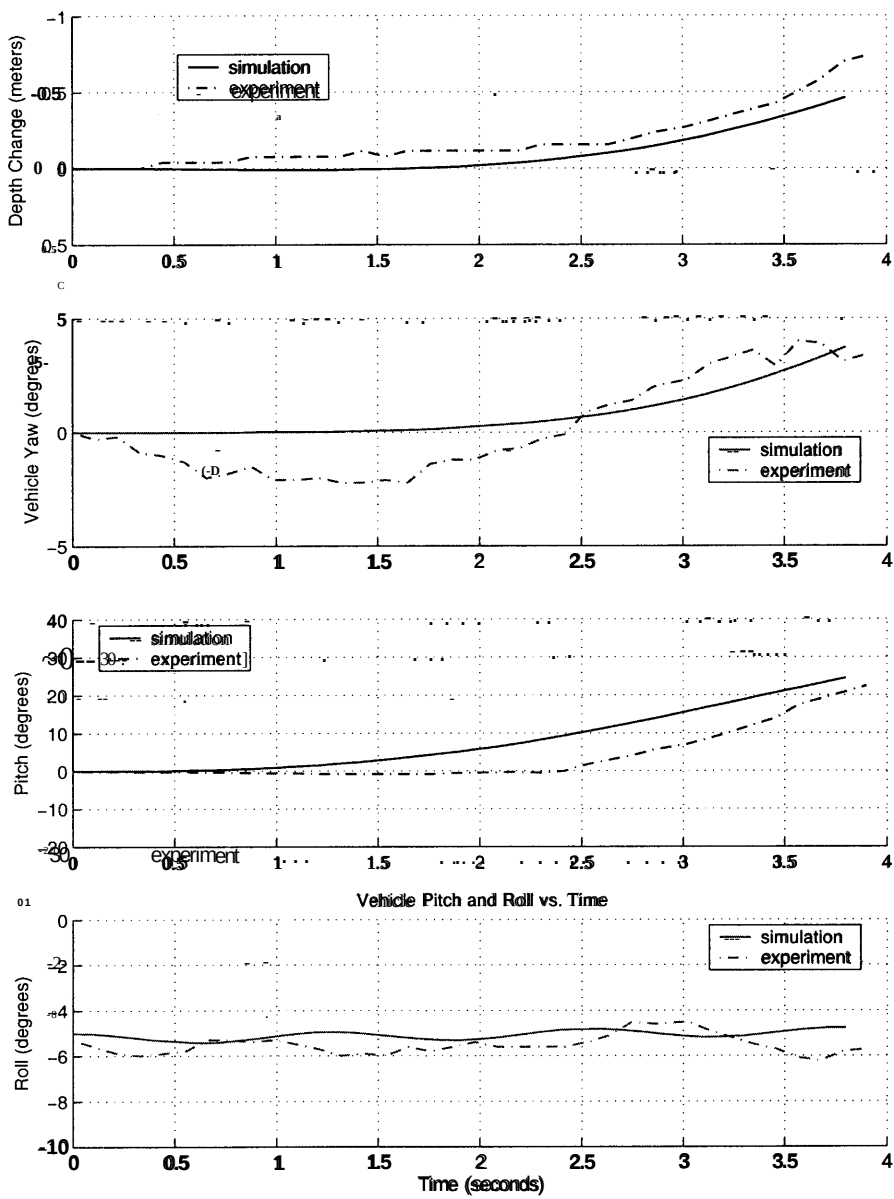


Figure 8-10 RMRCN Model Sensor data at vehicle pitching up Comparison plots for vehicle pitching up

8.4.2.2 车辆俯仰 Pitching Down

Figure 8-15 through 8-16 show the vehicle response to a step change in pitch range. A step command is input, then, in two seconds, the vehicle pitches down eight degrees of positive pitch fin for two seconds.

Figure 8-15 shows the depth change of about 0.6 meters and a pitch change of about 9.4 degrees which is close to the experimental data. The model pitch rate is slightly higher than the experimental data, but this could be due to a difference in initial conditions.

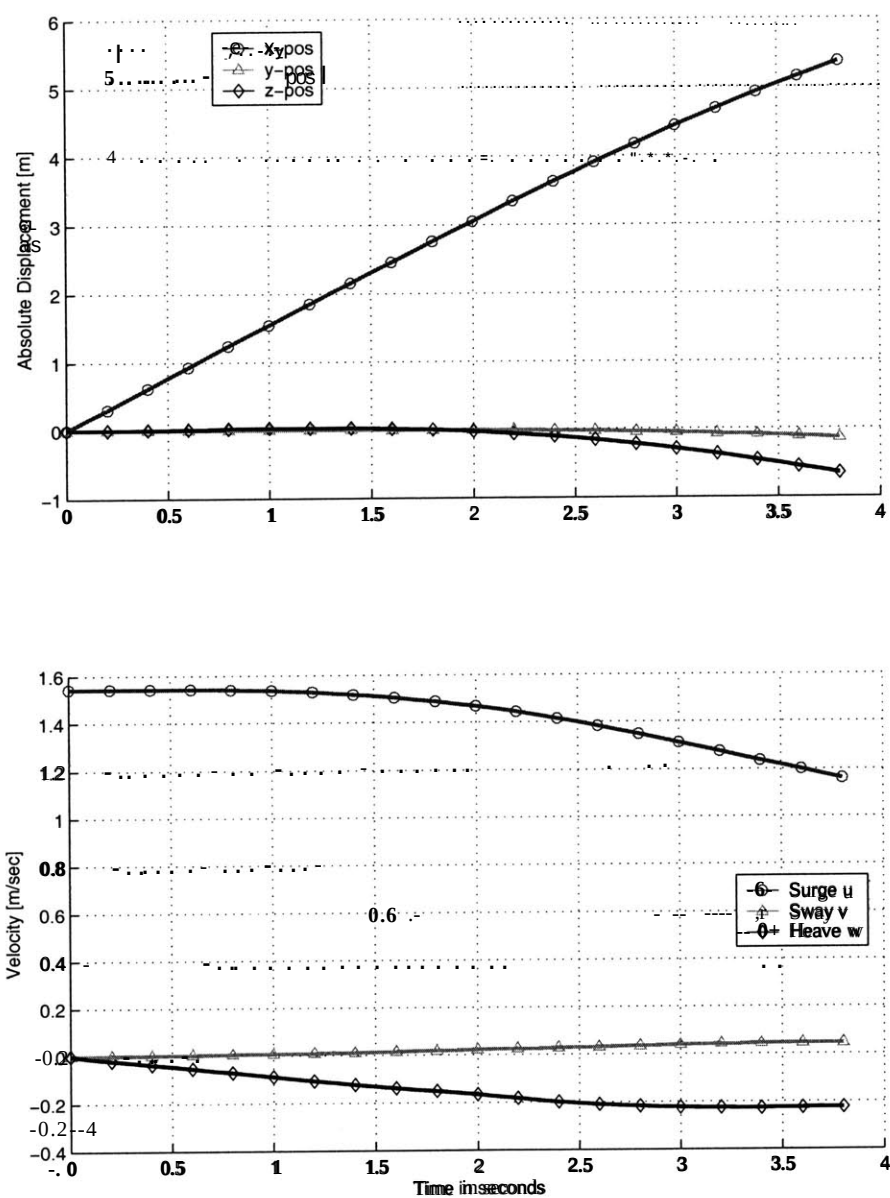


Figure 8-11 REMUS Simulator Data: Linear displacement and velocities for vehicle pitching down

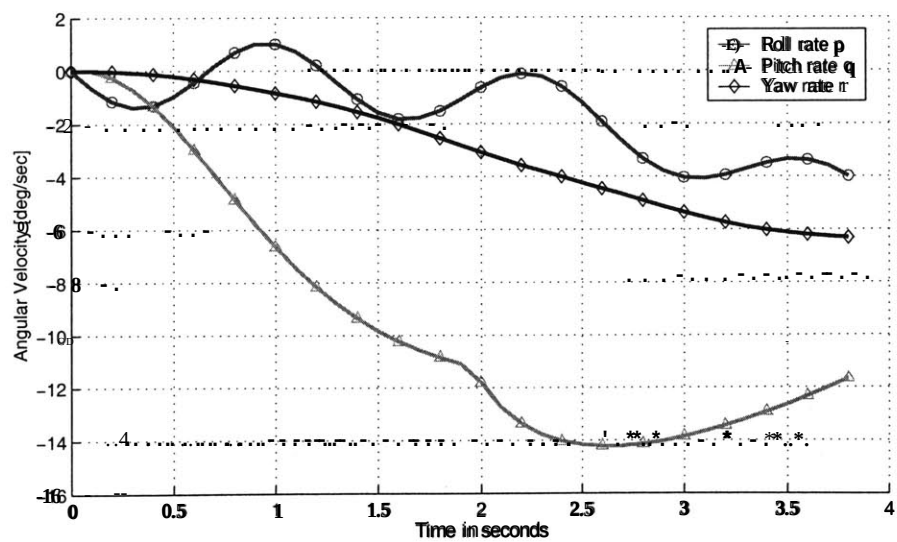
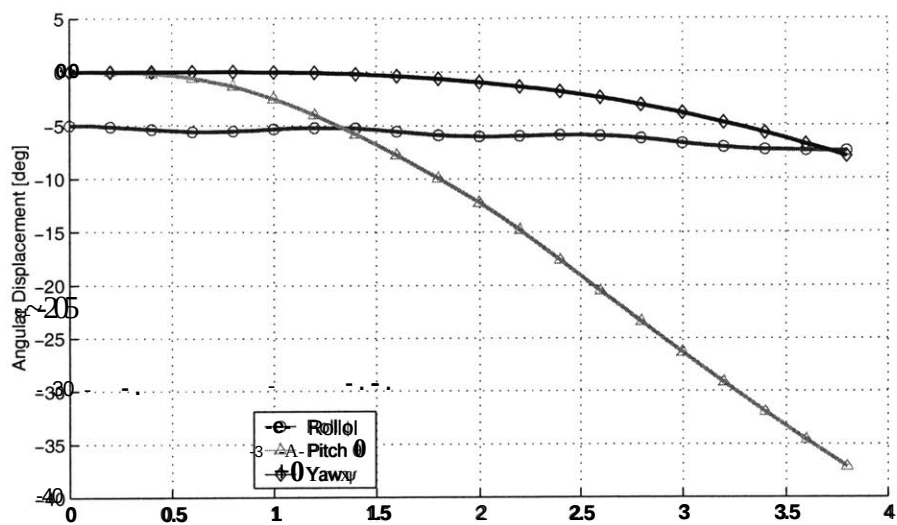


图 8-128-RENNES 模拟器数据 for 车辆俯仰向下的角位移和角速度 and velocities for vehicle pitching down

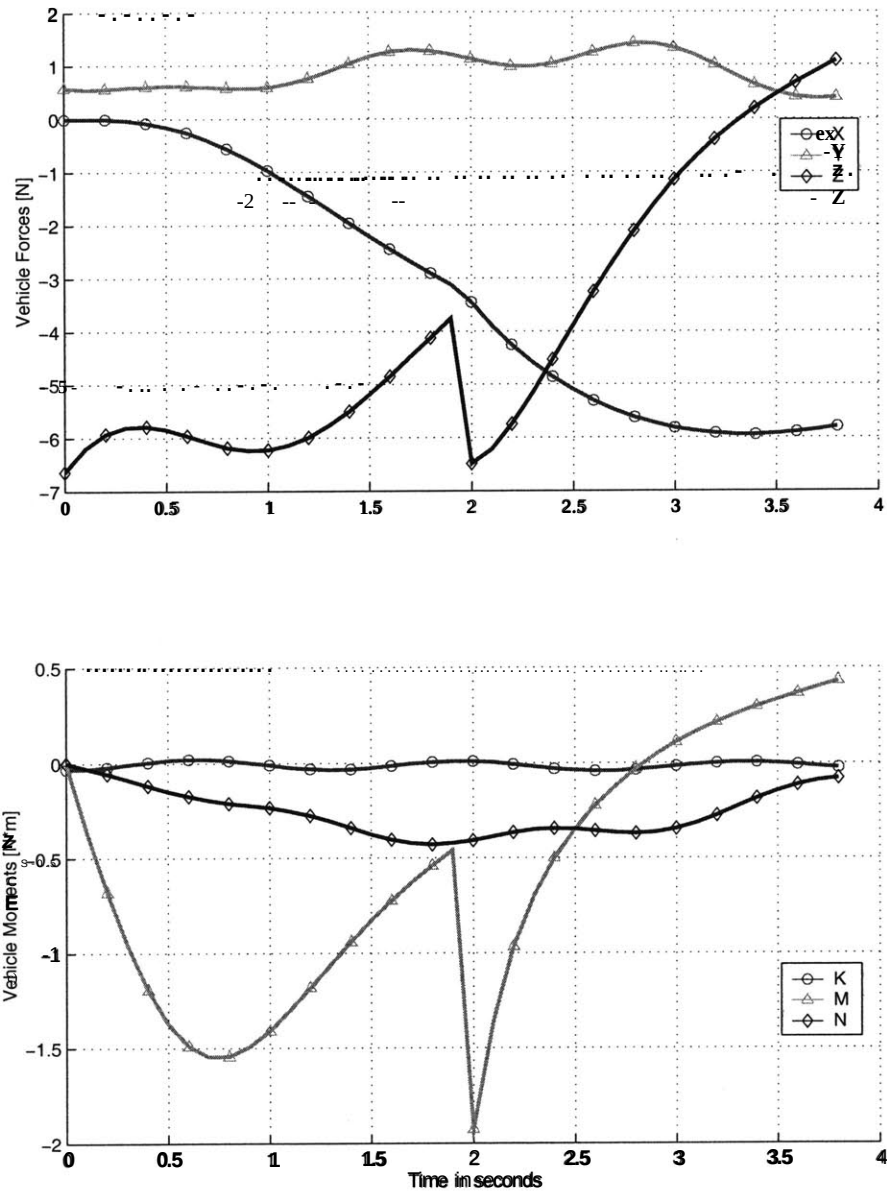


图8-138-IREMS 模拟器数据: 车辆俯仰下的力和力矩 for vehicle pitching down

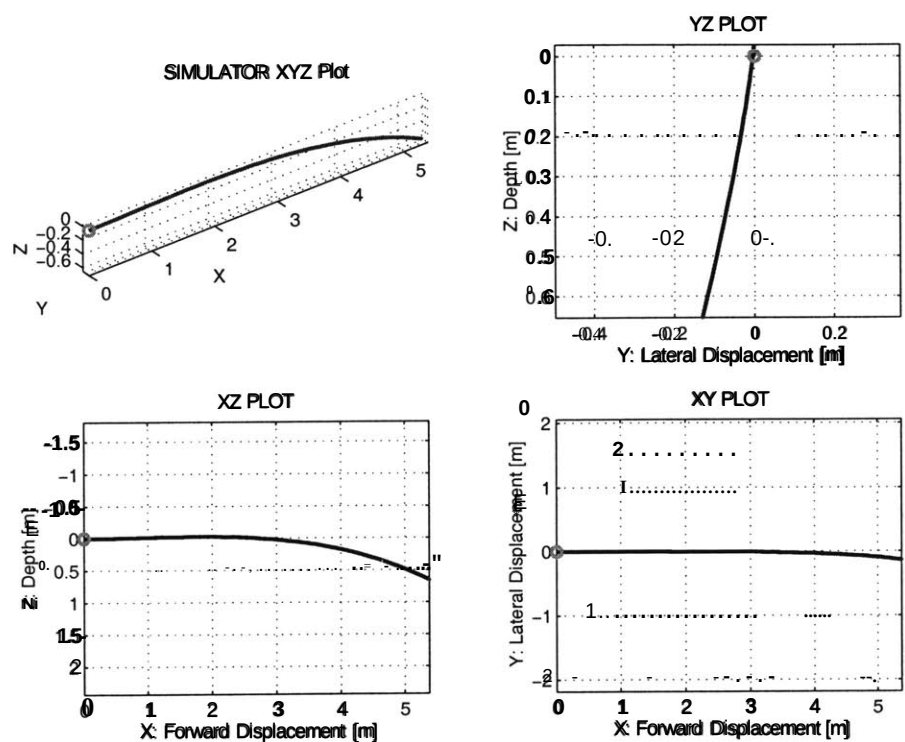


图8-14-1 IREPCUS仿真器数据：车辆俯仰下沉的车辆轨迹 for vehicle pitching down

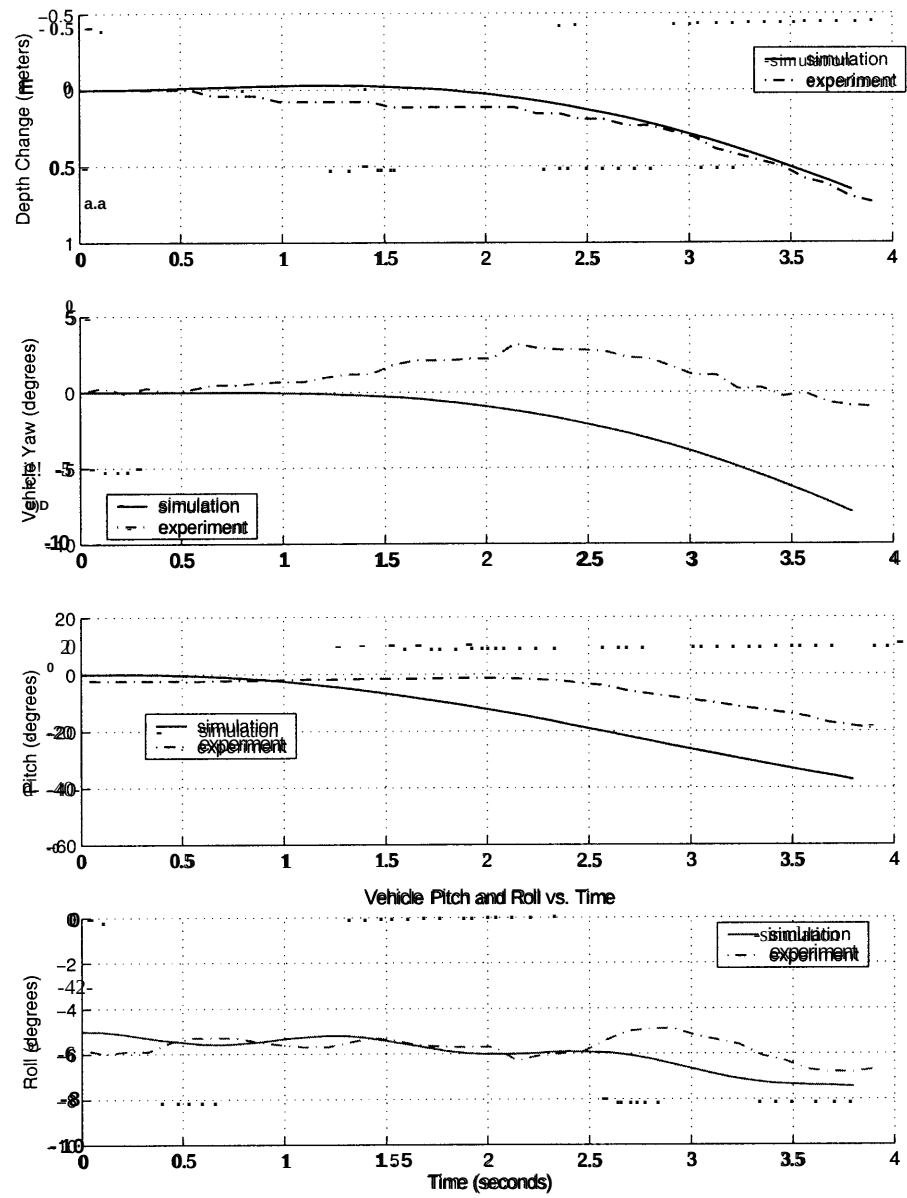


Figure 8-15 REMUS Model Simulator Vehicle Pitch and Roll Comparison plots for vehicle pitching down

Chapter 9

Linearized Depth Plane Model and Controller

This chapter describes the linearization of the vehicle equations of motion and the use of the model in the design of a pitch and depth loop PD depth controller. Finally, this chapter shows how real world effects such as environmental disturbances, sensor noise and actuator non-linearities can be added to the model.

Although this chapter only covers the depth plane model, the equations of motion are explained in sufficient detail to allow the development of a more sophisticated linearized, decoupled model in five degrees of freedom (disallowing vehicle roll).

9.1 Linearizing the Vehicle Equations of Motion

The vehicle equations of motion are described in Chapter 3. We will describe the linearization of the vehicle kinematics, rigid-body dynamics, and mechanics.

9.1.1 Vehicle Kinematics

The vehicle kinematics equations are developed in Section 3.1. The rotational coordinate transformation J is not defined for $\cos \theta = 0$. This will not be a problem, given that when we linearize the model we will be assuming small vehicle perturbations about $\theta = 0$.

As we assume the depth plane motion, we only need to consider the body forward velocity u , body heave velocity w , pitch rate q , and the earth relative vehicle forward position x , depth z , and pitch angle θ . We will set to zero all other velocities (v, p, r), and drop the equations for any out-of-plane terms.

Using these assumptions, Equation 3.4 leads to a fixed body and earth fixed vehicle velocities:

$$\begin{aligned}\dot{x} &= \cos \theta u + \sin \theta w \\ \dot{z} &= -\sin \theta u + \cos \theta w \\ \dot{\theta} &= q\end{aligned}\tag{9.1}$$

We will linearize these equations about the steady state conditions. In this case, u represents the steady-state forward velocity of the vehicle;

纵摇和横摇在零附近被线性化 about zero.

$$\begin{aligned} \mathbf{w} &= \mathbf{U} + \mathbf{w}' \\ \mathbf{w} &= \mathbf{w}' \\ \mathbf{q} &= \mathbf{q}' \end{aligned} \quad (9.2)$$

我们还将使用三角项的麦克劳林展开式 expansion of the trigonometric terms:

$$\begin{aligned} \sin \theta &= \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} + \dots \\ \cos \theta &= 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} + \dots \end{aligned} \quad (9.3)$$

应用这些线性化并去掉任何高阶项, 得到地面系和机体系速度之间的以下线性化运动学关系 linearized kinematic relationships between earth- and body-fixed velocities:

$$\begin{aligned} \dot{x} &= \dot{U} + \dot{\theta} w \\ \dot{z} &= -U\dot{\theta} + \dot{w} \\ \dot{\theta} &= \dot{q} \end{aligned} \quad (9.4)$$

9.1.2 车辆刚体动力学 Rigid-Body Dynamics

车辆运动方程在 Section 3 中进行了推导。在车辆动力学 Section 3. 我们将把刚体方程 (for rigid body) 简化为纯深度平面运动的描述。我们将把所有其他项, (v, p, r, y_g) , 变为零, 并丢弃 out-of-plane motion 的方程。我们将把所有其他项, (v, p, r, y_g) , 变为零, 并丢弃 out-of-plane motion 的方程。我们将把所有其他项, (v, p, r, y_g) , 变为零, 并丢弃 out-of-plane motion 的方程。

$$\begin{aligned} \sum \dot{\mathbf{X}} &= m(\dot{U} + w\dot{q} - x_g\dot{q}^2 + z_g\dot{q}) \\ \sum \dot{\mathbf{Z}} &= m(\dot{w} - U\dot{q} - z_g\dot{q}^2 - x_g\dot{q}) \\ \sum \dot{\mathbf{M}} &= I_y\dot{q} + m[z_g(\dot{U} + w\dot{q}) - x_g(\dot{w} - U\dot{q})] \end{aligned} \quad (9.5)$$

将方程 (9.5) 线性化的速度项 velocity 并去掉任何高阶项, 得到以下线性化运动方程 linearized equations of motion:

$$\begin{aligned} \sum \dot{\mathbf{X}} &= m[\dot{U} + z_g\dot{q}] \\ \sum \dot{\mathbf{Z}} &= m[\dot{w} - x_g\dot{q} - U\dot{q}] \\ \sum \dot{\mathbf{M}} &= I_y\dot{q} + m[z_g(\dot{U} - U\dot{q})] \end{aligned} \quad (9.6)$$

9.1.3 车辆机械 Mechanics

我们在车辆力学的假设中提出的假设完全相同。在线性化的车辆运动方程 (9.6) 中, 外部力和力矩 external forces and moments

$$\sum \mathbf{F}_{ext} = \mathbf{F}_{hydrostatic} + \mathbf{F}_{lift} + \mathbf{F}_{drag} + \mathbf{F}_{control} + \mathbf{F}_{disturbance}$$

以车辆系数来描述 in terms of vehicle coefficients. For example, linearized axial drag

$$\mathbf{F}_d = -\left(\frac{I^{-1}}{2} \rho a A_f U\right) \mathbf{u} = \mathbf{X} \mathbf{u} \implies \mathbf{X} = \frac{\partial \mathbf{F}_d}{\partial u} = -\frac{1}{2} \rho a A_f U$$

这些线性化系数基于理论方程和经验推导公式的结合。我们忽略向前由机体内力产生的前向力, 因为它是一个非线性项。that we neglect the forward force due to body lift as it is a nonlinear term.

9.2 线性化系数推导 Coefficient Derivation

推导车辆系数所需的各种参数要么包含在描述该系数的部分，要么列在附录中。 The various parameters necessary to determine the vehicle coefficients are either included in the section describing the coefficient, or are listed in the Appendix.

9.2.1 静力 Hydrostatics

在第4.1节中，提出了水静力作用力和力矩的非线性方程（见方程4.3）。我们将通过忽略高阶项来简化它。 We will simplify it by dropping higher-order terms, assuming that 然后我们会使用三角函数项的Maclaurin展开式（见方程9.3）。 We then drop the higher-order terms, as well as any resulting constant terms. This yields the following linearized hydrostatic equations:

X_θ = -(W - B)θ
M_θ = -(z_g W - z_b B)θ (9.7)

9.2.2 轴向阻力 Axial Drag

车辆轴向阻力可以用方程4.6表示。利用方程9.2给出的关系对非线性方程线性化，得到如下结果 given in Equation 9.2, results in the following:

X = -1/2 ρ C_d A_f (U + u)|U + u|
X = -1/2 ρ C_d A_f (U^2 + 2Uu + u^2) (9.8)

假设 Assuming

u << U, U > 0 (9.9)

去掉任何常数项后，得到以下线性化轴向阻力系数 and dropping any constant terms, the following linearized axial drag coefficient:

X_u = -ρ C_d A_f U (9.10)

9.2.3 横流阻力 Crossflow Drag

车辆横流阻力在第4.2.3节中进行了讨论。为对方程4.8中描述的二次横流阻力系数进行线性化，我们必须围绕零对升沉和俯仰扰动速度进行线性化。这可以通过对抛物线型速度平方曲线拟合斜率来实现，如图9-1所示。 coefficients described in Equation 4.8, we must linearize the heave and pitch perturbation velocities about zero. This is accomplished by fitting a slope to the parabolic velocity-squared curve, as show in Figure 9-1.

w^2 ≈ m_w w
q^2 ≈ m_q q (9.11)

在此参数化中使用的实际数值列于表9.1。最大预期升沉和纵摇速度的估计值取自现场数据。 The actual values used in this parameterization are given in Table 9.1. The estimated maximum expected heave and pitch velocity were taken from field data.

Table 9. 线性化速度参数 Velocity Parameters

Table with 4 columns: Parameter, Value, Units, Description. Rows include w_max, q_max, m_w, and m_q.

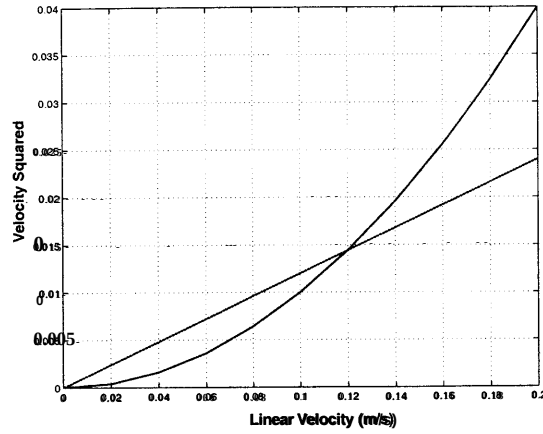


Figure 9.11 Port Velocity Linearization

Substituting these parameters into the following linear equations for crossflow drag:

$$\begin{aligned}
 Z_{ue} &= -\frac{1}{2} \rho c_{dc} v_w \int_{x_t}^{x_{b2}} 2R(x) dx = 2 \cdot \left(\frac{1}{2} \rho S_{fin} c_{df} m_w \right) \\
 M_{ue} &= -\frac{1}{2} \rho c_{dc} m_w \int_{x_t}^{x_{b2}} 2x R(x) dx = 2x_{fin} \cdot \left(\frac{1}{2} \rho S_{fin} c_{df} m_w \right) \\
 Z_{qe} &= -\frac{1}{2} \rho c_{dc} m_q \int_{x_t}^{x_{b2}} 2x |x| R(x) dx = 2x_{fin} |x_{fin}| \cdot \left(\frac{1}{2} \rho S_{fin} c_{df} m_q \right) \\
 M_{qc}^e &= -\frac{1}{2} \rho c_{dc} m_q \int_{x_t}^{x_{b2}} 2x^3 R(x) dx = 2x_{fin}^3 \cdot \left(\frac{1}{2} \rho S_{fin} c_{df} m_q \right)
 \end{aligned} \tag{9.12}$$

See Table A.2 for the limits of integration.

9.2.4 Added Mass Mass

Vehicle added mass is discussed in Section 4.3 through the linearized velocity B and substituted into linearized velocities and dropping the out-of-plane and higher-order terms, the non-linear equations (see Equation 4.15) reduce to the following:

$$\begin{aligned}
 \dot{X}_A &= \dot{X}_u \dot{\theta} + Z_q m_q \dot{q} \\
 \dot{Z}_A &= \dot{Z}_u \dot{\theta} + Z_q \dot{q} - X_A U_q \\
 \dot{M}_A &= \dot{M}_w \dot{w} + M_q \dot{q} - (Z_{\dot{w}} - X_{\dot{w}}) U_w - Z_q U_q
 \end{aligned} \tag{9.13}$$

Linearized added mass is given by the equation:

$$\dot{X}_{\dot{u}} = -m \dot{v}_{11} \tag{9.14}$$

并且横向前增加的质量由以下方程给出: the equations:

$$\begin{aligned} Z_{\dot{w}} &= - \int_{x_f}^{x_f} m_{22}(x) dx - \int_{x_f}^{x_{f2}} m_{22}f(x) dx - \int_{x_{f2}}^{x_{b2}} m_{22}(x) dx \\ M_{\dot{w}} &= Z_{\dot{q}} = \\ &= - \int_{x_f}^{x_f} x m_{22}(x) dx - \int_{x_f}^{x_{f2}} x m_{22}f(x) dx - \int_{x_{f2}}^{x_{b2}} x m_{22}(x) dx \\ M_{\dot{q}} &= - \int_{x_f}^{x_f} x^2 m_{22}(x) dx - \int_{x_f}^{x_{f2}} x^2 m_{22}f(x) dx - \int_{x_{f2}}^{x_{b2}} x^2 m_{22}(x) dx \end{aligned} \quad (9.15)$$

See Appendix A.2 for details of integration.

The remaining cross terms result from added mass coupling:

$$\begin{aligned} X_{\dot{q}w} &= Z_{\dot{q}} m_{q1} \\ Z_{\dot{q}w} &= -X_{\dot{w}} U \\ M_{\dot{w}q} &= -(Z_{\dot{w}} - X_{\dot{q}}) U \\ M_{\dot{q}w} &= -Z_{\dot{q}} U \end{aligned} \quad (9.16)$$

The added mass cross-term, $M_{\dot{w}q}$, is also called the **Munk Moment**, and relates to the pitch moment experienced by a body in ideal, inviscid flow at an angle of attack.

9.2.5.5 Vehicle Body Lift Force and Moment

The 4th order linear equations for vehicle body lift are discussed in Section 4.4. Linearizing the velocity relationships according to equation 9.2 and substituting into Equation 4.27 the relationships given above, we are left with the following, linearized equations for vehicle body lift:

$$Z_L = -\frac{1}{2} \rho d^2 c_{yd\beta} \dot{w} U \quad (9.17)$$

which results in the body lift coefficient:

$$Z_{wL} = -\frac{1}{2} \rho d^2 c_{yd\beta} U \quad (9.18)$$

and body lift moment:

$$M_{wL} = -\frac{1}{2} \rho d^2 c_{gd\beta} x_{cg} U \quad (9.19)$$

9.2.6 Fin Lift

Vehicle fin lift is discussed in Section 4.5

On dropping the velocity terms from equation 4.40, equations for effective fin velocity (see Equation 4.40) leads to the following:

$$\begin{aligned} u_{fin} &= u \\ w_{fin} &= w - x_{fin} \dot{q} \end{aligned} \quad (9.20)$$

The effective angle can be expressed as

$$\delta_e = \delta_s + \beta_e \quad (9.21)$$

where δ_s is the fin zero plane angle, β_e is the effective angle of attack of the fin zero plane, as shown

在Figure 4.2中这个有效角度表示为

$$\beta_e = \frac{w_{fin}}{U_{fin}} \approx \frac{1}{U} (w - x_{fin} q) \quad (9.22)$$

按照方程(9.22)和4.44 for 4.45 中的速度线性化后, 得到以下一组增升力系数: the following sets of fin lift coefficients:

$$\begin{aligned} Z_{\delta_s} &= -\rho c_{L\delta} S_{fin} U^2 \\ Z_{w_f} &= -\frac{1}{2} \rho c_{L\delta} S_{fin} U \\ Z_{q_f} &= \frac{1}{2} \rho c_{L\delta} S_{fin} x_{fin} U \end{aligned} \quad (9.23)$$

和鳍阻力矩系数: coefficients:

$$\begin{aligned} M_{\delta_s} &= \rho c_{L\delta} S_{fin} x_{fin} U^2 \\ M_{w_f} &= \frac{1}{2} \rho c_{L\delta} S_{fin} x_{fin} U \\ M_{q_f} &= -\frac{1}{2} \rho c_{L\delta} S_{fin} x_{fin}^2 U \end{aligned} \quad (9.24)$$

9.2.7 联合项 Combined Terms

车辆在外流平面上的力和力矩之和可以表示为: the sum of the forces and moments on the vehicle can be expressed as:

$$\begin{aligned} \sum X &= X_{\dot{\theta}} + X_u u + X_q q + X_{\dot{\theta}} \theta \\ \sum Z &= Z_{\dot{\psi}} + Z_{\dot{q}} + Z_w w + Z_q q + Z_{\delta_s} \delta_s \\ \sum M &= M_{\dot{\psi}} + M_{\dot{q}} + M_u u + M_q q + M_{\theta} \theta + M_{\delta_s} \delta_s \end{aligned} \quad (9.25)$$

where

$$\begin{aligned} Z_w &= Z_{we} + Z_{wl} + Z_{wf} \\ M_w &= M_{we} + M_{wl} + M_{we1} + M_{wl2} + M_{wf} \quad (9.26) \\ Z_q &= Z_{qe} + Z_{ql} + Z_{qf} \\ M_q &= M_{qe} + M_{ql} + M_{qf} \end{aligned}$$

这些组合项的数值见表9.2. The values of the combined terms are given in Table 9.2.

9.2.8 线性化系数 Linearized Coefficients

那个线性化系数的最终值能表示为, coefficients are given below in Table 9.3. the final values for the linearized coefficients are given below in Table 9.3.

9.3 线性化运动方程 Linearized Equations of Motion

我们现在将结合前几章中推导的方程, 来推导运动的线性化方程. We will now combine the equations developed in the preceding chapters to develop the linearized equations of motion.

Table 9.2.9. 组合线性化系数 Combined Linearized Coefficients

Parameter	Value	Units	Description
$Z_{\dot{w}}$	-1.57e+001	kg/s	Crossflow Drag
$Z_{\dot{w}}$	-3.45e+001	kg · m/s	Body Lift
$Z_{\dot{w}}$	-1.60e+001	kg · m/s	Fin Lift
$Z_{q\dot{e}}$	+1.20e-001	kg · m/s	Crossflow Drag
$Z_{q\dot{a}}$	+1.44e+000	kg · m/s	Added Mass Cross Term
$Z_{q\dot{f}}$	-1.12e+001	kg · m/s	Fin Lift
$M_{\dot{w}}$	-4.00e+001	kg · m/s	Crossflow Drag
$M_{\dot{w}}$	+5.34e+001	kg · m/s	Added Mass Cross Term
$M_{\dot{w}}$	-1.11e+001	kg · m/s	Body Lift
$M_{\dot{w}}$	-1.12e+001	kg · m/s	Fin Lift
$M_{q\dot{e}}$	-2.15e+000	kg · m ² /s	Crossflow Drag
$M_{q\dot{a}}$	+2.97e+000	kg · m ² /s	Added Mass Cross Term
$M_{q\dot{f}}$	-7.68e+000	kg · m ² /s	Fin Lift

9.3.11 Equations of Motion

将方程 9.6 和 9.25 结合得到以下线性化车辆运动方程 the following linearized vehicle equations of motion:

$$\begin{aligned}
 (m - X_{\dot{w}})\dot{w} + m\dot{z}_g - X_{\dot{w}}\dot{w} - X_{\dot{w}}\dot{w} - X_{\dot{w}}\dot{w} - X_{\dot{w}}\dot{w} &= 0 \\
 (m - Z_{\dot{w}})\dot{w} - (m\dot{x}_g + Z_{\dot{w}})\dot{q} - Z_{\dot{w}}\dot{w} - (m\dot{U} + Z_{\dot{w}})\dot{q} - Z_{\dot{w}}\dot{q} &= Z_{\dot{w}}\delta_s \\
 m\dot{z}_g - (m\dot{x}_g + M_{\dot{w}})\dot{w} + (M_{\dot{w}} - M_{\dot{q}})\dot{q} & \\
 - M_{\dot{w}}\dot{w} + (m\dot{x}_g U - M_{\dot{q}})\dot{q} - M_{\dot{q}}\dot{\theta} &= M_{\dot{w}}\delta_s
 \end{aligned} \tag{9.27}$$

如果我们假设 z_g 相对于其他项较小, the other terms, we can decouple the resulting equations of motion:

$$\begin{aligned}
 (m - Z_{\dot{w}})\dot{w} - (m\dot{x}_g + Z_{\dot{w}})\dot{q} - Z_{\dot{w}}\dot{w} - (m\dot{U} + Z_{\dot{w}})\dot{q} &= Z_{\dot{w}}\delta_s \\
 - (m\dot{x}_g + M_{\dot{w}})\dot{w} + (M_{\dot{w}} - M_{\dot{q}})\dot{q} & \\
 - M_{\dot{w}}\dot{w} + (m\dot{x}_g U - M_{\dot{q}})\dot{q} - M_{\dot{q}}\dot{\theta} &= M_{\dot{w}}\delta_s
 \end{aligned} \tag{9.28}$$

Similarly, 方程 9.4 的运动方程简化为 from Equation 9.4 reduce to:

$$\begin{aligned}
 \dot{z} &= wU - U\theta \\
 \dot{\theta} &= q
 \end{aligned} \tag{9.29}$$

Table 9: 线性化操纵系数 Linearized Maneuvering Coefficients

Parameter	Value	Units	Description
X_0	$+8.90e+000$	$\text{kg} \cdot \text{m}/\text{s}^2$	Hydrostatic
X_u	$-1.35e+001$	kg/s	Axial Drag
X_a	$-9.33e+001$	kg	Added Mass
$X_{\dot{q}}$	$-5.76e+001$	$\text{kg} \cdot \text{m}/\text{s}$	Added Mass Cross Term
Z_w	$-6.66e+001$	kg/s	Combined Term
Z_q	$-9.67e+000$	$\text{kg} \cdot \text{m}/\text{s}$	Combined Term
Z_b	$-3.55e+001$	kg	Added Mass
Z_4	$-1.93e+000$	$\text{kg} \cdot \text{m}$	Added Mass
$Z_{\dot{z}}$	$-5.06e+001$	$\text{kg} \cdot \text{m}/\text{s}^2$	Fin Lift
M_0	$-5.77e+000$	$\text{kg} \cdot \text{m}^2/\text{s}^2$	Hydrostatic
M_w	$+3.07e+001$	$\text{kg} \cdot \text{m}/\text{s}$	Combined Term
$M_{\dot{q}}$	$-6.87e+000$	$\text{kg} \cdot \text{m}^2/\text{s}$	Combined Term
$M_{\dot{w}}$	$-1.93e+000$	$\text{kg} \cdot \text{m}$	Added Mass
$M_{\dot{q}}$	$-4.88e+000$	$\text{kg} \cdot \text{m}^2$	Added Mass
$Z_{\dot{z}}$	$-3.46e+001$	$\text{kg} \cdot \text{m}^2/\text{s}^2$	Fin Lift

9.3.2 四项态向量 Four-Term State Vector

我们将方程 9.28 和 9.29 合并成如下结合更简便的 Equations 9.28 and 9.29 into a single matrix form, as follows:

$$\begin{bmatrix} -m\dot{x}_g + X_u & -(m\dot{x}_g + Z_q) & 0 & 0 \\ -(m\dot{x}_g + M_u) & I\dot{y} - M_{\dot{q}} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{u} \\ \dot{q} \\ \dot{z} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} Z_w & m\dot{U} + Z_q q & 0 & 0 \\ M_w'' & -m\dot{q}U + M_q & 0 & M_0 \\ 1 & 0 & 0 & -U \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} u \\ q \\ z \\ \theta \end{bmatrix} + \begin{bmatrix} Z_b \\ M_b \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} \dot{z} \\ \dot{\theta} \\ 0 \\ 0 \end{bmatrix} \quad (9.30)$$

给定状态向量 state vector

$$\mathbf{x} = [\mathbf{w} \quad \mathbf{q} \quad \mathbf{z} \quad \theta]^T \quad (9.31)$$

和输入向量 input vector

$$\mathbf{u} = [\delta_s]^T \quad (9.32)$$

我们可以将方程 9.30 写成

$$\mathbf{M}\dot{\mathbf{x}} - \mathbf{C}_d\mathbf{x} = \mathbf{D}\mathbf{u} \quad (9.33)$$

or

$$\dot{\mathbf{x}} = \mathbf{M}^{-1}\mathbf{C}_d\mathbf{x} + \mathbf{M}^{-1}\mathbf{D}\mathbf{u} \quad (9.34)$$

通常用符号表示的 represented using the notation

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \quad (9.35)$$

将第 4 章中开发并在附录 D 列出的系数值代入, 我们得到 Chapter 4 and listed in Appendix D, we arrive

在以下矩阵中:

$$\mathbf{A} = \mathbf{M}^{-1} \mathbf{C}_d \equiv \begin{bmatrix} -2.38 & +1.26 & +0.00 & +0.04 \\ +4.23 & -1.12 & +0.00 & -0.70 \\ +1.00 & +0.00 & +0.00 & -1.54 \\ +0.00 & +1.00 & +0.00 & +0.00 \end{bmatrix}$$

$$\mathbf{B} = \mathbf{M}^{-1} \mathbf{D} \equiv \begin{bmatrix} -1.38 \\ -3.84 \\ +0.00 \\ +0.00 \end{bmatrix} \quad (9.36)$$

9.3.3 Three-term State Vector

假设升沉速度相比其他项较小, 我们可以将运动方程进一步简化为如下形式, 我们可将运动方程进一步简化为如下形式:

$$\begin{bmatrix} \mathbf{M} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} \end{bmatrix} \begin{bmatrix} \mathbf{q} \\ \dot{\mathbf{z}} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} -M_q & 0 & -M_\theta \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{q} \\ \mathbf{z} \\ \theta \end{bmatrix} = \begin{bmatrix} M_{\delta_s} \\ 0 \\ 0 \end{bmatrix} \quad (9.37)$$

我们可以简单地简化为:

$$\begin{bmatrix} \dot{\mathbf{q}} \\ \dot{\mathbf{z}} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \frac{M_q}{I_{yy}-M_q} & 0 & \frac{M_\theta}{I_{yy}-M_q} \\ 0 & 0 & -U \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{q} \\ \mathbf{z} \\ \theta \end{bmatrix} + \begin{bmatrix} \frac{M_{\delta_s}}{I_{yy}-M_q} \\ 0 \\ 0 \end{bmatrix} \delta_s \quad (9.38)$$

再次代入第4章中得出的系数值, 我们在附录D, 和式(9.39)中列出, 并应用该形式:

$$\dot{\mathbf{x}} = \mathbf{A} \mathbf{x} + \mathbf{B} \mathbf{u} \quad (9.39)$$

我们得到以下矩阵:

$$\mathbf{A} = \begin{bmatrix} -0.82 & +0.00 & -0.69 \\ +0.00 & +0.00 & -1.54 \\ +1.00 & +0.00 & +0.00 \end{bmatrix}$$

$$\mathbf{B} = \begin{bmatrix} -4.16 \\ +0.00 \\ +0.00 \end{bmatrix} \quad -4.16 \quad (9.40)$$

9.4 控制系统设计 System Design

我们现在将查看一个简单车辆控制器的设计, 使用控制状态向量模型开发的状态方程(参见方程9.38)。这个示例控制器, 与实际车辆控制器类似, 由一个内部比例-微分(PD)俯仰回路和一个外部比例深度回路组成。我们将依次讨论每个控制器的设计。

9.4.1 车辆传输函数 Transfer Functions

设计车辆控制系统的第一步是推导车辆传输函数。首先, 我们希望推导俯仰回路的传输函数, 将输入舵面角度与输出车辆俯仰角相关联。通过对方程9.38取拉普拉斯变换, 我们可以得到输出车辆俯仰角与输入舵面角度之间的关系。

将此开环传递函数表示为 transfer function as:

$$G_{\theta\theta}(s) \equiv \frac{\theta(s)}{\delta_s(s)} \equiv \frac{\frac{M_{\theta\delta}}{I_{yy} - M_{\dot{q}}}}{s^2 - \frac{M_{\dot{q}}}{s - I_{yy} - M_{\dot{q}}} s - \frac{M_{\theta}}{I_{yy} - M_{\dot{q}}}} \quad (9.41)$$

接下来, 我们想要找到外层深度回路的传递函数, 它将输入车辆俯仰角 θ 与输出车辆深度 z 关联起来。实际上, 与外层深度回路相比, 内层俯仰回路的响应速度足够快, 因此我们可以将期望车辆俯仰角 θ^i 视为与实际车辆俯仰角相同。再次对车辆状态方程 (方程 9.38) 进行拉普拉斯变换, 我们得到了所需的开环传递函数:

$$G_z(s) = \frac{z(s)}{\theta(s)} = -\frac{U}{s} \quad (9.42)$$

9.4.2 控制律 Control Law

我们现在将为内环和外环设计控制律。如本章开头所述, 我们将设计两个 As 控制器 (PD) 内环和比例外环。此外, 内环的控制律可以表示为比例-微分 (PD) inner loop, and a proportional outer loop. The control law, then, for the inner loop can be expressed as:

$$\frac{\delta_s(s)}{\theta(s)} \equiv -K_p(\tau_d s + 1) \quad (9.43)$$

其中

$$e_{\theta} = \theta_d - \theta \quad (9.44)$$

K_p 是比例增益, τ_d 是微分时间常数 (以秒为单位) derivative time constant in seconds.

由于尾舵角和车辆俯仰角之间符号约定的差异, 比例增益应用了负号。不同的尾舵角会产生相反的力矩, 使车辆俯仰 (负俯仰速度) pitch angle. Positive stern plane angle will generate a negative moment about the y-axis, forcing the vehicle to pitch down (negative pitch rate).

外环的控制律可以表示为 the outer loop can be expressed as:

$$\frac{\theta(s)}{e_z(s)} = K_p \quad (9.45)$$

其中

$$e_z = z_d - z \quad (9.46)$$

而 K_p 是比例增益 proportional gain.

我们可以将闭环控制系统表示为 a block diagram:

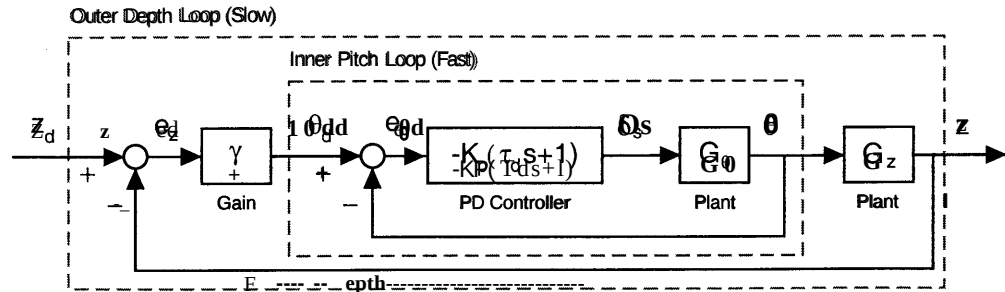


图 9.26 9 深度平面控制系统框图 Control System Block Diagram

9.4.3 控制器设计程序 Controller Design Procedure

在进入选择车辆控制器增益的具体内容之前,我们将首先回顾一般的控制器设计程序。Before getting into specifics of selecting the vehicle controller gains, we will first review a general controller design procedure.

此过程假设系统具有二阶响应。鉴于我们选择使用三阶状态向量,这对于仿射作开显成立的。对于高阶系统,这一假设同样有效,前提是高频极点至少比两个主导极点距离s平面原点近五倍以上。The process assumes the system has a second order response. Given that we have chosen to work with a three-state vector, this assumption is also valid for higher-order systems, provided the higher order poles are at least five times further from the origin of the s-plane than the two dominant poles.

我们将设计我们的控制器,使其具有特定的二阶响应,以自然频率 ω_n 和阻尼比 ζ 的形式,其中,对于二阶系统 ratio ζ , where, for a second order system:

$$G(s) = \frac{K}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (9.47)$$

超调百分比与稳态时间或峰值时间 Percent Overshoot and Settling or Peak Time

我们在选择系统响应时的主要考虑因素将是百分比超调, $\%OS$, 以及峰值或整定时间, T_p and T_s . 有关这些术语的解释,请参见 NN-2 [25] for a graphical explanation of these terms.

我们想要的阻尼比是超调百分比的函数,并由以下方程给出:

$$\zeta = \frac{-\ln\left(\frac{\%OS}{100}\right)}{\left[\pi^2 + \ln^2\left(\frac{\%OS}{100}\right)\right]^{\frac{1}{2}}} \quad (9.48)$$

表 9.4 给出了一个数值范围,也可以从极点-零点图上绘制恒定阻尼比的线。Table 9.4 gives a range of values. One can also plot lines of constant damping ratio on the pole-zero diagram.

表 9.4.9. 超调百分比和阻尼比 Percent Overshoot and Damping Ratio

%OS	5	10	15	20	25	30
ζ	0.690	0.591	0.517	0.456	0.404	0.358

给定所需的阻尼比,我们所需的固有频率可以通过以下方程求得: Given the desired damping ratio, our desired natural frequency can be found through the equation:

$$\omega_n = \frac{\pi}{T_p \sqrt{1 - \zeta^2}} \quad (9.49)$$

或者

$$\omega_n = \frac{4}{\zeta T_s} \quad (9.50)$$

所需极点 Desired Poles

从方程 9.41 阶传递函数,我们现在可以找到所需的极点位置 through the quadratic formula:

$$s_{1,2} = -\zeta\omega_n \pm \frac{1}{2}\sqrt{4\zeta^2\omega_n^2 - 4\omega_n^2} \quad (9.51)$$

9.4.4 开环控制器增益 Open-loop Controller Gains

将表 9.4 中的系数值代入方程 9.41 得到以下开环传递函数: Equation 9.41 results in the following open-loop transfer function:

$$G_\theta(s) = \frac{-3118}{s^2 + 1.09s + 0.52} \quad (9.52)$$

以及以下开环极点:

$$s_{1,2} = -0.55 \pm 0.47i \quad (9.53)$$

Figure 9-3 shows the open-loop step response.

鉴于REMUS车辆在深水中运行此方法承受显著的深度超调，我在设计车辆俯仰控制器时选择可以depth参数shoot, I chose the following parameters in designing the vehicle pitch controller:

$$\begin{aligned} \%OS &= 0.05 \\ T_p &= 0.75 \text{ seconds} \end{aligned} \quad \begin{aligned} (9.54) \\ (9.54) \end{aligned}$$

这些数值得出了阻尼比 ζ 以及自然频率 ω_n , 如图 9-5 根轨迹图例所示。 ω_n , shown in the legend of the root-locus plot on Figure 9-5.

有许多系统的方法可以生成所需的控制器增益。在这种情况下，我设定了微分时间常数 τ_d 并使用根轨迹图来找到所需的增益 K_p 生成的控制增益也显示在 Figure 9-5 的图例中。find the necessary gain K_p . The resulting controller values are also shown in the legend of Figure 9-5.

俯仰回路传递函数加上控制器 H_θ 所得到的闭环响应如图 9-6 所示。 H_θ , is shown in Figure 9-6.

9.4.5 深度回路控制器增益 Controller Gains

用于寻找深度回路控制器增益的方法与前节的俯仰类似。增益深度传递函数 G_z 添加了一个极点。The极零图所示 transfer function G_z adds a third pole, as shown in the pole-zero plot in Figure 9-7.

为了确保俯仰回路响应比深度回路响应快得多，我们必须确保俯仰回路的极点至少比深度回路的极点，离原点近五倍。 $\text{ensure that the pitch loop poles end up at least five times further away from the origin than the depth loop pole.}$

这是通过使用根轨迹选择来确定的。适当的比例增益 K_p 实现的 ζ 得到的数值可以在 Figure 9-8 所示的根轨迹图的图例中找到。由此得到的 K_p 可以在 Figure 9-8 的图例中看到。the root-locus plot shown in Figure 9-8. The resulting closed loop response can be seen in Figure 9-9.

9.5 现实世界现象 Real-World Phenomena

尽管该控制器似乎表现出理想的性能特征，但在现实世界中并没有完美实现。例如，如果控制器遇到足够大的深度误差，没有任何东西可以阻止它发出荒谬的俯仰角和尾舵命令，角度可能超过 180 度。换句话说，该控制器假设传递函数关系在无限范围内都是线性的。实际上，尾舵在有效鳍角大于大约 180 度时就会失速。同样，REMUS 车辆在俯仰角大于 30 度时会关闭，因为软件假设车辆任务发生了灾难性的错误。 $\text{In other words, as designed, the controller assumes the transfer function relationships to be linear out to infinity. In reality, the stern planes will stall at an effective fin angle of greater than about 12 degrees. Similarly, the REMUS vehicle shuts down at greater than 30 degrees of pitch angle, as the software assumes that something has gone disastrously wrong with the vehicle mission.}$

此外，模拟假设车辆的传感器没有噪声，并且车辆没有经历未建模的干扰。 $\text{Furthermore, the model assumes the vehicle sensors have no noise and the vehicle experiences no unmodeled disturbances. In reality, this is rarely the case.}$

将传递函数离散化并尝试将这些现实世界的影响纳入车辆模型是一个有趣的练习。图 9-10 包含一些在 Figure 9-10 中包含的俯仰角和尾舵命令的图，以及俯仰传感器噪声和环境干扰。 $\text{Figure 9-10 shows a block diagram which incorporates some of these effects into the vehicle model. Figure 9-10 shows a block diagram which incorporates some of these effects into the vehicle model.}$ $\text{Figure 9-11 and 9-14 show the results of this kind of a simulation. In Figure 9-11, the model is run without disturbances or actuator saturation. Notice that the controller commands nonsensical vehicle fin angles in excess of 180 degrees. In reality, the vehicle fins stall at an angle of attack less than 15 degrees.}$

在 Figure 9-11 模型包含了鳍角饱和、随机俯仰传感器噪声以及节垂直水流的影响。 $\text{Figure 9-11 model includes fin angle saturation, random pitch sensor noise, and the effects of vertical flow.}$ 注意，响应较慢，并且鳍片表现出相当大的颤动。 $\text{Notice that the vehicle response is slower, and that the fins exhibit a considerable amount of "flutter".}$

9.6 控制器实现 Controller Implementation

在本章中，我们展示了基于车辆动力学线性化模型的车辆深度控制系统的发展。尽管在仿真中控制器似乎能够达到期望的响应，但仍有待观察它是否在实际车辆上同样有效。 $\text{In this chapter, we have shown the development of a vehicle depth control system based on a linearized model of the vehicle dynamics. Although the controller appears to achieve the desired response, it remains to be seen if it would work as well on the actual vehicle.}$

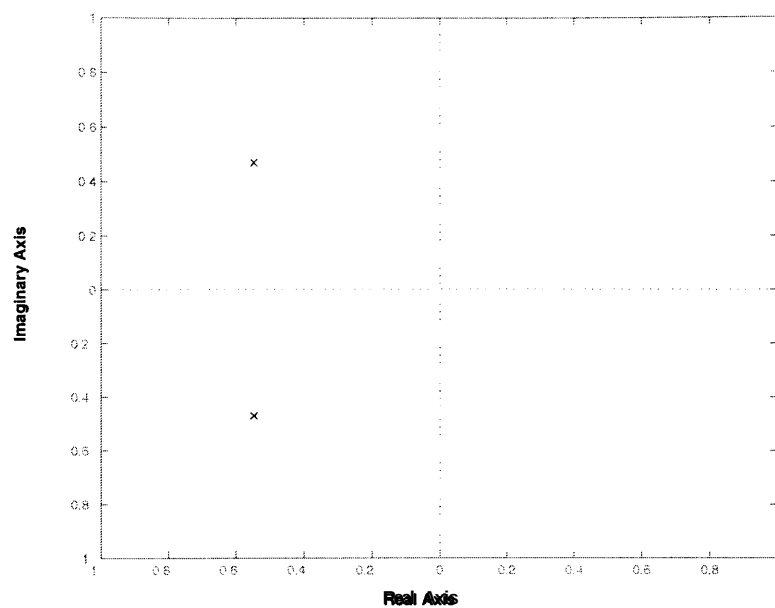


Figure 9-3-9 Pole-Zero Plot

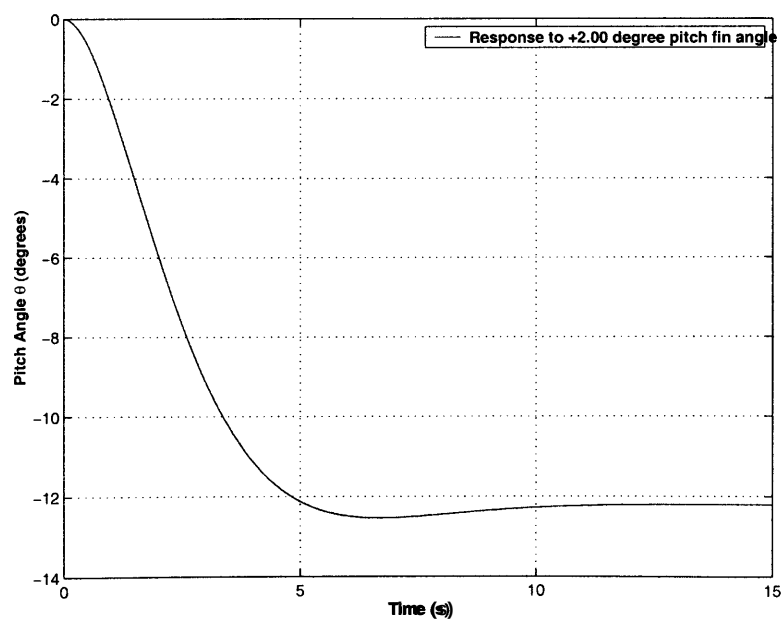


Figure 9-4-4 Open-loop Step Response

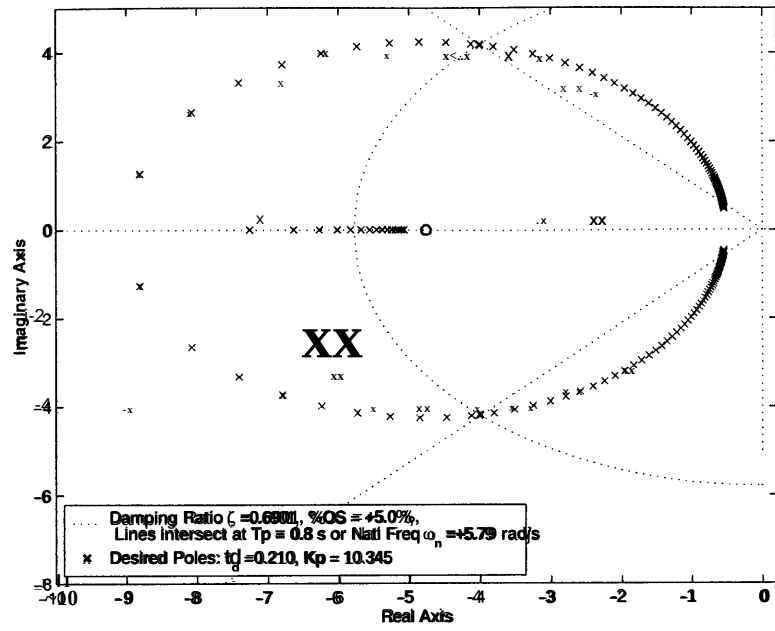


Figure 9-5: Root Locus Plot

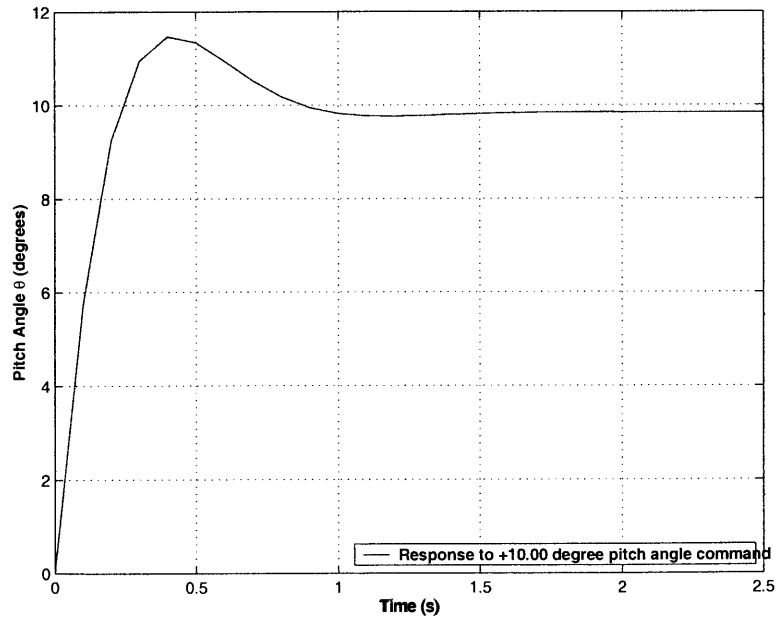


Figure 9-6: Closed-Loop Step Response

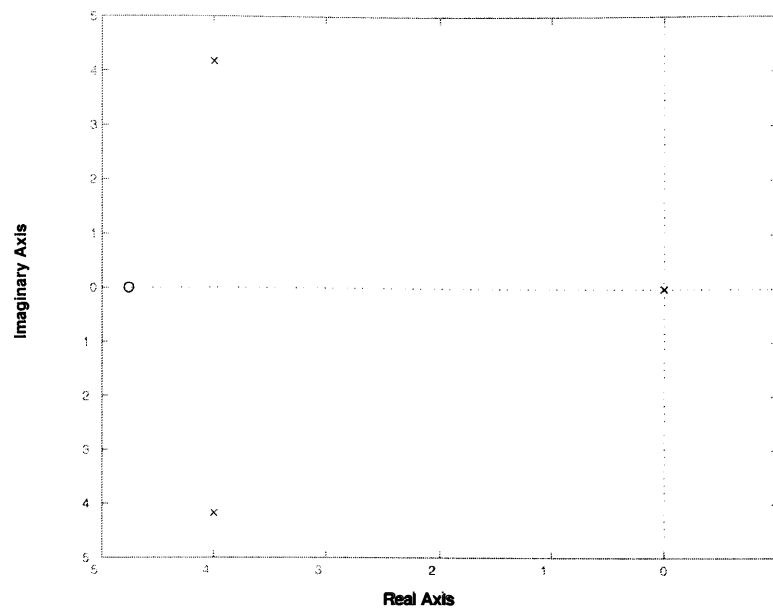


Figure 9-7-9 G: GH Pole-Zero Plot

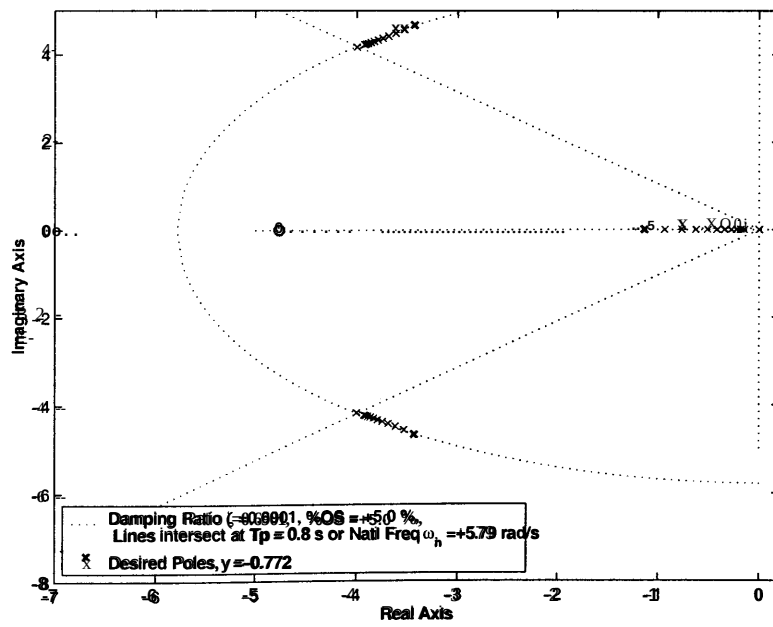


Figure 9-8-9 G: GH Locus Plot

测试控制器的下一步是将图9-10中的线性化深度平面模型替换为前面章节中开发的非线性六自由度车辆模型。这将使我们能够评估控制器在小角度和小加速度的线性化范围之外，处理车辆行为的能力。

最后一步是在实际的海上车辆上测试深度平面控制器。

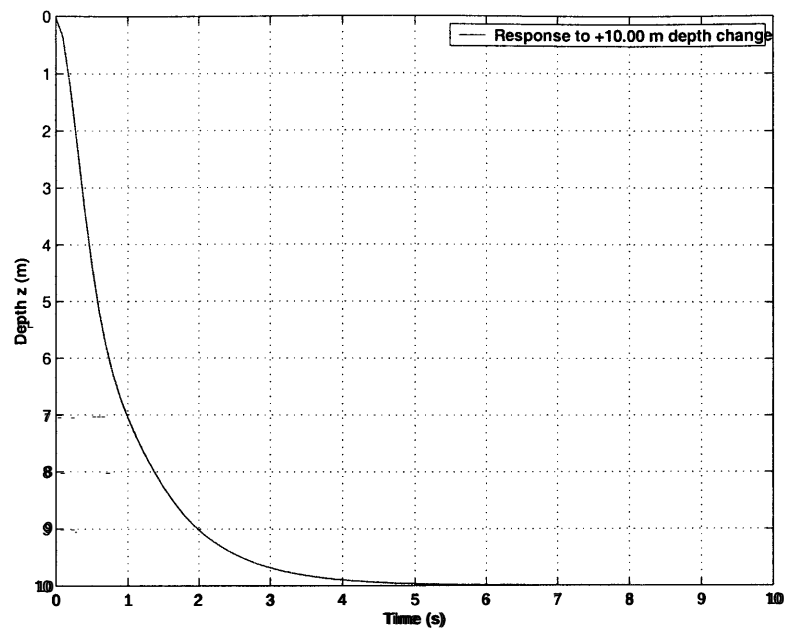


Figure 9-9: CTD Closed Loop Step Response

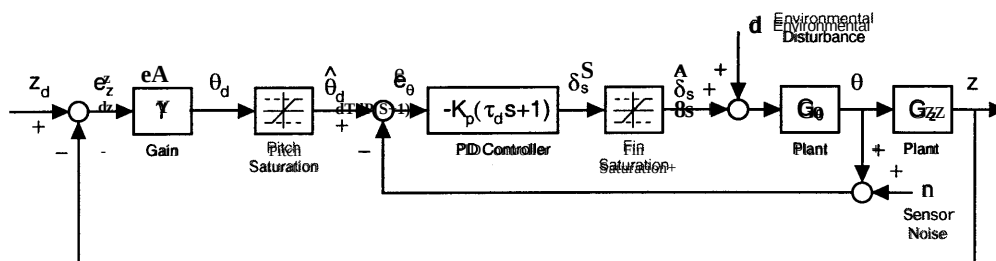


Figure 9-10: Depth Control System Block Diagram with Limits, Environmental Disturbance, and Sensor Noise

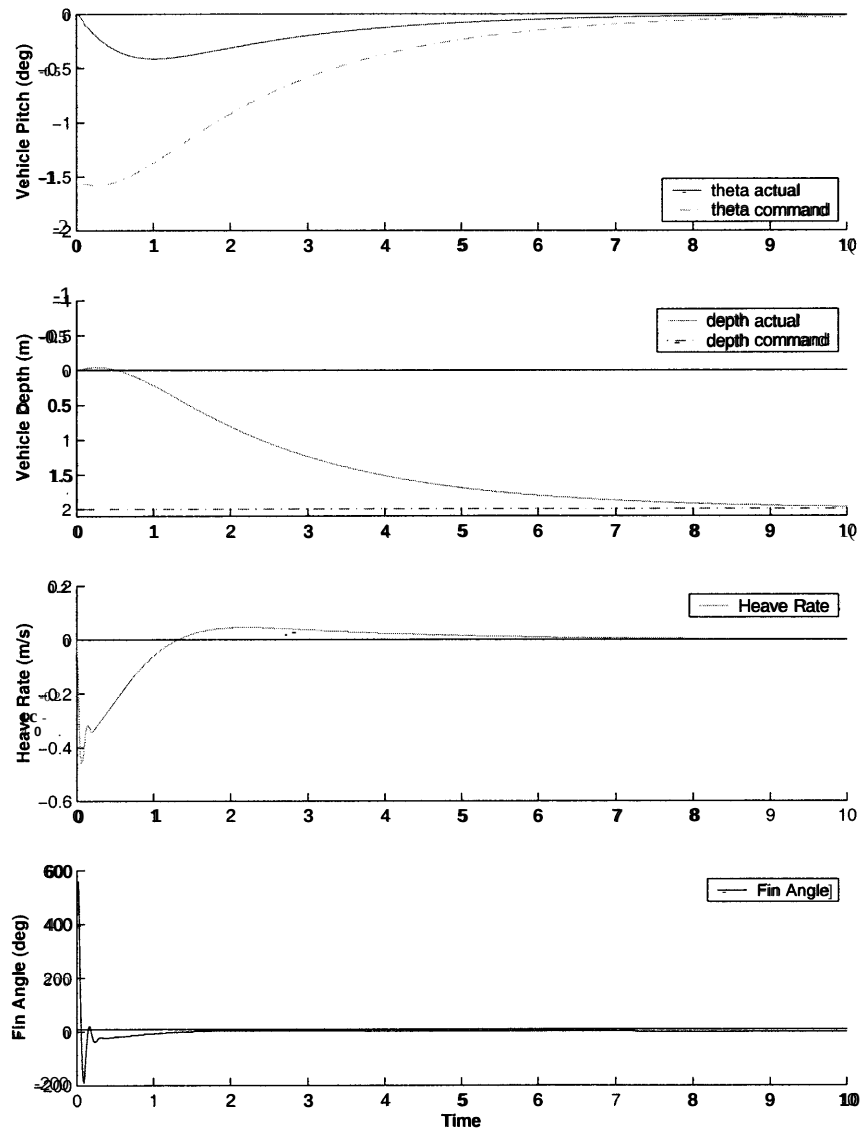


Figure 9- Vehicle Simulation based on basic block diagram

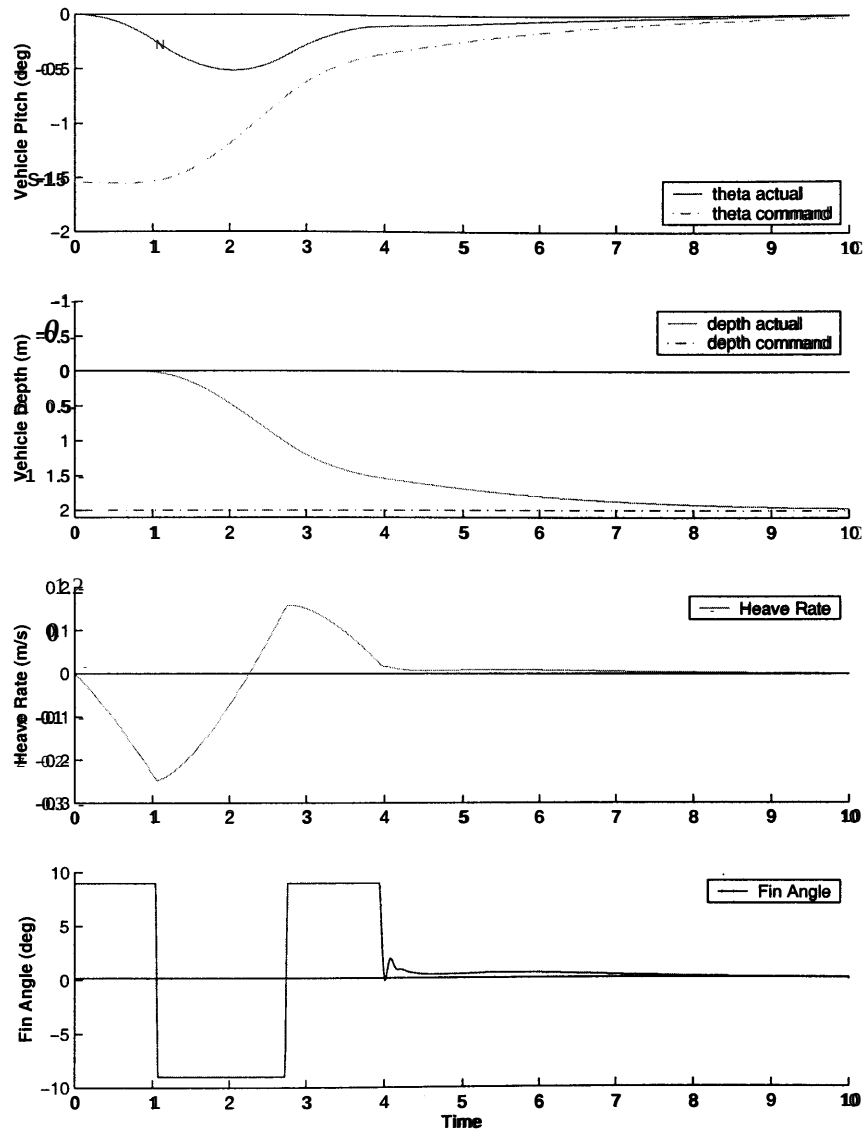


Figure 12-14 Vehicle Simulation with fin angle saturation

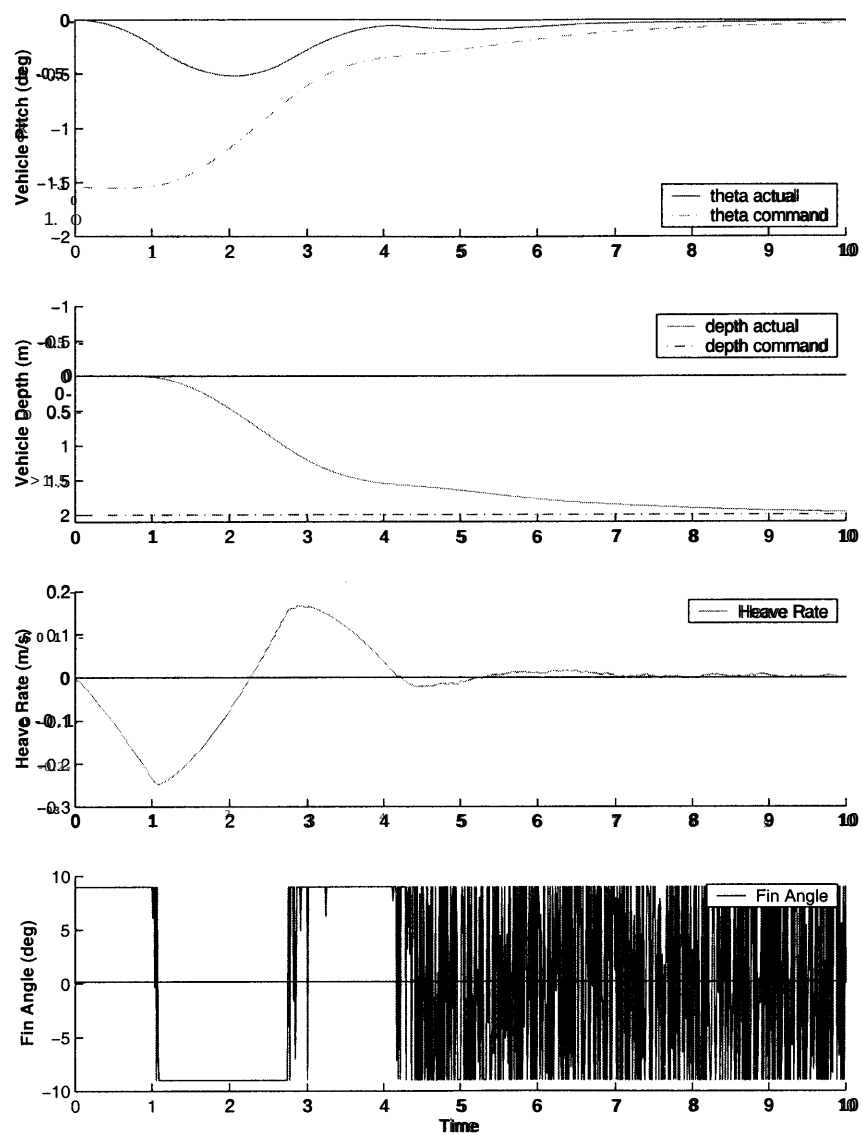


Figure 9-43 Vehicle Simulation with fin angle sensor noise

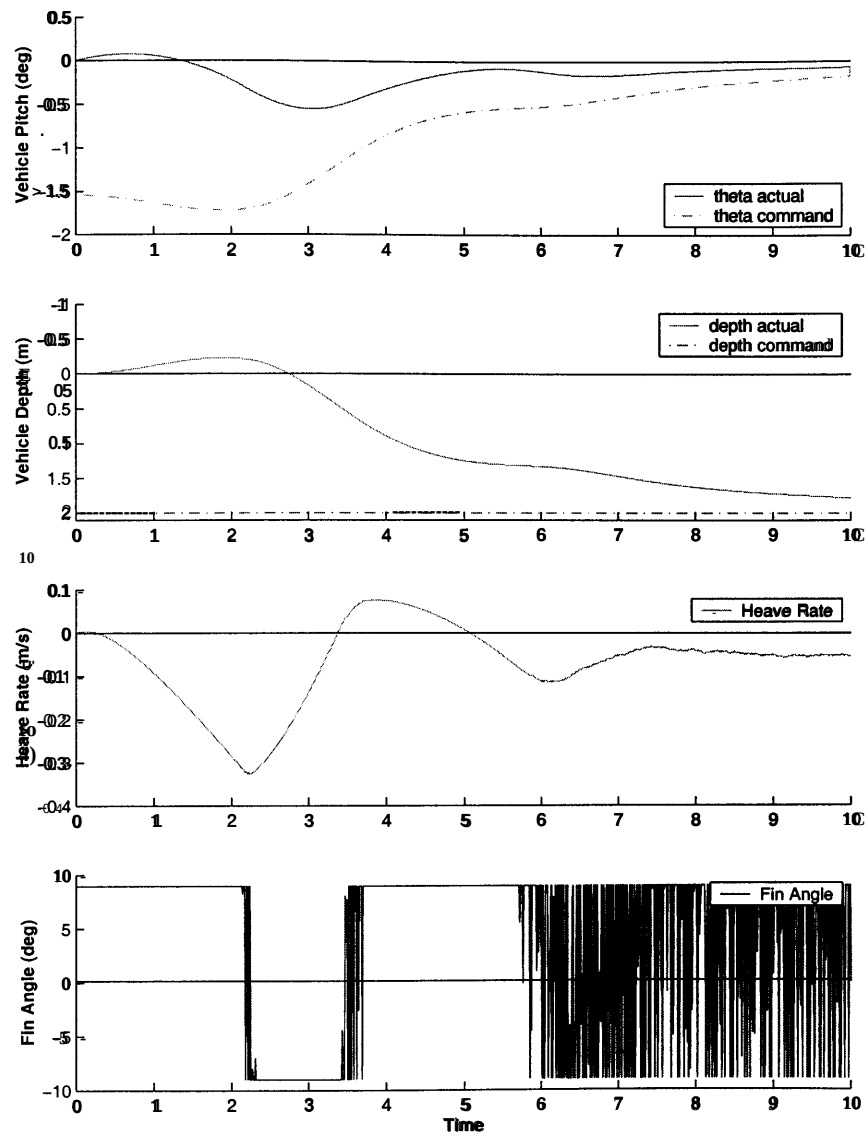


Figure 9-4 车辆俯仰饱和、随机纵倾传感器噪声以及1节垂直水流
 Figure 9-4 Vehicle Saturation, random pitch sensor noise, and 1 knot vertical current

Chapter 10

Conclusion

在论文中,我们展示了自主水下航行器动力学数学模型的开发过程。我们研究了验证该模型性能的方法。最后,我们已经看到这样的模型如何应用于航行器控制系统的开发。 developed methods for validating the performance of this model. Finally, we have seen how such a model can be applied to the development of a vehicle control system.

In the following sections, we outline a series of recommendations for expanding upon this work.

10.1 Expanded Tow Tank Measurements

在 the previous chapter, the author pointed out the nonlinearity of the most significant error is in the way coefficients are used to model the vehicle moving at some angle of attack, as shown in Figure 10-1. The fluid effects are

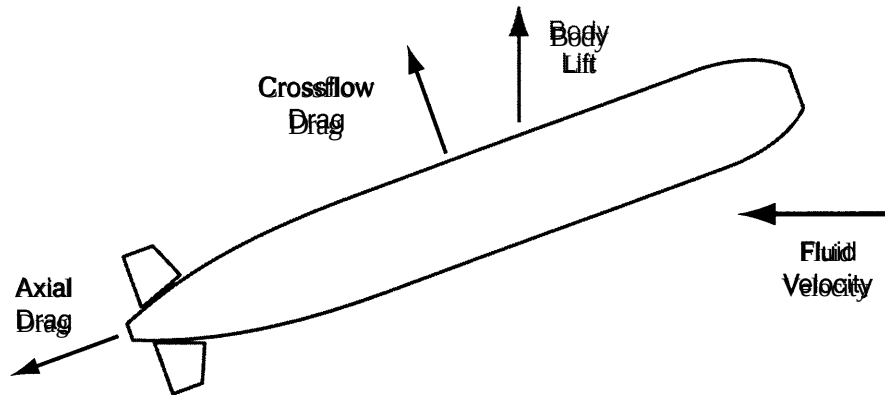


Figure 10-4: 10-ton vehicle Forces on the vehicle at an angle of attack

body lift up into the vehicle body and flow drag, and vehicle body lift up. The most difficult to estimate body lift.

The semi-empirical Brachdon [10] and Nilsen [24] methods使用的计算模型中系数的半经验方法。在本论文的研究过程中发现,结果相差 [23],数量级。因此,该实验测量车辆在迎角移动时的力和力矩,将对验证经验估计非常重要。It will therefore be important to experimentally measure forces and moments on the vehicle moving at an angle of attack, in order to verify the empirical estimates.

10.2 海上未来实验 Experiments at Sea

This is derived from the author's experience and consultation with experts in the field.

10.2.1 改进的车辆仪表 Improved Vehicle Instrumentation

在第 7 章中描述的实验中收集的车辆运动学数据, 在记录的车辆状态数量和测量的准确性方面都有限。这使得判断模型比较的有效性变得极为困难。those measurements. This made it extremely difficult to judge the validity of the model comparison.

在未来的实验中, 作者, 在第 7 章中描述的标准 REMUS 传感器基础 EM 增加了 sensors 惯性测量单元 Section 7.3 with IMU AHRS (动态测量单元姿态与航向参考系统)-AHRS (Dynamic Measurement Unit - Attitude and Heading Reference System)。该仪器可以输出三轴的磁力方向、加速度和角速度。通过该仪器, 我们将能够准确地测量或推导出以下车辆状态。From this instrument, we will be able to accurately measure or derive the following vehicle states.

$$\mathbf{x} = [\dot{u} \ \dot{v} \ \dot{w} \ u \ v \ w \ p \ q \ r \ \phi \ \theta \ \psi]^T \quad (10.1)$$

这将显著提高我们特别是测量车辆初始条件的能力, 以及总体测量车辆运动的能力。In particular, and the vehicle motion in general.

10.2.2 车辆参数的测量 Measurement of Vehicle Parameters

除了纵向重心 z_{cg} 是在每次实验前使用天平测量的之外, 作者对车辆质量分布的估算只是使用车辆重量清单计算的。同样, 车辆浮心的估算是使用 vehicle 船体形状计算的, were calculated using the vehicle weight list. Similarly, the estimates for the vehicle center of buoyancy were calculated using the Myring hull shape.

为了减少未来实验中的不确定性, 有必要通过实验测量这些数值, 最好在每次实验之前进行。experimentally, preferably before each experiment.

10.2.3 车辆运动的隔离 Vehicle Motion Isolation

由于模型验证的实验延迟响应数据的一个响应是, 对车辆初始条件的了解不齐全, 在进行实验时, 作者无法在鳍角变化之前测量车辆的加速度和角速度。At the time of the experiments, the author had no way of measuring the vehicle accelerations and angular rates prior to the changes in fin angles.

这些车辆运动来源于两个方面: 车辆控制输入和外部干扰。在未来的实验中, 应尽量减少这些运动。In future experiments, every effort should be made to minimize these motions.

为了最小化车辆控制动作, 理解车辆的稳态条件非常重要。under REMUS 这样的回 steady 航行器在无人控制操作时, 可能是稳定的。使用惯性测量单元, 可以确定导致车辆直线和平稳飞行的螺旋桨转速和固定鳍角。it will be possible to identify the propeller RPM and fixed fin angles which result in straight and level vehicle flight. These settings should then be used at the start of every experimental run.

为了尽量减少环境干扰, 实验可以在已知无水流区域进行, 并且在足够深的水域以避免自由表面相互作用。at sufficient depth to avoid free-surface interactions. Two locations meeting these criteria would be deep lakes and flooded sinkholes.

10.3 基于控制器的模型比较 Model Comparison Based on Controller

该车辆模型在开发车辆控制系统方面特别有用。in developing 将模型输出与在开环操作期间收集的, 的车辆数据进行比较。the model output to vehicle data collected during open-loop maneuvers. it would be more useful to compare the model to the vehicle behavior during closed-loop control. 这意味着需要, 要将车辆控制器作为模型代码的接口。与使用诸如车辆舵角等执行器状态不同, 模型输入将是命令状态, 例如期望深度或期望航向。Rather than actuator states such as the vehicle fin angles δ_s and δ_r , the model inputs would instead be the commanded states, such as desired depth z_d or desired heading ψ_d .

10.4 车辆传感器模型 Vehicle Sensor Model

开发车辆传感器模型将使我们能够在不必改进车辆传感器的情况下提高车辆性能。作为车辆拖曳水拖实验和实验的车辆 sensors. As part of the vehicle tow tank experiments and experiments

在海上的精密车辆惯性测量可以用于校准其他车辆传感器并估算它们的动态响应。vehicle sensors and estimate their dynamic response.

10.5 改进的基于系数的Efficient-Based Model

Mission planning for the shallow water vehicle task relies on the knowledge of the performance limits of these vehicles in terms of water depth and sea state. A map of this two-dimensional space for shallow water is illustrated in Figure 10-2. At present, these limits are not known.

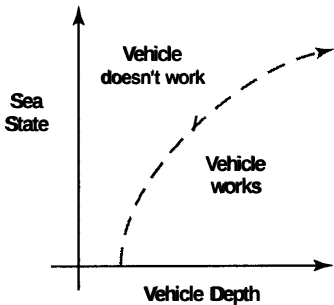


Figure 10-2 Vehicle performance function of depth and sea state

A vehicle model based on transfer functions for vehicle dynamics could provide an effective method for simulating the dynamics of underwater vehicles near the surf zone. The transfer functions for such a vehicle model would themselves be functions of the operating state of the vehicle, the water depth and the sea state.

These transfer functions could be derived from existing data sets and wave data or numerical vehicle models [26, 13, 34, 21].

Given such a model and a method for simulating the random motion of a vehicle, by predicting the probabilistic deviation of an underwater vehicle from a given desired trajectory, control system designers and mission planners could use this stochastic analysis to determine the operating limits of their vehicles.

Appendix A

Tables of Parameters

Table 1A. STB REMUS Vehicle Parameters

Parameter	Value	Units	Description
ρ	+1.03e+003	kg/m ³	Seawater Density
A_f	+2.85e-002	m ²	Hull Frontal Area
A_p	+2.26e-001	m ²	Hull Projected Area (xz plane)
S_w	+7.09e-001	m ²	Hull Wetted Surface Area
∇	+3.15e-002	m ³	Estimated Hull Volume
W	+2.99e+002	N	Measured Vehicle Weight
B	+3.08e+002	N	Measured Vehicle Buoyancy
B_{est}	+3.17e+002	N	Estimated Hull Buoyancy
$x_{cb}(est)$	+5.54e-003	m	Est. Long. Center of Buoyancy
C_d	+3.00e-001	n/a	REMUS Axial Drag Coeff.
C_{dc}	+1.10e+000	n/a	Cylinder Crossflow Drag Coeff.
$C_{y\beta}$	+1.20e+000	n/a	Hoerner Body Lift Coeff.
x_{cp}	-3.221e-001	n/a	Center of Pressure
α	+3.59e-002	n/a	Ellipsoid Added Mass Coeff.

Table 2A. Integration Limits for Limits of Integration

Parameter	Value	Units	Description
x_t	-7.221e-001	m	Aft End of Tail Section
x_{t2}	-2.18e-001	m	Forward End of Tail Section
x_f	-6.85e-001	m	Aft End of Fin Section
x_{f2}	-6.11e-001	m	Forward End of Fin Section
x_b	+4.37e-001	m	Aft End of Bow Section
x_{b2}	+6.10e-001	m	Forward End of Bow Section

Table 3A. Center of Buoyancy and Vehicle Nose

Parameter	Value	Units
x_{cb}	-6.11e-001	m
y_{cb}	+0.00e+000	m
z_{cb}	+0.00e+000	m

Table A: 相对于原点的重力 wrt Origin at CB

Parameter	Value	Units
x_{cg}	+0.00e+000	m
y_{cg}	+0.00e+000	m
z_{cg}	+1.96e-002	m

Table A: AERDES 参数 Parameters

Parameter	Value	Units	Description
S_{fin}	+6.65e-003	m ²	Planform Area
b_{fin}	+8.57e-002	m	Span
$x_{finpost}$	-6.33e-001	m	Moment Arm wrt Vehicle Origin at CB
α_{max}	+1.36e+001	deg	Maximum Fin Angle
a_{fin}	+5.14e+000	m	Max Fin Height Above Centerline
C_{mean}	+7.47e-002	m	Mean Chord Length
t	+6.54e-001	n/a	Fin Taper Ratio (Whicker-Felmer)
c_{df}	+5.58e-001	n/a	Fin Crossflow Drag Coefficient
AR_e	+2.21e+000	n/a	Effective Aspect Ratio
a	+9.00e-001	n/a	Lift Slope Parameter
CLQ	+3.12e+000	n/a	Fin Lift Slope

Appendix B

Tables of Combined Non-Linear Coefficients

Note that all coefficients are for the STC-RAMUS ship hull only. The values in Table B.1 are different from the values in Table 4.2 and 4.132. These values include the correction factors described in Section 8.1.2.

Table B.1: Non-Linear Force Coefficients

Parameter	Value	Units	Description
X_{2u}	-1.662e+0000	kg/m	Cross-flow Drag
X_{xi}	-9.30e-0001	kg	Added Mass
X_{uq}	-3.55e-0001	kg/rad	Added Mass Cross-term
X_{uq}	-1.93e+000	kg -m/rad	Added Mass Cross-term
X_{ur}	+3.55e+001	kg/rad	Added Mass Cross-term
X_{rr}	-1.93e+000	kg -m/rad	Added Mass Cross-term
X_{prop}	+3.86e+000	N	Propeller Thrust
Y_{vi}	-1.31e+0033	kg/m	Cross-flow Drag
Y_{rr}	+6.32e-001	kg -m/rad ²	Cross-flow Drag
Y_{uv}	-2.86e-0001	kg/m	Body Lift Force and Fin Lift
Y_{vi}	-3.55e-0001	kg	Added Mass
Y_{r}	+1.93e+000	kg -m/rad	Added Mass
Y_{ur}	+5.22e+000	kg/rad	Added Mass Cross Term and Fin Lift
Y_{wp}	+3.55e+001	kg/rad	Added Mass Cross-term
Y_{wpq}	+1.93e+000	kg -m/rad	Added Mass Cross-term
Y_{wdr}	+9.64e+000	kg/(m - rad)	Fin Lift Force
Z_{ww}	-1.31e+0022	kg/m	Cross-flow Drag
Z_{qq}	-6.32e-0001	kg -m/rad ²	Cross-flow Drag
Z_{uv}	-2.86e-0001	kg/m	Body Lift Force and Fin Lift
Z_{vi}	-3.55e-0001	kg	Added Mass
Z_{r}	-1.93e+000	kg -m/rad	Added Mass
Z_{uq}	-5.22e-0000	kg/rad	Added Mass Cross-term and Fin Lift
Z_{ur}	-3.55e-0001	kg/rad	Added Mass Cross-term
Z_{rp}	+1.93e+000	kg/rad	Added Mass Cross-term
Z_{wdr}	-9.64e-0008	kg/(m - rad)	Fin Lift Force

Table B: 北线号系数 Moment Coefficients

Parameter	Value	Units	Description
K_{pp}	-11.30e-0011	$\text{kg} \cdot \text{m}^2/\text{rad}^2$	Rolling Resistance
K_p	-7.004e-0002	$\text{kg} \cdot \text{m}^2/\text{rad}$	Added Mass
$K_{p\phi p}$	-5.43e-0001	$\text{N} \cdot \text{m}$	Propeller Torque
M_{uv}	+3.18e+000	kg	Cross-flow Drag
M_{qq}	-1.88e+002	$\text{kg} \cdot \text{m}^2/\text{rad}^2$	Cross-flow Drag
M_{uw}	+2.40e+001	kg	Body and Fin Lift and Munk Moment
M_{av}	-1.93e+000	$\text{kg} \cdot \text{m}$	Added Mass
$M_{\dot{q}}$	-4.88e+000	$\text{kg} \cdot \text{m}^2/\text{rad}$	Added Mass
M_{uq}	-2.00e+000	$\text{kg} \cdot \text{m}/\text{rad}$	Added Mass Cross Term and Fin Lift
M_{vp}	-1.93e+000	$\text{kg} \cdot \text{m}/\text{rad}$	Added Mass Cross Term
M_{rp}	+4.86e+000	$\text{kg} \cdot \text{m}^2/\text{rad}^2$	Added Mass Cross-term
M_{uads}	-6.15e+000	kg/rad	Fin Lift Moment
N_{qv}	-3.18e+000	kg	Cross-flow Drag
N_{rr}	-9.40e+001	$\text{kg} \cdot \text{m}^2/\text{rad}^2$	Cross-flow Drag
N_{uv}	-2.40e+001	kg	Body and Fin Lift and Munk Moment
$N_{\dot{q}}$	+1.93e+000	$\text{kg} \cdot \text{m}$	Added Mass
$N_{\dot{r}}$	-4.88e+000	$\text{kg} \cdot \text{m}^2/\text{rad}$	Added Mass
N_{uF}	-2.00e+000	$\text{kg} \cdot \text{m}/\text{rad}$	Added Mass Cross Term and Fin Lift
N_{vp}	-1.93e+000	$\text{kg} \cdot \text{m}/\text{rad}$	Added Mass Cross Term
N_{rp}	-4.86e+000	$\text{kg} \cdot \text{m}^2/\text{rad}^2$	Added Mass Cross-term
N_{uadr}	-6.15e+000	kg/rad	Fin Lift Moment

Appendix C

Tables of Non-linear Coefficients by Type

Note that all coefficients are calculated for the S-10 hull profile. Unlike those given in Tables 4.2 and 4.3, these values include the correction factors described in Section 8.1.2. 中给出的不同

Table C.1: Axial Drag Coefficient

Parameter	Value	Units
X_{U_u}	-1.622e+000	kg/m

Table C.2: Transverse Drag Coefficients

Parameter	Value	Units
$Y_{\dot{w}}$	-1.331e+003	kg/m
$Y_{\dot{\delta}_l}$	+6.32e+001	kg · m/rad ²
$Z_{\dot{w}}$	-1.331e+002	kg/m
$Z_{\dot{q}_d}$	-6.32e+001	kg · m/rad ²
$M_{\dot{w}}$	+3.18e+000	kg
$M_{\dot{\delta}_l}$	-1.888e+002	kg · m ² /rad ²
$N_{\dot{w}}$	-3.18e+000	kg
$N_{\dot{\delta}_l}$	-9.40e+001	kg · m ² /rad ²

Table C.3: Rolling Resistance Coefficient

Parameter	Value	Units
K_{pp}	-1.330e+001	kg · m ² /rad ²

Table C4: C机体Body和力矩系数Moment Coefficients

Parameter	Value	Units
Y_{uv}	-2.886e+001	kg/m
Z_{zu}	-2.886e+001	kg/m
M_{aoub}	-4.47e+000	kg
N_{aoub}	+4.47e+000	kg

Table C55: Added Mass Coefficients

Parameter	Value	Units
X_a	-9.30e+001	kg
X_b	+0.00e+000	kg
X_c	+0.00e+000	kg
X_p	+0.00e+000	kg · m/rad
X_q	+0.00e+000	kg · m/rad
X_r	+0.00e+000	kg · m/rad
Y_u	+0.00e+000	kg
Y_v	-3.55e+001	kg
Y_w	+0.00e+000	kg
Y_p	+0.00e+000	kg · m/rad
Y_q	+0.00e+000	kg · m/rad
Y_r	+1.93e+000	kg · m/rad
Z_u	+0.00e+000	kg
Z_v	+0.00e+000	kg
Z_w	-3.55e+001	kg
Z_p	+0.00e+000	kg · m/rad
Z_q	-1.93e+000	kg · m/rad
Z_r	+0.00e+000	kg · m/rad
K_b	+0.00e+000	kg · m
K_c	+0.00e+000	kg · m
K_{cb}	+0.00e+000	kg · m
K_p	-7.00e-0002	kg · m ² /rad
K_q	+0.00e+000	kg · m ² /rad
K_r	+0.00e+000	kg · m ² /rad
M_a	+0.00e+000	kg · m
M_b	+0.00e+000	kg · m
M_c	-1.93e+000	kg · m
M_p	+0.00e+000	kg · m ² /rad
M_q	-4.88e+000	kg · m ² /rad
M_r	+0.00e+000	kg · m ² /rad
N_a	+0.00e+000	kg · m
N_b	+1.93e+000	kg · m
N_c	+0.00e+000	kg · m
N_p	+0.00e+000	kg · m ² /rad
N_q	+0.00e+000	kg · m ² /rad
N_r	-4.88e+000	kg · m ² /rad

Table C: 附加质量交叉项系数 Cross-term Coefficients

Parameter	Value	Units
X_{q_q}	+0.00e+000	kg/rad
$X_{u_{qq}}$	-3.55e+001	kg/rad
$X_{u_{qr}}$	-1.93e+000	kg · m/rad
$X_{u_{rr}}$	+3.55e+001	kg/rad
X_{p_p}	+0.00e+000	kg · m/rad
$X_{r_{rr}}$	-1.93e+000	kg · m/rad
$X_{u_{rr}}$	+0.00e+000	kg/rad
$X_{u_{dr}}$	+0.00e+000	kg/rad
$X_{u_{qq}}$	+0.00e+000	kg/rad
$X_{p_{pq}}$	+0.00e+000	kg · m/rad
$X_{q_{qr}}$	+0.00e+000	kg · m/rad
Y_{u_r}	+0.00e+000	kg/rad
Y_{o_p}	+0.00e+000	kg/rad
Y_{rra}	+0.00e+000	kg · m/rad
Y_{rp}	+0.00e+000	kg/rad
Y_{rp}	+0.00e+000	kg · m/rad
Y_{u_p}	+0.00e+000	kg/rad
$Y_{u_{rr}}$	+0.00e+000	kg/rad
$Y_{u_{ra}}$	-9.30e-001	kg/rad
Y_{u_p}	+3.55e+001	kg/rad
Y_{p_q}	+1.93e+000	kg · m/rad
Y_{q_r}	+0.00e+000	kg · m/rad
$Z_{u_{qq}}$	+0.00e+000	kg/rad
$Z_{u_{qa}}$	+0.30e-001	kg/rad
$Z_{u_{qa}}$	+0.00e+000	kg · m/rad
Z_{rp}	-3.55e+001	kg/rad
Z_{rp}	+1.93e+000	kg/rad
Z_{rp}	+0.00e+000	kg · m/rad
Z_{rp}	+0.00e+000	kg/rad
Z_{rp}	+0.00e+000	kg/rad
Z_{rp}	+0.00e+000	kg/rad
Z_{rp}	+0.00e+000	kg/rad
Z_{rp}	+0.00e+000	kg · m/rad
Z_{rp}	+0.00e+000	kg · m/rad

Table C 附加质量、质量 K-Moment 交叉项系数 Coefficients

Parameter	Value	Units
K_{uu}	+0.00e+000	kg
K_{uq}	+0.00e+000	kg· m/rad
K_{uu}	+0.00e+000	kg
K_{uq}	+0.00e+000	kg· m/rad
K_{qq}	+0.00e+000	kg· m ² /rad ²
K_{vv}	+0.00e+000	kg
K_{vr}	+0.00e+000	kg· m/rad
K_{vp}	+0.00e+000	kg· m/rad
K_{rr}	+0.00e+000	kg· m ² /rad ²
K_{pp}	+0.00e+000	kg· m ² /rad ²
K_{uv}	+0.00e+000	kg
K_{qv}	+0.00e+000	kg
K_{vr}	+0.00e+000	kg· m/rad
K_{vp}	+0.00e+000	kg· m/rad
K_{ur}	+0.00e+000	kg· m/rad
K_{qr}	+0.00e+000	kg· m/rad
K_{qh}	+0.00e+000	kg· m ² /rad ²
K_{qr}	+0.00e+000	kg· m ² /rad ²

Table C.8. Added Mass M , N-Moment N , and Cross-term Coefficients

Parameter	Value	Units
M_{uq}	+0.00e+000	kg · m/rad
M_{uqa}	+1.93e+000	kg · m/rad
M_{uu}	+0.00e+000	kg
M_{wqa}	+0.00e+000	kg
M_{uwa}	+3.46e+001	kg
M_{ur}	+0.00e+000	kg · m/rad
M_{vp}	-1.93e+000	kg · m/rad
M_{pp}	+0.00e+000	kg · m ² /rad ²
M_{rp}	+0.00e+000	kg · m ² /rad ²
M_{rp}	+4.86e+000	kg · m ² /rad ²
M_{uv}^a	+0.00e+000	kg
M_{un}	+0.00e+000	kg
M_{up}	+0.00e+000	kg · m/rad
M_{ur}	+0.00e+000	kg · m/rad
M_{pqw}	+0.00e+000	kg · m/rad
M_{ur}	+0.00e+000	kg · m/rad
M_{pq}	+0.00e+000	kg · m ² /rad ²
M_{qr}	+0.00e+000	kg · m ² /rad ²
N_{as}	+0.00e+000	kg
N_{bu}	+0.00e+000	kg
N_{aq}	+0.00e+000	kg · m/rad
N_{wq}	+0.00e+000	kg · m/rad
N_{qq}	+0.00e+000	kg · m ² /rad ²
N_{aa}	+0.00e+000	kg
N_{ar}	+0.00e+000	kg · m/rad
N_{bp}	+0.00e+000	kg · m/rad
N_{rp}	+0.00e+000	kg · m ² /rad ²
N_{pp}	+0.00e+000	kg · m ² /rad ²
N_{uwa}	-3.46e+001	kg
N_{vw}	+0.00e+000	kg
N_{up}	+0.00e+000	kg · m/rad
N_{urq}	+1.93e+000	kg · m/rad
N_{wp}	-1.93e+000	kg · m/rad
N_{uq}	+0.00e+000	kg · m/rad
N_{pq}	-4.86e+000	kg · m ² /rad ²
N_{qr}	+0.00e+000	kg · m ² /rad ²

Table C.9. Propeller Terms

Parameter	Value	Units
X_{prop}	+3.86e+000	N
K_{prop}	-5.43e+001	N · m

Table C.10. Control Fin Coefficients

Parameter	Value	Units
Y_{uadr}	+9.64e+000	kg/(m·rad)
Z_{uuds}	-9.64e+000	kg/(m·rad)
M_{udsds}	-6.15e+000	kg/rad
N_{uadr}	-6.15e+000	kg/rad
Y_{uvf}	-9.64e+000	kg/m
Z_{uvf}	-9.64e+000	kg/m
Y_{urf}	+6.15e+000	kg/rad
Z_{uqf}	-6.15e+000	kg/rad
M_{uvff}	-6.15e+000	kg
N_{uvff}	+6.15e+000	kg
M_{uqf}	-3.93e+000	kg·m/rad
N_{urf}	-3.93e+000	kg·m/rad

Appendix D

线性化模型参数表
Linearized Model
Parameters

Note that all coefficients are for the STD REMUS hull profile.

Table D.1: Linearized Combined Coefficients			
Parameter	Value	Units	Description
Z_{dc}	-1.57e+001	kg/s	Crossflow Drag
Z_b	-3.45e+001	kg · m/s	Body Lift
Z_{if}	-1.64e+001	kg · m/s	Fin Lift
Z_{qc}	+1.20e-001	kg · m/s	Crossflow Drag
Z_{qa}	+1.44e+000	kg · m/s	Added Mass Cross Term
Z_{qf}	-1.12e+001	kg · m/s	Fin Lift
M_{wc}	-4.03e-001	kg · m/s	Crossflow Drag
M_{wa}	+5.34e+001	kg · m/s	Added Mass Cross Term
M_{wf}	-1.11e+001	kg · m/s	Body Lift
M_{syf}	-1.12e+001	kg · m/s	Fin Lift
M_{qc}	-2.11e+000	kg · m ² /s	Crossflow Drag
M_{qa}	+2.97e+000	kg · m ² /s	Added Mass Cross Term
M_{qf}	-7.68e+000	kg · m ² /s	Fin Lift

Table D 线性运动系数 Maneuvering Coefficients

Parameter	Value	Units	Description
X_0	+8.90e+000	kg · m/s ²	Hydrostatic
X_u	-1.35e+001	kg/s	Axial Drag
$X_{\dot{u}}$	-9.30e+001	kg	Added Mass
$X_{\dot{v}}$	-5.78e+001	kg · m/s	Added Mass Cross Term
Z_w	-6.66e+001	kg/s	Combined Term
Z_q	-9.67e+000	kg · m/s	Combined Term
$Z_{\dot{u}}$	-3.55e+001	kg	Added Mass
$Z_{\dot{v}}$	-1.93e+000	kg · m	Added Mass
$Z_{\dot{\delta}_s}$	-5.00e+001	kg · m/s ²	Fim Lift
M_0	-5.77e+000	kg · m ² /s ²	Hydrostatic
M_w	+3.07e+001	kg · m/s	Combined Term
M_q	-6.87e+000	kg · m ² /s	Combined Term
$M_{\dot{u}}$	-1.93e+000	kg · m	Added Mass
$M_{\dot{v}}$	-4.88e+000	kg · m ²	Added Mass
$M_{\dot{\delta}_s}$	-3.44e+001	kg · m ² /s ²	Fim Lift

Appendix E

MATLAB Code

E.1 Vehicle Simulation

REMUS模拟器程序是用MATLAB编写的。第一个程序REMUS_SSM.m加载车辆初始条件SSM，loads the vehicle initial conditions and tracks the vehicle state. The second program, REMUS.m, calculates the new vehicle accelerations based on the vehicle state and control inputs.

E.1.1 REMUS_SSM.m

```
% REMUS_SSM 车辆模拟器 Simulator

% 输入 + coeffs.mat - 由 COEFFS.M 生成，通常用于每次运行
% + vdata.mat - 由 COEFFS.M 生成，通常用于每次运行
% + coeffs.mat - generated by COEFFS.M, typically for each run
% + vdata.mat - generated by COEFFS.M, typically for each run

清除 % 清除所有变量 clear variables
clc ;

disp(sprintf('REMUS 动力学模拟器 (Simulator) by Timothy Prester, MIT/WHOI\n\n')) ;
disp(sprintf('注意：模型使用 Xs REMUS 尺寸。 \n\n', vehicle)) ;
disp(sprintf(' Timothy Prester, MIT/WHOI\n\n')) ;
load vehicle_type ;
disp(sprintf(' NOTE: Model using %s REMUS dimensions.\n\n', vehicle)) ;

% 检查系数、初始条件以及控制输入向量，并设置运行选项
% Check coeffs, initial conditions, and control input vector
if choose-setup == 'y'
    choose_setup = input(' Run set-up (y/n):','s') ; if choose_setup == 'y'
        sim_setup ;
    else
        showall ;
    end
end

% 输出标志
% Output flags
step = 1 ; show-speed run-savedata = 0 ; run-plots Xchoose
% setup = 1 ; = 0 showpos = 0 chooseint = 0 ; = 0 ;
show_step = 1 ; show_speed = 0 ; show_pos = 0 ;
run_savedata = 0 ; run_plots = 0 ; choose_int = 0 ;
% choose_setup = 1 ;
```

```

%
% 设置积分方法: 基本欧拉' 改进欧拉' 四阶龙格-库塔'; intmethods = {'eul
% 'imp-euler' 'rkutta'}; 如果 chooseint 显示(sprintf(' 积分方法:')) 对 i = 1:size(int-list,2) ;
显示(sprintf(' Basic Euler method is chosen')) 结束 Euler 输入(' Fourth Order') 否则 d=3 结
束 Euler 结束 char(intmethods) = {'euler' 'imp_euler' 'rkutta' }; if
choose_int
disp(sprintf(' Integration Method:\n')) ;
for i = 1:size(int_list,2) ;
disp(sprintf(' %i - %s', i, char(int_list(i)))) ;
end
d = input('\n Enter a number: ');
else
d = 3 ;
end int_method = char(int_methods(d)) ;

% 检查工作目录
cd_outputs ;

% 创建数据文件名称
yy = d(1); mo = d(2); dd = d(3); hh = d(4); mm = d(5);
ss = d(6); 日期字符串 = datestr(datenum(yy,mo,dd),1) 时间字符串 = datestr(datenum
(yy,mo,dd,hh,mm,ss),13) XX 生成随机文件名 X [dummy, file-string, dummy, du
mmy] = fileparts(tempname); disp(sprintf(' 当前模拟器数据文件:')); Xs *.mat; Xfile
% generate random filename
% tempname, input(sprintf(' 输入数据文件名称, s'), fileparts(tempname), O); file-str
ing = sprintf('%s\%s', tempstr(1:2), tempstr(4:6), tempstr(10:11), tempstr(13:14), tem
pstr(16:17)); disp(sprintf(' 数据文件已保存为 %s\Xs.mat', cd, file-string));
% file_string = input(sprintf('\nEnter name for data file: '), 's') ;
temp_str = datestr(now,0) ; file_string =
strcat('sim-', temp_str(1:2), temp_str(4:6), temp_str(10:11), '-', ...
temp_str(13:14), temp_str(16:17)) ;
disp(sprintf('\nData file saved as\n %s\%s.mat', cd, file_string));

% 实验给定值/ASSIGNMENT VALUE 输入向量
% -----
% 加载模型输入 del 在 SIMSEG 中生成 SIM 输入向量; X 数据来自 F
load inputs.mat; % data 时间步长 IN 初始状态; Xs 数据来自
INITIALCONDITIONS.M 上述内容
load initial_state ; % data from INITIAL_CONDITIONS.M on above

pitch_max=90 ;

% 运行模型 %X----- X 初始化步
数和存储矩阵 如果 strcmp(intmethod,'euler') n_steps = size(ui,2) 否则步数 = size(ui,
% initialize number of steps and storage matrix
n_steps = size(ui,2)-1; disp(sprintf('n 模拟器运行中.
..')); steps = size(ui,2) ;
else
n_steps = size(ui,2)-1 ;
end output_table = zeros(n_steps, size(x,1)+size(ui,1)+7);
disp(sprintf('\n Simulator running...'));

% 主程序 PROGRAM= 1:n_steps, X 打印 当前步骤 用于 错误检查 if show_step == 1
如果 i = 1:n_steps disp(sprintf('步骤完成: X02d XX', i/n_steps*100)); 结束
% Print current step for error checking
if show_step == 1
if ~rem(i*10,n_steps)
disp( sprintf(' Steps Completed : %02d %% ', i/n_steps*100));
end

```

```
% 计算当前状态current_state(x(n))以秒为单位的值, 0 and time_i, in 4 seconds
output_table(i,21) = (4-i)*time_step;
output_table(i,21) = (i-1)*time_step;
```

```
% Calculate forces, add iterations
% xdot(i) = f(x(i),u(i)) [xdot,forces] = remus(
% **,CALLS REMUS.M
%
% xdot(i) = f(x(i),u(i))
[xdot,forces] = remus(x,ui(:,i)');
```

```
% Store forces at step n, 15:20) = [f
output_table(i,15:20) = [forces'] ;
```

```

如果 strcmp(int_method,'euler') 欧拉积分用于计算新状态 x(n+1) XX x(i+1)=x(i)+dx/dt*delta_t X
%% EULER INTEGRATION to calculate new states x(n+1)
X注意：将旧状态覆盖为新状态，循环顶部保存 x=x+(xdot.*时间步长)；elseif strcmp(int_method,
%% x(i+1)=x(i)+dx/dt*delta_t
'impuler') XX 改进的欧拉积分用于计算新状态 k1_vec=x+(xdot.*时间步长)；k2_vec=remus(k1vec
%% NOTE: overwriting old states with new states, saving back at the top of the loop
,ui(:,i+1)) x=x+(0.5*时间步长.*(xdot+k2_vec) elseif strcmp(int_method,'rkutta') XX 龙格-库塔近似
x=x+(xdot.*time_step)；
elseif strcmp(int_method,'impuler') 应近似计算 k2,k3 的 ui 值 XX 即 (ui(:,i)+ui(:,i+1))/2 k1vec
= %IMPROVED EULER INTEGRATION step calculate (new states)；k3_vec=remus(x+(0.5.*时
间步长.*k2_vec),(ui(:,i)+ui(:,i+1))./2)；k4_vec=remus(x+(时间步长.*k3_vec), ui(:,i+1)) x=x+时
间步
x=x+0.5.*time_step.*(xdot+k2_vec)；
elseif strcmp(int_method,'rkutta')
%% RUNGE-KUTTA APPROXIMATION to calculate new states
%% NOTE: ideally, should be approximating ui values for k2,k3
%% ie (ui(:,i)+ui(:,i+1))/2
k1_vec=xdot；
k2_vec=remus(x+(0.5.*time_step.*k1_vec), ((ui(:,i)+ui(:,i+1))./2)')；
k3_vec=remus(x+(0.5.*time_step.*k2_vec), ((ui(:,i)+ui(:,i+1))./2)')；
k4_vec=remus(x+(time_step.*k3_vec), ui(:,i+1)')；
x=x+time_step/6.*(k1_vec+2.*k2_vec+2.*k3_vec+k4_vec)；
% k1_vec=xdot；
% k2_vec=remus(x+(0.5.*time_step.*k1_vec), ((ui(:,i)+ui(:,i+1))./2)')；
% k3_vec=remus(x+(0.5.*time_step.*k2_vec), ui(:,i)')；
% k4_vec=remus(x+(time_step.*k3_vec), ui(:,i)')；
% x=x+time_step/6.*(k1_vec+2.*k2_vec+2.*k3_vec+k4_vec)；
end

```

```
%-----  
% 模型系数矩阵以数值加载到 vehicle_parameters_loaded_in_REMUS_M  
% 模型系数矩阵以数值加载到 vehicle_parameters_loaded_in_REMUS_M，加载车辆类型、加载  
% 质量矩阵，加载车辆参数 vehicle_type ; load inv_mass_matrix ; load  
vehicle_coeffs ;
```

115


```
load vehicle_coeffs ; % non-zero vehicle coefficients only
```

```
% 输出标志 flags
show_forces = 10 ;
```

```
% 获取并检查状态变量和控制输入X获取状态变量
```

```
% -----  
% Get state variables  
u = x(1); x1 = x(2); w = x(3); p = x(4); q = x(5); r = x(6); phi = x(10); theta = x(11); psi = x(12);  
x2 = x(2); r = x(6); phi = x(10); theta = x(11); psi = x(12);
```

```
% 获取控制输入old_inputs=ui(1);deltar=ui(2)
delta_s = ui(1) ; delta_r = ui(2) ;
```

```
% (检查控制输入是否饱和) (use delta later)
if abs(delta_r) > delta_max * delta_max 结束如
果 delta_r > delta_max * delta_max
end
delta_r = sign(delta_r) * delta_max ;
end
```

```
% 初始化坐标变换矩阵的元素 coordinate system transform matrix
% -----
e1 = cos(phi); c2 = cos(theta); t1 = cos(psi); s1 = sin(phi); s2 = sin(theta);
s3 = sin(psi); t2 = tan(theta);
```

```
% Set up force from equations of motion
%
X = -(W/B)*sin(theta)*u*cos(alpha)*(Kq*(Xq-m)*wq + (Xqq +
m*g)*q^2 ...
+ ((Xv+m)*r*xr + (Xvrg+2m*xg)*vr^2 - g)*p*qXprop - zg*p*r ...
+ Xprop ;
```

$$Y = (W-B)*\alpha\beta(\text{cheat})*\sin(\text{pribs}(y)) + Yvv*r*abs(y) + Yrr*r*abs(r) + Yuv*u*v \dots + (Ywp+m)*w*p + (Yur+n)*u*r + (m*xg*xg)*q*(y*p) - (Ypg)*p*xg + p*p*du \dots + u^2*\text{delta}_r;$$
[illegible]
$$\begin{aligned} K = & -(g^2 w^2 y^2 B^2) * (d_1 d_2 (\theta) \sin(\phi) \cos(\phi w) z^2 B) * \cos(\theta) * \sin(\phi) \dots + Kpp * p^a \\ & (g^2 w^2 z^2 B^2) * \cos(\theta) * \sin(\phi) z g * u^r + Kprop \\ & + Kpp * p_{abs}(p) - (Izz - Iyy) * q^r - (m * z g) * w^p + (m * z g) * u^r + Kprop ; \end{aligned}$$
[illegible]

```

N = -(xgqwwxrb)Biescdet(henq)fsin(gni)yb*Biygn(hvya)B)*sin(theta(v)+.
Nr*(Nur*(y+da(v))+.Nr*xpab(s)lxt)Nwq*(Nwp-m*xg)*w*p+(Nur+m*x
g)*u*(Np(Nuadr*xxl)*p;q+(Nwp-m*xg)*w*p+(Nur+m*xg)*u*r...
+Nuadr*u^2*delta_r;

```

$$FORCES \neq [XKYMZNQ' M N] ;$$

```

ACCELERATIONS(1)*X+ Minv(1,2)*Y+ Mflv(1,3)*Z+ Milv(1,4)*K+ Milv(1,5)*M+ Miflv(1,6)*N
Miflv(2,1)*X+ Minv(2,2)*Y+ Mflv(2,3)*Z+ Milv(2,4)*K+ Milv(2,5)*M+ Miflv(2,6)*N
+ Minv(3,2)*X+ Minv(3,3)*Z+ Milv(3,4)*K+ Milv(3,5)*M+ Miflv(3,6)*N
Y+ Minv(4,3)*Z+ Milv(4,4)*K+ Milv(4,5)*M+ Miflv(4,6)*N
) *Z+ Minv(5,4)*K+ Minv(5,5)*M+ Miflv(5,6)*N
)*Z+ Minv(6,4)*K+ Minv(6,5)*M+ Miflv(6,6)*N
)*K+ Minv(6,5)*M+ Miflv(6,6)*N
c3+s1*s2*s3)*v + (c1*s2*s3-c3*s1)*w
c2*c3+c1*c2*c3) *v + (c1*s2*s3-c3*s1)*w
s3*c2*u + (c1*c3+s1*s2*s3)*v + (c1*s2*s3-c3*s1)*w
-s2*u + c2*s1*v + c1*c2*w
p + s1*t2*q + c1*t2*r
c1*q - s1*r
s1/c2*q + c1/c2*r] ;

```

Appendix F

Example REMUS Mission File

以下是个REMUS任务文件示例 取自1999年7月27日进行的论文现场实验 本次实验的具体目标是测量: The goals of this particular experiment were to measure:

- the vehicle behavior with forced fins 车辆对尾舵平面和舵角阶跃变化的响应 车辆转弯半径与稳态舵角的关系
- the vehicle response to step changes in stern plane and rudder angle
- the vehicle turn radius as a function of steady-state rudder angle
- the vehicle roll offset as a function of propeller RPM

请参阅第4.7.3节以了解任务编程的详细信息。

F.1 REMUS Mission Code

[Types of objectives] 延迟>秒后完成 事件等待=等待标志被设置 死推算=将车辆死推算到 LAT/LON 设置深度=等待车辆达到指定深度后继续 使用声呐=使用声呐将车辆回标到 LAT/LON 测试声呐=在选定通道生成测试声呐 等待应答器=等待应答器将车辆回标到 LAT/LON 将车辆推进器=等待车辆推进器旋转后继续 指南针校准=对 PNI 指南针进行水中校准 长基线=使用与两个应答器的测距导航 车辆 LBL 行=

Dead reckoning=Relies on the vehicle's LAT/LON goal

Set position=Sets the position of the vehicle to a LAT/LON goal

Wait depth=Waits until the vehicle is deeper than a depth before continuing

Transponder Home=Uses a transponder to home the vehicle to a LAT/LON goal

Test ping=Generates a test ping on the selected channel.

Wait transponder=Waits until acquires the selected transponder

Dock=Docks the vehicle

Undock=Undocks the vehicle

ATS diagnostic=Creates ATS matlab file for diagnostic purposes

Surface=Surfaces the vehicle

Selftest=Performs offline vehicle diagnostics

Wait prop=Waits until the vehicle's prop is spun before continuing

Compass cal=Does an in water calibration of the PNI compass

Long Baseline=Uses range to 2 transponders to navigate the vehicle

LBL rows=Uses LBL nav to mow the lawn

[Objective] 设置 位置 目的地 名称

Set position 目的地 经度

Destination longitude=

Offset direction=131

Offset distance (Meters)=0

Offset Y axis (Meters)=0

```

【Objective】等待 属性 必需 转速=60
type=wait 深度传感器=否
Required rpm=60
Auto calibrate depth sensor=NO

【Objective】到达深度 (20) START 向 DEPTH (20) AND READING 线130 地 #####
type=wait 保持当前指令=否 直接 (俯仰 方向舵 推进器)=无 转速
Delay 150.0 seconds=120.0 航向指令=130.0
Keep current commands=NO
Direct (pitch rudder thruster)=none
RPM command=1500.0
Depth command=2.0
Heading command=130.0

【Objective】制水# CONTROLLED#LEV类型FLIGHT器延#####
type=wait 保持当前指令=否 直接 (俯仰 方向舵 推进器)=无 转速
Delay 150.0 seconds=120.0 航向指令=130.0
Keep current commands=NO
Direct (pitch rudder thruster)=none
RPM command=1500.0
Depth command=2.0
Heading command=130.0

【Objective】清零 所有ZERO#ALL#RINS#####
type=wait 保持当前指令=否 直接 (俯仰 方向舵 推进器)=无 转速
Delay 150.0 seconds=120.0 航向指令=130.0
Keep current commands=NO
Direct (pitch rudder thruster)=pitch rudder
Direct pitch command=0
Direct rudder command=0
RPM command=1500.0

【Objective】深度 2米的IMC器类DEPT时延
type=wait 保持当前指令=否 直接 (纵倾 方向舵 推进器)=无 转速
Delay 150.0 seconds=120.0 航向指令=130.0
Keep current commands=NO
Direct (pitch rudder thruster)=none
RPM command=1500.0
Depth command=2.0
Heading command=130.0

【Objective】俯仰####2PITCH DOWN#2###READING 时#####
type=wait 保持当前指令=否 直接 (俯仰、方向舵、推进器)=直
Delay 150.0 seconds=120.0 航向指令=130.0 转速指令=1500.0
Keep current commands=NO
Direct (pitch rudder thruster)=pitch
Direct pitch command=10
Heading command=130.0
RPM command=1500.0

```

【Objective】定时器到TIMER TO DEPTH 6M
 type=Timer 保持当前命令=否 直接（俯
 Delay in seconds=90.0 深度命令
 Keep current commands=NO
 Direct (pitch rudder thruster)=none
 RPM command=1500.0
 Depth command=6.0
 Heading command=130.0

【Objective】仰 2# 航向类型定时 HEADING
 type=Timer 保持当前命令=否 直接（俯仰 方向舵
 Delay in seconds=10.0 航向命令=1
 Keep current commands=NO
 Direct (pitch rudder thruster)=pitch
 Direct pitch command=-10
 Heading command=130.0
 RPM command=1500.0

【Objective】定时器到TIMER TO DEPTH 2M
 type=Timer 保持当前命令=否 直接（俯仰
 Delay in seconds=90.0 深度命
 Keep current commands=NO
 Direct (pitch rudder thruster)=none
 RPM command=1500.0
 Depth command=2.0
 Heading command=130.0

【Objective】下俯##航向#DOWN#4类型HEADING #####
 type=Timer 指令=否 直接（俯仰 方向舵 推进器）= 俯仰 直接 俯
 Delay in seconds=6.0 转速 指令=1500.0
 Keep current commands=NO
 Direct (pitch rudder thruster)=pitch
 Direct pitch command=20
 Heading command=130.0
 RPM command=1500.0

【Objective】深度6#定时器到TIMER TO DEPTH 6M
 type=Timer 保持当前指令=否 直接控制（俯
 Delay in seconds=90.0 无 转速指令=1500.0
 Keep current commands=NO
 Direct (pitch rudder thruster)=none
 RPM command=1500.0
 Depth command=6.0
 Heading command=130.0

【Objective】仰 4# 航向类型定时 HEADING
 type=Timer 保持当前指令=否 直接（俯仰方向舵
 Delay in seconds=90.0 航向 指令=-20 航向 指令=1
 Keep current commands=NO
 Direct (pitch rudder thruster)=pitch
 Direct pitch command=-20
 Heading command=130.0
 RPM command=1500.0

【Objective】定时器到TIMER TO DEPTH 3M
 type=Timer =90.0 保持当前命令=否
 Delay in seconds=90.0
 Keep current commands=NO

```

Direct (pitch rudder thruster)=none e
RPM command=1500.0深度命令
Depth command=3.0
Heading command=130.0

[Objective]俯仰定时器DOWN#6#HEADING ##### #
type=Timer定时器延迟(秒)=3.0保持当前命令=否
Delay (秒)=3.0直接(俯仰)=俯仰直接俯仰
Keep current commands=NO
Direct (pitch rudder thruster)=pitch
Direct pitch command=20
Heading command=130.0
RPM command=1500.0

[Objective]定时器到深度DEPT#6#
type=Timer=90.0保持当前命令=否直接(
Delay (秒)=90.0直接(俯仰)=无
Keep current commands=NO
Direct (pitch rudder thruster)=none e
RPM command=1500.0深度命令
Depth command=3.0
Heading command=130.0

[Objective]向上俯仰PTC航向类型HEAD#1#
type=Timer保持当前命令=否直接(俯仰方向舵
Delay (秒)=4.0直接(俯仰)=俯仰直接俯仰命令=-20航向命令=1
Keep current commands=NO
Direct (pitch rudder thruster)=pitch
Direct pitch command=-20
Heading command=130.0
RPM command=1500.0

[Objective]到深度TIMER类型DEPT#6#
type=Timer单位=90.0保持当前指令=否直接
Delay (秒)=90.0直接(俯仰)=无
Keep current commands=NO
Direct (pitch rudder thruster)=none e
RPM command=1500.0深度命令
Depth command=3.0
Heading command=130.0

[Objective]舵轮#1#RUDDER#PORT, FIXED PITCH ##### #
type=Timer定时器延迟(秒)=20.0保持当前命令=否直接(
Delay (秒)=20.0直接(俯仰)=俯仰直接俯仰命令=10直接舵命
Keep current commands=NO
Direct (pitch rudder thruster)=pitch rudder
Direct pitch command=10
Direct rudder command=40
RPM command=1500.0

[Objective]定时器到深度3M, DEPT#3M, HEADING 130 零
type=Timer定时器延迟(秒)=120.0保持当前命令
Delay (秒)=120.0直接(俯仰)=无转速命
Keep current commands=NO
Direct (pitch rudder thruster)=none
RPM command=1500.0
Depth command=3.0
Heading command=130.0

```

[Objective] 向舵机 RUDDER 输出距离 STEER 并延迟 10 秒
 type=Timer 保持当前指令=否 直接 (桨距方向舵推进器)=桨距
 Delay 直接桨距指令=20.0 直接方向舵指令=-40 转速指令=1500.
 Keep current commands=NO
 Direct (pitch rudder thruster)=pitch rudder
 Direct pitch command=0
 Direct rudder command=-40
 RPM command=1500.0

[Objective] 定时器 # 深度 DEPTH 航向 HEADING 类型=HEADING 延迟 130
 type=Timer 保持当前指令=否 直接 (俯仰 舵机 推进器) =
 Delay 指令=seconds=3.0 航向指令=130.0
 Keep current commands=NO
 Direct (pitch rudder thruster)=none
 RPM command=1500.0
 Depth command=3.0
 Heading command=130.0

[Objective] 方向舵 RUDDER 深度 DEPTH
 type=Timer 保持当前指令=否 直接 (俯仰 舵
 Delay) in seconds=120.0 深度指令=3.0 转速
 Keep current commands=NO
 Direct (pitch rudder thruster)=rudder
 Direct rudder command=0
 Depth command=3.0
 RPM command=1500.0

[Objective] 螺旋桨 SPIN DOWN PROP 1250- #####
 type=Timer 保持当前指令=否 直接 (俯仰 方向舵 推进
 Delay 无转 seconds=120.0 深度指令=3.0 航向指令=130.0
 Keep current commands=NO
 Direct (pitch rudder thruster)=none
 RPM command=1250.0
 Depth command=3.0
 Heading command=130.0

[Objective] N 螺旋桨 SPIN DOWN PROP 1000
 type=Timer 保持当前指令=否 直接
 Delay 无转 seconds=120.0 RPM 指令=1000.
 Keep current commands=NO
 Direct (pitch rudder thruster)=none
 RPM command=1000.0
 Depth command=3.0
 Heading command=130.0

[Objective] 螺旋桨 SPIN DOWN PROP 750
 type=Timer 保持当前指令=否 直接 (俯仰 舵机 推进器)
 Delay 无转 seconds=120.0 深度指令=750.0 深度
 Keep current commands=NO
 Direct (pitch rudder thruster)=none
 RPM command=750.0
 Depth command=3.0
 Heading command=130.0

[Objective] N 螺旋桨 SPIN DOWN PROP 500
 type=Timer 保持当前命令=否 直接
 Delay 无转 seconds=120.0
 Keep current commands=NO
 Direct (pitch rudder thruster)=none

Speed command=50.0度命令
Depth command=30.0
Heading command=130.0

[Object type]
type=END

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